JURALCO

JURALCO VIKING®
BALUSTRADE SYSTEM

ISSUE 1-25 v1

Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

Auckland Council
Te Kaunihera o Tāmaki Makaurau

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Viking® Aluminium Balustrade System
Supplied to:	Auckland Council
Client / designers name:	Josh Harrison
Description of work: *E.g. New dwelling*	12mm thick Glass Balustrade installation for External Balcony area using Channel System
Address of property:	83 Speights Road, St Heiliers, Auckland 1071

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☑ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ *please specify*
Location of deck / balcony / stairs:	☐ Internal	☑ External	☐ *please specify*		
Deck / balcony / stairs:	☑ New	☐ Existing	Cladding:	☑ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☑ Toughened	☑ Toughened laminated	☐ Toughened laminated with stiff interlayer
Thickness:	12mm (mm)	Width of pane:	0.5-1.8 (m)
Height above floor:	(m)	Height above fixing:	(m)

Note: If barrier extends more than 1.2m above fixing point, test reports must be provided

Supplier:	Royal Glass

PART 4: BARRIER DETAILS

System name:	Viking® Aluminium Balustrade System
Drawings (ref page number)	
Engineers name:	Lautrec Technology Group
Hardware (type):	Aluminium Clamps

Drawings <u>must</u> show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES *(tick all boxes that apply)*

- [x] Continuous interlinking rail (show how fixed to building and each pane of glass)
- [ ] Continuous interlinking top cap (show how fixed to building and each pane of glass)
- [ ] Clamps, brackets or point fixings (show how fixed to building and each pane of glass)
- [ ] Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass)
- [ ] Glass with stiff interlayer
- [ ] Two or three glass edges supported by continuous clamps
- [ ] Structural post to fix interlinking rail to deck
- [ ] Other (specify)

Engineers name:	Lautrec Technology Group
Drawings (ref page number)	
Comments:	

PART 6: COVER SHEET COMPLETED BY

Name:	Roxy Huang
Signature:	
Date:	14/05/2025
Contact number:	0800 769 254
Email:	info@royalglass.co.nz

NAME:

ADDRESS:

DATE: **CONSENT AUTHORITY: Auckland Council**

THIS PS1 DOES NOT CONSTITUTE A SITE SPECIFIC REVIEW BY THE ENGINEERING COMPANY BELOW AND IS ONLY APPLICABLE FOR STANDARD FIXINGS AND DESIGN PARAMETERS AS DETAILED IN THE RELEVANT MANUAL

PRODUCER STATEMENT – PS1 DESIGN

BUILDING CODE CLAUSE(S): B1, F2, F4, F9 **JOB NUMBER:** 1392-1803

ISSUED BY: Lautrec Technology Group Limited
(Engineering Design Firm)
TO: Juralco Aluminium Building Products Ltd
(Owner/Developer)
TO BE SUPPLIED TO: All Building Consent Authorities in NZ
(Building Consent Authority)
IN RESPECT OF: Juralco Viking® Balustrade System
(Description of Building Work)
AT: N/A (all locations within NZ Wind Zone)
(Address, Town/City)
LEGAL DESCRIPTION: **N/A** ☑

We have been engaged by the owner/developer referred to above to provide *(Extent of Engagement)*: Specific engineering design for structural components only in respect of the requirements of the Clause(s) of the Building Code specified above for Part only, as specified in the Schedule, of the proposed building work.

The design carried out by us has been prepared in accordance with:

- ☑Compliance documents issued by the Ministry of Business, Innovation & Employment *(Verification method/acceptable solution)* B1/VM1 and/or;
- ☑Alternative solution as per the attached Schedule.

The proposed building work covered by this producer statement is described on the drawings specified in the Schedule, together with the specification, and other documents set out in the Schedule.

On behalf of the Engineering Design Firm, and subject to:

- Site verification of the following design assumptions: assumed adequate support structure by others.
- All proprietary products meeting their performance specification requirements;

I believe on reasonable grounds that:

- the building, if constructed in accordance with the drawings, specifications, and other documents provided or listed in the Schedule, will comply with the relevant provisions of the Building Code and that;
- the persons who have undertaken the design have the necessary competency to do so.

I recommend the **CM1** level of **construction monitoring**.

I, *(Name of Engineering Design Professional)* Kevin Brown, am:

- ☑CPEng number 140404

and hold the following qualifications BE, CMEngNZ, CPEng, IntPE(NZ), MBA

The Engineering Design Firm holds a current policy of Professional Indemnity Insurance no less than $200,000
The Engineering Design Firm **Is** a member of ACE New Zealand.

SIGNED BY *(Name of Engineering Design Professional)*: Kevin Brown
(Signature below):

Kevin Brown

ON BEHALF OF *(Engineering Design Firm)*: Lautrec Technology Group Limited Date: 05/02/2026

Note: This statement has been prepared solely for the Building Consent Authority named above and shall not be relied upon by any other person or entity. Any liability in relation to this statement accrues to the Engineering Design Firm only. As a condition of reliance on this statement, the Building Consent Authority accepts that the total maximum amount of liability of any kind arising from this statement and all other statements provided to the Building Consent Authority in relation to this building work, whether in tort or otherwise, is limited to the sum of $200,000.

This form is to accompany **Form 2 of the Building (Forms) Regulations 2004** for the application of a Building Consent.

SCHEDULE to PS1

Please include an itemised list of all referenced documents, drawings, or other supporting materials in relation to this producer statement below:

Structural Maintenance Schedule refer to the relevant Juralco product manuals

B2 Letter

Engineering Drawings	**Refer to product manual Juralco Viking® Balustrade System Issue 1-25**
Engineering Calculations	**Refer to PS1 Report - Juralco Viking® Balustrade System_11.02.25**

Alternative Solution
The design carried out by Lautrec Technology Group Limited has been prepared in accordance with the following alternative solutions:

NZS1170.2:2021 Incorporating Amendment No. 1
AS/NZS 1664.1:1997
NZS AS 1720.1:2022
NZS 3101:2006
NZS 3404:1997
EOTA TR 049 and EN1992-4:2018 for post-installed anchor design
NZS 4223.3:2016
NZS 4223.4:2008
NZS 8500:2016

Lautrec Technology Group Ltd.
Professional Consulting Engineers

11/02/2025

To the Building Official,

B2 COMPLIANCE - JURALCO VIKING® BALUSTRADE SYSTEM
at ***occupancy categories A, B, E, and C3;*** at ***any location in New Zealand that falls within the scope of this PS1 - see supporting PS1 report for scope***

We have been asked to provide a PS1 for Clause B2 of the Building Code - Structural Durability

We are not able to provide this because there is no effective verification method for B2 contained within the New Zealand Building Code.

As these systems can be installed in a variety of settings, including internal and exposed environments, it is not deemed practical to specify durability requirements for the sub structure. Timber treatments, mild steel corrosion protection coatings, and concrete and masonry covers are therefore up to the building designer to specify in accordance with the relevant recognised standards.

However, we can confirm that for the structural elements shown in the attached documentation:

Material	**Means of compliance**	**Details**
Juralco Viking® Balustrade System - Aluminium and Stainless Steel	Alternative Solution	Stainless steel components conform to SS316. Aluminium extrusions, components, and hardware conform to 6060 T5. Non-structural cladding has a durability requirement not less than 50 years for this project. Note that the industry proven life of Aluminium, Stainless Steel, and Glass would satisfy a 50-year design life requirement. We note that this is on a time to first maintenance basis.
Connections - Stainless Steel fixings	Alternative Solution	All bolt and screw fixings within Juralco Viking® Balustrade System shall be minimum SS316. Non-structural fixings that are difficult to replace has a durability requirement not less than 50 years for this project. Note that the industry proven life of Stainless Steel would satisfy a 50-year design life requirement.

It is assumed that these structural elements are fixed to adequate structures by others.

Minor tea staining may occur in coastal environments. Refer to Juralco Viking® Balustrade System manual issue 1-25 v1 for the supplier's maintenance requirements.

Yours faithfully,

Managing Director
Kevin Brown
BE, CMEngNZ, CPEng, IntPE(NZ), MBA

Complies With AS/NZS 1170:2002, NZS 4223.3.2016, NZ Building Code B1, B2, F2 ,F4 and F9
Complies with French Standard NF P01-013 (1988-08)

Viking® Balustrade is for Occupancy types A, A Other and C3 only
Occupancy Types as per AS/NZ 1170.1.2002.

Code	Type of Occupancy for part of the building or structure	Specific Uses	Glass
A	Domestic and Residential activities	All areas within or serving exclusively one dwelling including stairs, landings etc, but excluding external balconies and edges of roofs.	6mm and 10mm Toughened Glass
A Other, C3	Areas without obstacles for moving people and not susceptible to over crowding	Stairs, landings, external balconies, edges of roofs etc.	

Note 1	**All for 6mm and 10mm, Toughened Glass. Glass must have a minimum strength of 100MPa. All edges polished**
Note 2	**Juralco Balustrade Systems building code compliance documentation requires all balustrade installations are to be completed in accordance with the requirements of our authorised installer certification.**
Note 3	**All Semi Frameless glass Balustrades must have a Top or Side mounted handrail to conform to NZS 4223.3.2016.**

masterspec partner
Section 4852JB

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Juralco Aluminium Building Products Ltd (JABP)

Specifications for Juralco Viking® Balustrade System

1.Scope

- This specification details the documents the Juralco Viking® Balustrade System refers to in relation to the New Zealand Building Code, the manufacturer's documents, products used in the System, requirements in relation to fixing and surface finishings.

2. NZBC Compliance

- The Juralco Viking® Balustrade System has been reviewed by Lautrec Technology Group Ltd, to demonstrate compliance with the structural requirements of the New Zealand Building Code and AS/NZS 1170 : 2002 occupancy A, A Other and C3. NZS 3604 Wind Zones, up to and including Low, Medium, High, Very High and Extra High.
- The Structural Engineering design includes the requirements of B1 Structure, B2 Durability, F2 Hazardous Materials and F4 Safety from falling, from the Building Code.
- Verification Method B1 / VM1, B2/AS1, F4 / AS1
- All glass used in the Juralco Viking® Balustrade System must conform to AS/NZS 2208.
- Complies with NZS 4223.3.2016

3. Manufacturer's Documents

- The Juralco Viking® Balustrade System manual details all extrusions and components used for the fabrication and installation/fixing of the system.
- A Producer Statement 1(Design) is available.
  Copies of the above documents are available from:
  Juralco Aluminium Building Products Ltd
  48 Bruce McLaren Rd, Henderson, Auckland
  Phone 09 478 8018 Fax 09 478 7883 Email specify@juralco.co.nz
- Any deviation from the standard fabrication or installation/fixing must be accompanied by a site specific PS1 with site specific calculations and drawings

4. Products

- Only extrusions, components and hardware supplied by or specified by JABP may be used in the Juralco Viking® System
- Aluminium extrusions, components and hardware – unless specified are manufactured to 6060 T5 or T6 specifications
- Stainless Steel components, hardware, fixings – all components to 316 grade
- Glass - all glass used in the Juralco Viking® Balustrade System must conform to the specifications as listed in the Juralco Viking® manual with each panel conforming to AS/NZS 2208 as confirmed by the Safety Stamp detailing the manufacturer's description and licence number.

5.Surface Finishing

- Juralco Aluminium Building Products Ltd is a Dulux Registered Applicator site, registration number 2101. JABP uses only Dulux branded powder coating materials
- Dulux Duralloy® powder coating systems are suitable for properties greater than 100m from high tide level AAMA 2603 performance. Residential buildings, 3 levels max. Warranty 10 yrs
- Dulux Duralloy Plus® powder coating systems are suitable for properties greater than 10m from high tide level. AAMA 2603 performance. Residential and Light commercial buildings, 3 levels max Warranty 15 yrs
- Dulux Duratec® powder coating systems are suitable for properties greater than 10m from high tide level AAMA2603 and 2604 performance. All Residential and Commercial buildings. Warranty 25 yrs

6. Installation and Fixing

- The Juralco Viking® Balustrade System must only be installed in accordance with the Juralco Viking® Balustrade System manual
- Any deviation from that specified in the Juralco Viking manual must only be in accordance with the site specific PS1 with site specific calculations and drawings listing the non standard details
- The Juralco Viking® Balustrade System must only be fabricated/installed by a Juralco approved fabricator
- Upon completion of the installation the fabricator must supply the owner with a PS3 (Construction)

Important information - Powder Coating systems.

Powdercoat Systems The new standard Dulux powder coating system used by Juralco is Duralloy Plus®. Also Duralloy® and Duratec®. All as per specs above. Juralco Powder coated prices are for Duralloy Plus® and Duralloy® (same pricing). Duratec® prices on application.

Attachment to structures A PVC Tape or similar material spacer must be used to separate powder coated aluminium items from all concrete and steel structures. Failure to do so can lead to the chemicals in the structure affecting the powder coating, leading to corrosion.

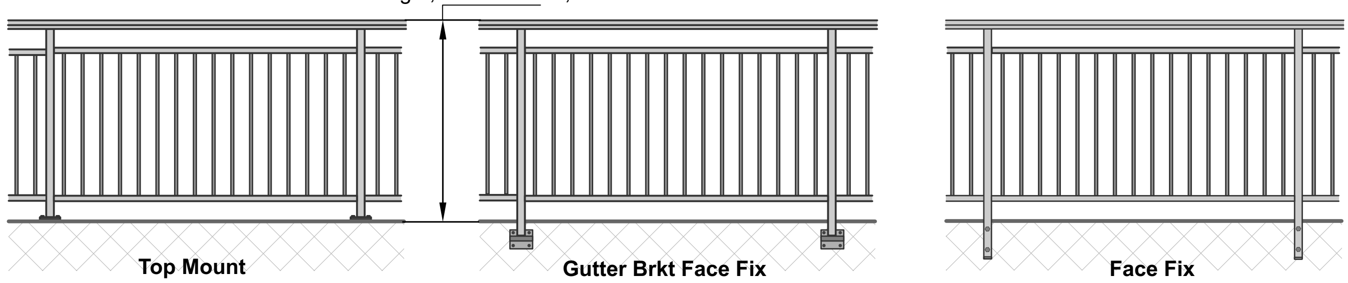
Swimming Pools The chlorinated water in swimming pools can cause the deterioration of powder coated surfaces, leading to corrosion of the underlying surface. It is recommended that Powder coated surfaces be 1200mm min from a pool.

Care The Dulux powder coating warranty period is conditional upon the surface being maintained in accordance with the Dulux 'Care and Maintenance Instructions'. Download from Dulux or refer to the back page of this manual.

For 17mm Baluster

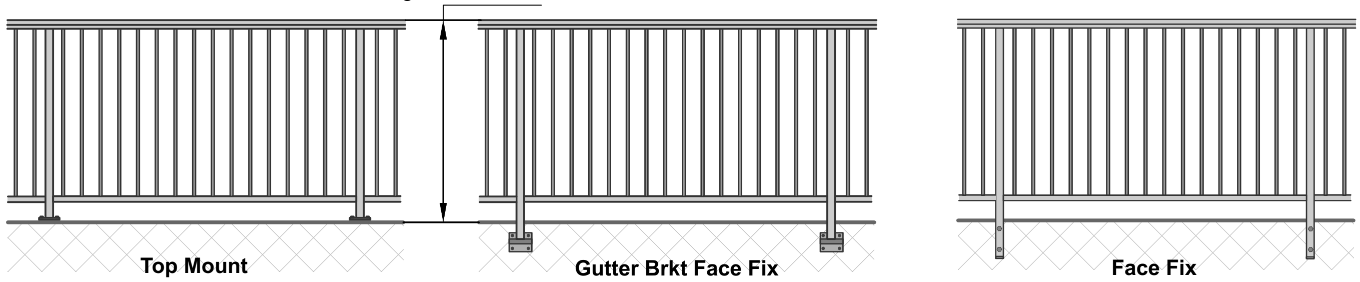
17mm Baluster - Split Rail. Handrail + Top and Bottom Rail.

Balustrade Height, 1000mm min,1275mm max above FFL.



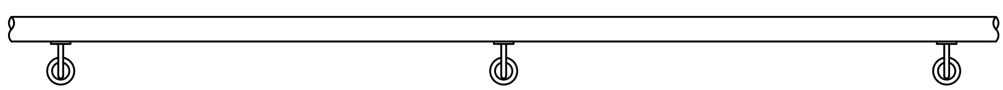
17mm Baluster - Full Height. Handrail + Bottom Rail.

Balustrade Height, 1000mm min,1275mm max above FFL.



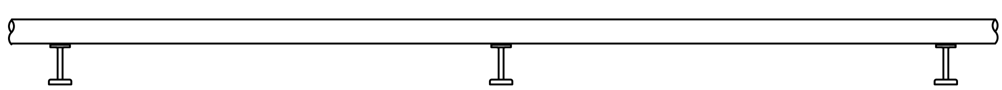
For JA/160 50mm dia Aluminium Tube + wall Brackets

Handrail Bracket JVB116
(Curved, Side Fix)



Handrail - Wall mount. Could be Horizontal or for Stairway

Handrail Bracket JVB117
(80mm Top fix)

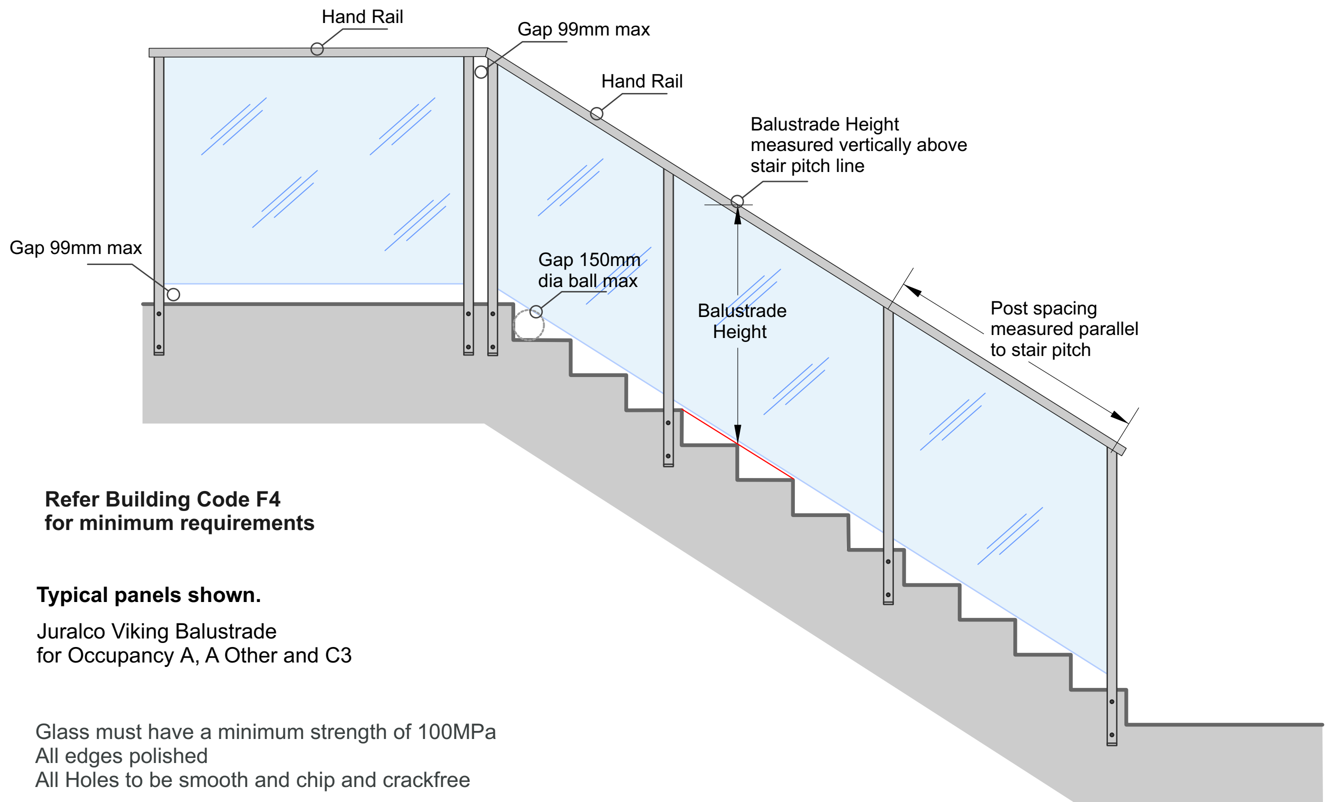


Handrail - Nib Wall mount. Could be Horizontal or for Stairway

Juralco Viking® Balustrade System Stair Setouts, Construction

Viking Balustrade
Face Fix shown

Typical Stair installation
Fixings into Single or Double Joists/Stiffeners, Concrete or Steel



Refer Building Code F4
for minimum requirements

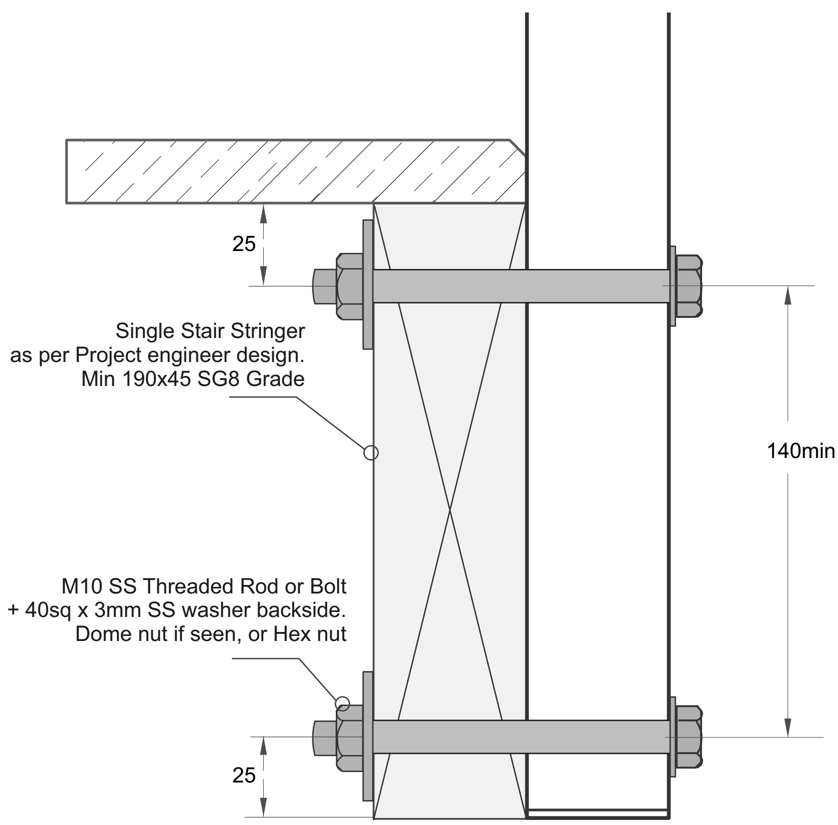
Typical panels shown.

Juralco Viking Balustrade
for Occupancy A, A Other and C3

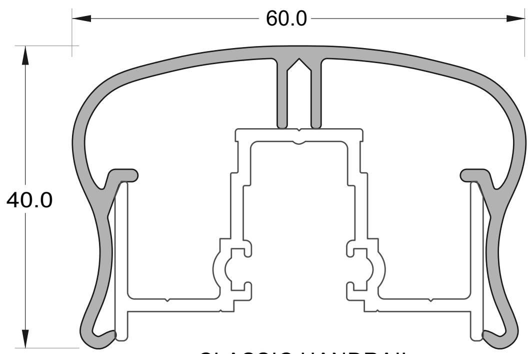
Glass must have a minimum strength of 100MPa
All edges polished
All Holes to be smooth and chip and crackfree

Viking Balustrade
Stair Stringer Detail

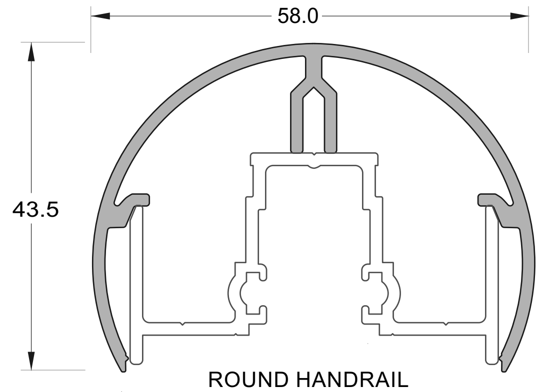
Stair structure to be designed by others to resist Balustrade actions as per NZS1170.1 Table 3.3
Applicable to Occupancy A only unless Project Engineer ensures it can support appropriate balustrade loads



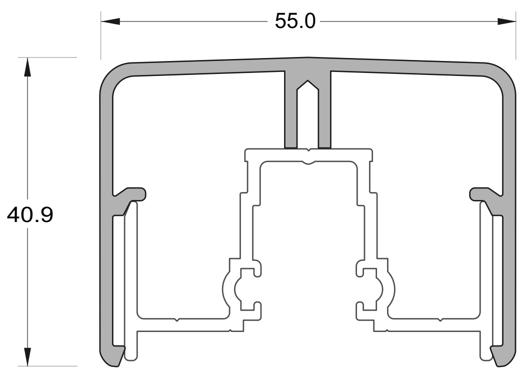
Handrails below suitable for Private and Common Stairways, but <u>NOT</u> suitable for Accessible Stairways



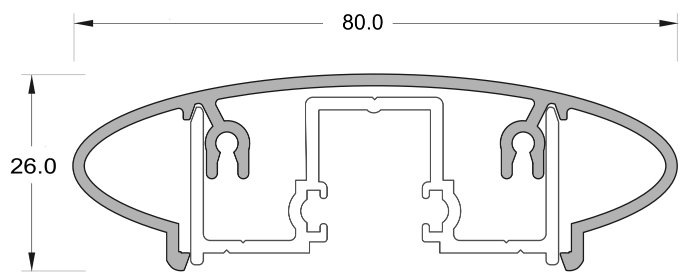
CLASSIC HANDRAIL
JVB/001/5.8



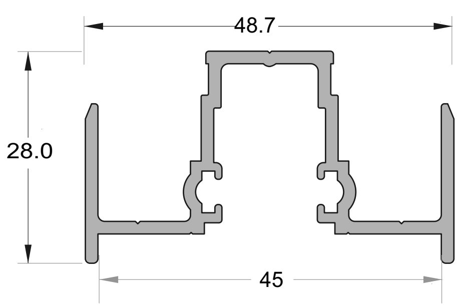
ROUND HANDRAIL
JVB/009/5.8



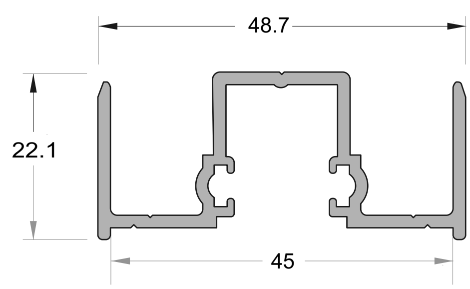
RECTANGULAR HANDRAIL
JVB/016/5.8



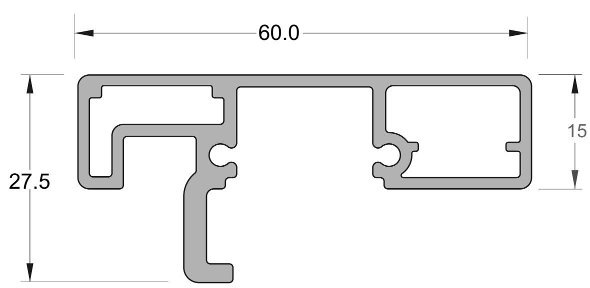
ELLIPTICAL HANDRAIL
JVB/024/5.8



TOP RAIL
JVB/004/5.8



ELLIPTICAL TOP RAIL
JVB/023/5.8

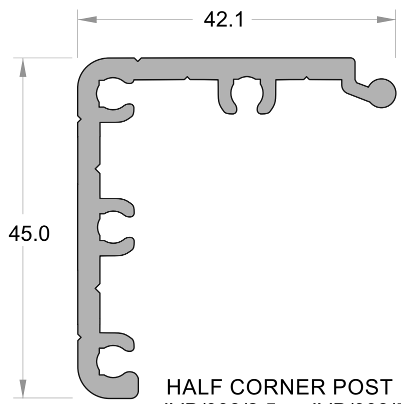


INTERLINKING RAIL
JEB/222/5.8

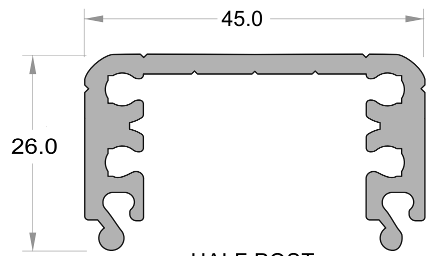
Note - For use with JEC 29 Handrail Bracket
- Infill clips not used with the JEB 222

Juralco Viking® Balustrade System

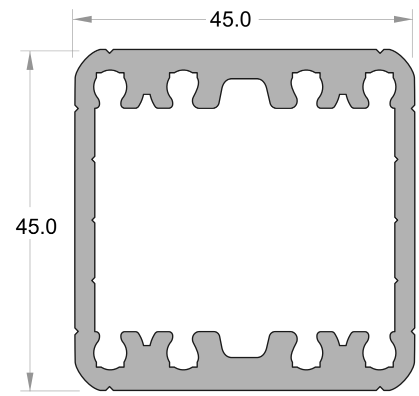
Extrusions



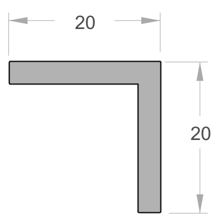
HALF CORNER POST
JVB/008/2.5 JVB/008/5



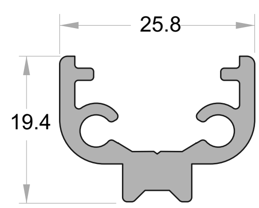
HALF POST
JVB/007/2.5 JVB/007/5



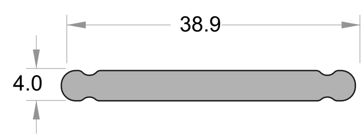
ONE PIECE POST
JVB/015/2.5 JVB/015/5



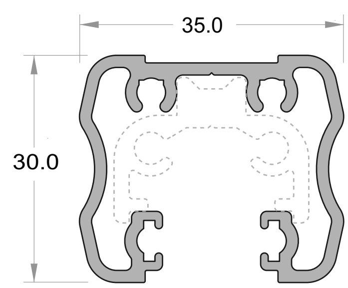
WATER DEFLECTION ANGLE
JA118/5



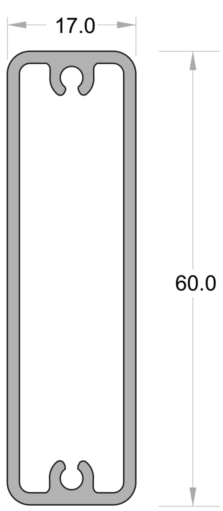
SPIGOT
JVB/017/5
(Fits Bottom Rail)



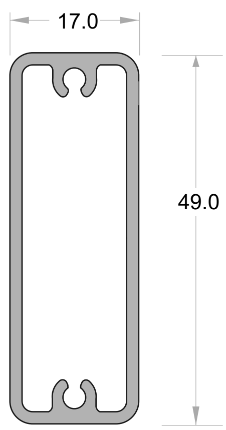
POST STIFFENER
JVB/013/5



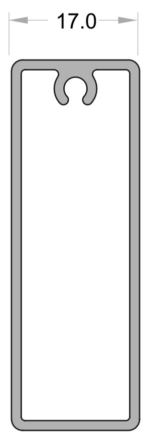
BOTTOM RAIL
JVB/002/5.8



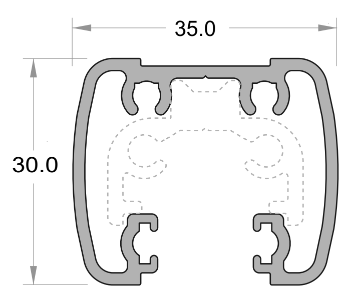
INFILL
60mm SLAT
JVB/019/5.3



INFILL
49mm SLAT
JVB/018/5.35

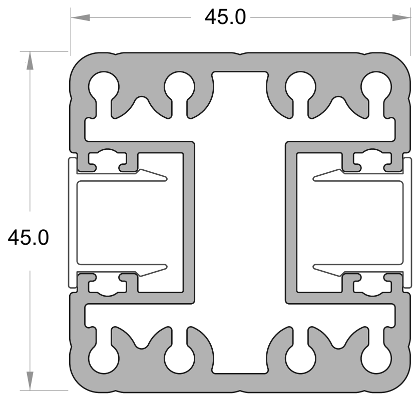


INFILL
49mm SLAT
(Single Screw)
JVB/030/5.35

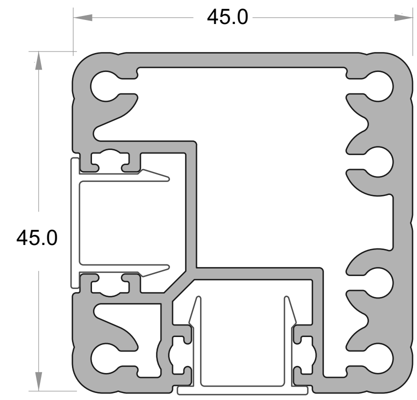


BOTTOM RAIL
JVB/022/5.8w

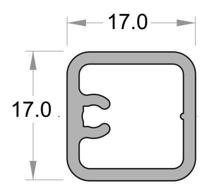
Extrusions



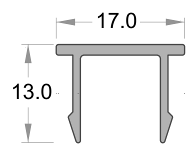
FRAMELESS POST
JVB/011/2.5 JVB/011/5



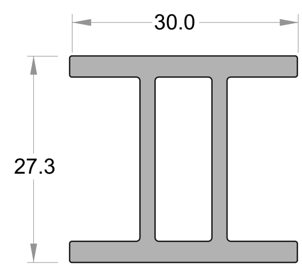
FRAMELESS CORNER POST
JVB/012/2.5 JVB/012/5



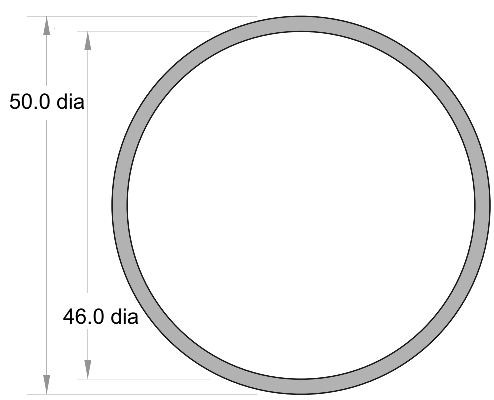
BALUSTER
JVB/005/5.35



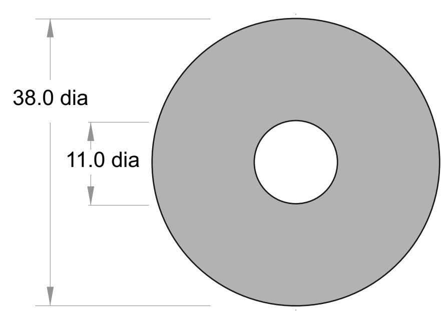
FRAMELESS POST CAP
JVB/006



SEMI FRAMELESS
POST STIFFENER
JVB/020/5



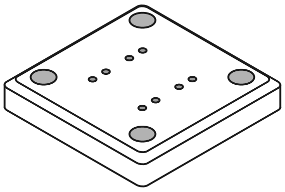
HANDRAIL
JA160/5



SPACER LENGTH
JVB/125/1000

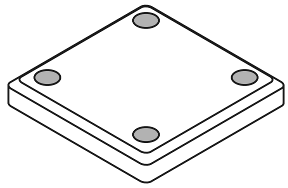
Components

Base Plate
JVB100



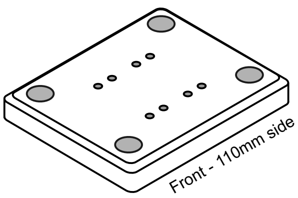
100mm x 100mm x 12mm - 4 x holes

Base Plate
JVB100/Blank



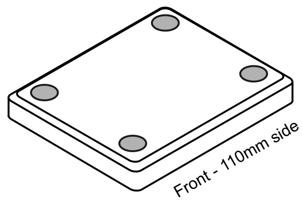
100mm x 100mm x 12mm - 4 x holes

Base Plate
JVB121



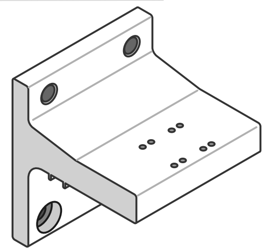
110mm x 90mm x 12mm - 4 x holes

Base Plate
JVB121/Blank



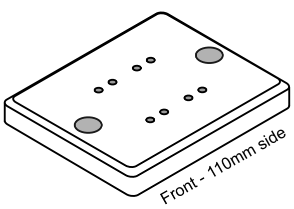
110mm x 90mm x 12mm - 4 x holes

Gutter Bracket
JVB137/45



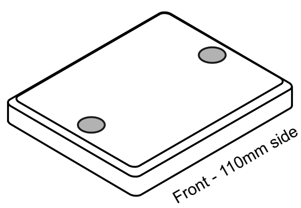
120mm wide x 135mm high

Base Plate
JVB101



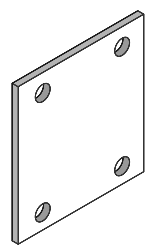
110mm x 90mm x 12mm - 2 x holes

Base Plate
JVB101/Blank



110mm x 90mm x 12mm - 2 x holes

Gutter Bracket Spacer
JEC139

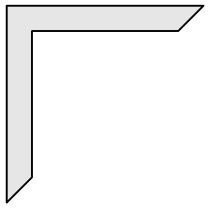


135mm x 120mm x 5mm - 4 x holes

Juralco Viking® Balustrade System

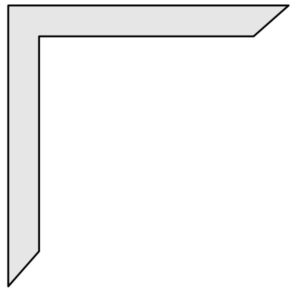
Components

Handrail Corner Joiner 90deg
JVB150



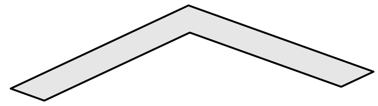
Legs 70mm long x 9mm wide
x 3mm thick

Handrail Corner Joiner 90deg
JVB151



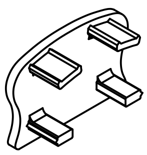
Legs 100mm long x 11mm wide
x 3mm thick

Handrail Corner Joiner 135deg
JVB152

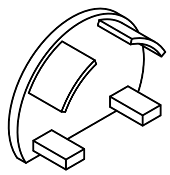


Legs 70mm long x 11mm wide
x 3mm thick

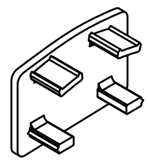
Classic End Cap
JVB105



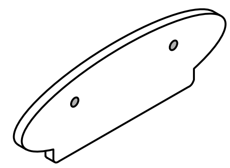
Round End Cap
JVB113



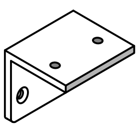
Rectangular End Cap
JVB129



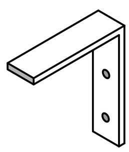
Elliptical End Cap
JVB140



Top Rail Wall Bracket
4.5mm thick JVB103

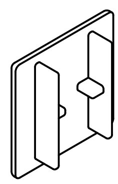


Glass Setting Angle
JVB118

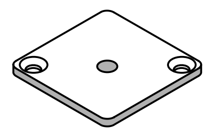


30mm x 30mm Angle x 12mm wide

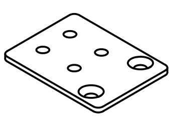
Frameless Post-top Cap
JVB115



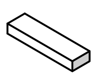
Post End Cap
JVB104



Bottom Rail Wall Bracket
3mm thick JVB102

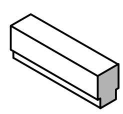


4mm Glass Setting Block
Part No JVB119



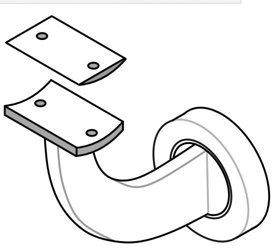
4mm x 8mm x 30mm long

Glazing Block 50mm
JVB108



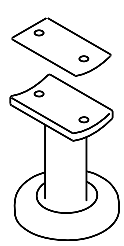
17mm x 9mm x 350mm long

Handrail Bracket JVB116
(Curved, Side Fix)



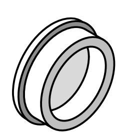
Satin or SCC finishes

Handrail Bracket JVB117
(80mm Top fix)



Satin or SCC finishes

50.8mm dia Handrail End Cap
JGS 132

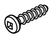


SCC finishes

Juralco Viking® Balustrade System

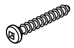
Components

PK SCREW SS
JVBScrew/10x6



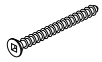
Top Rail
to Handrail
Pan SQ drive

PK SCREW SS
JVBScrew/25x6



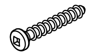
Bottom Rail
to Baluster
Pan SQ drive

PK SCREW SS
JVBScrew/38x6



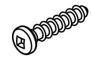
Top Rail
to Baluster
C/S SQ drive

PK SCREW SS
JVBScrew/25x8



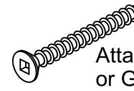
Post to
Bottom Rail
Pan SQ drive

PK SCREW SS
JVBScrew/25x10



Top Rail
to Post
Pan SQ drive

PK SCREW SS
JVBScrew/50x10

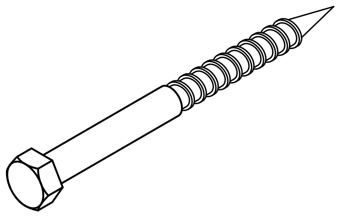


Attaching Post to Base plate
or Gutter Bracket
C/S SQ drive. High Tensile

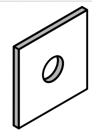
COACHSCREW SS
JVBCoach/M10x160

COACHSCREW SS
JVBCoach/M10x180

COACHSCREW SS
JVBCoach/M10x200



Square Washer
JVBSqwsh/40x40x3



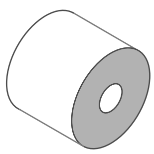
SS 40mm sq x 3mm

Round Washer
JVBWasher/21x2

SS 21mmdiax11x2mm thick

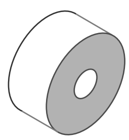
All For M10 SS
Coachscrews
or Bolts

Face Fix Spacer
JVB125/30mm



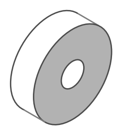
Aluminium
38mm dia
x 30mm long.

Face Fix Spacer
JVB125/15mm



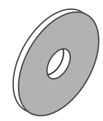
Aluminium
38mm dia
x 15mm long.

Face Fix Spacer
JVB125/10mm



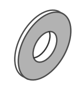
Aluminium
38mm dia
x 10mm long.

EPDM Washer
JVB126



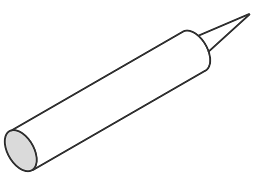
For M10 Fastener
38mm dia
x 3mm

EPDM Washer
JVB130



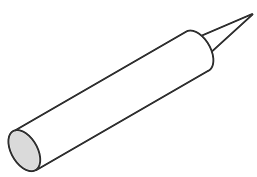
For M10 Fastener
20mm dia
x 1.6mm

SIKA Supergrip
JECSUPERGRIP30



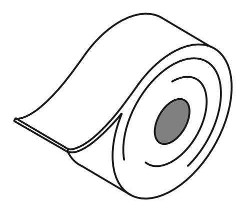
For All Coachscrews fixings

Rhodorsil V60 Clear Silicone
H/RTV419098



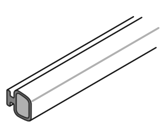
Construction Silicone

Foam Tape 100mm x 4.8mm
JVB/FTAPE100



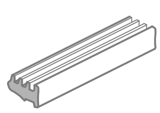
Base Plate Separator
for Concrete and Steel
Single sided 100mm wide x 15.2mt Roll

BULB SEAL
JVB/Bulbseal



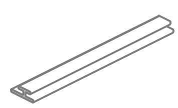
300m Roll, Black

GLAZING WEDGE
JVBWedge/Brown



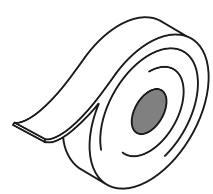
6mm Glass Wedge 75m Roll, Black

BACKING WEDGE
JVB/BackWedge500



500m Roll, Black

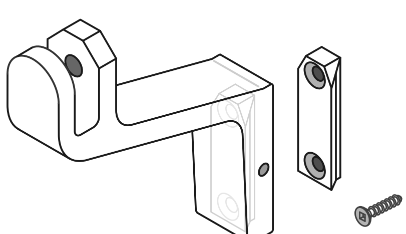
Foam Tape 38mm x 4.8mm
JVB/FTAPE38



Face fix Post Separator
for Concrete and Steel
Single sided 38mm wide x 15.2mt Roll

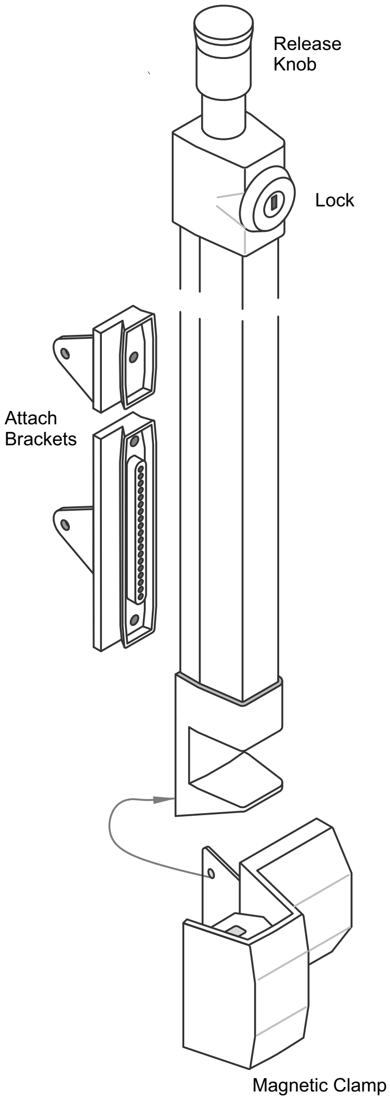
HANDRAIL BRACKET KIT
JEC 29/Kit

Includes 2 x grub screws
(For use with Interlinking
Rail Extrusion)



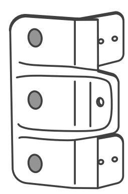
Use JVB/Screw 10gx20 C/S

Gate Components

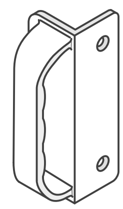


Latch - 540mm high OA

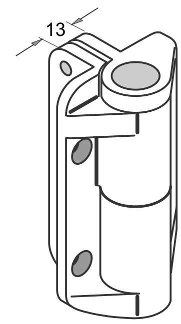
MAGNETIC POOL GATE LATCH
Nylon, Black
JEF/APL



SOFT GATE STOP
Nylon, Black.
Part No JEF/GS



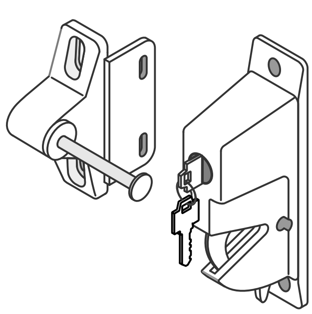
GATE HANDLE
Nylon, Black.
Part No JEF/GH



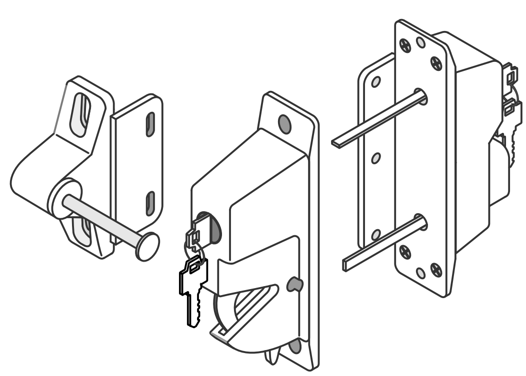
Suitable for Pool gates
Max Self Closing 45kg

ADJUSTABLE TENSION
HINGE, LEGS
JEF/AHHD (Pairs)

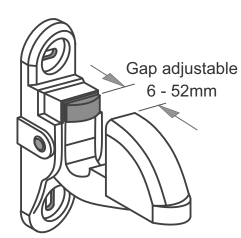
These Latches not suitable for Pool Gates



UNIVERSAL DROP LATCH
AND STRIKER
Nylon, Black
JEF/DL

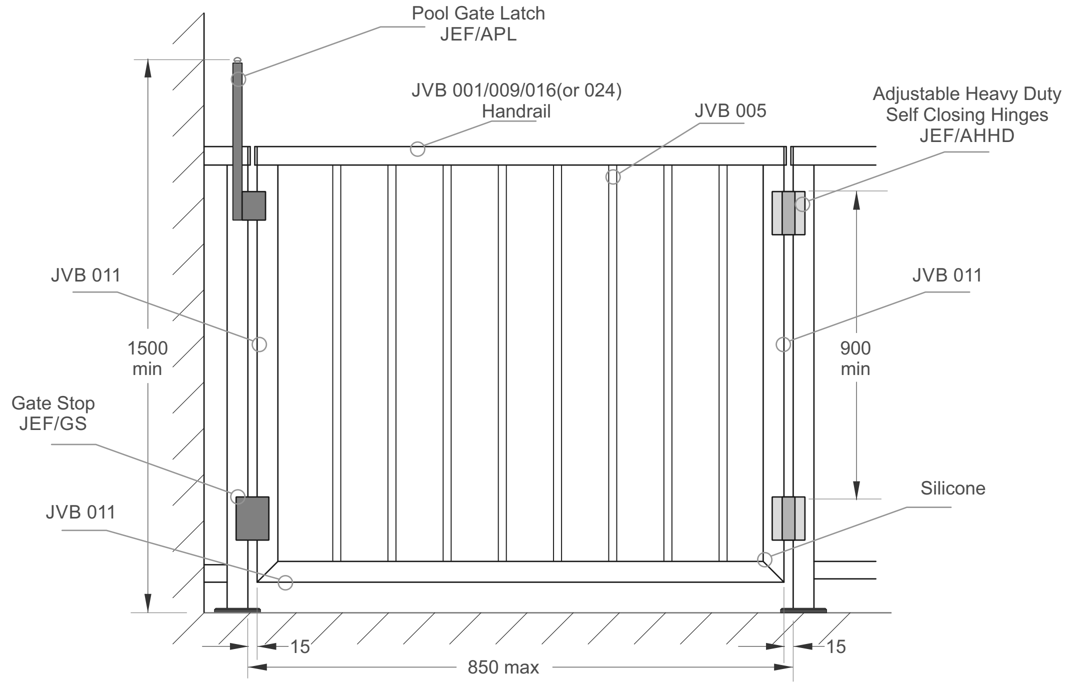


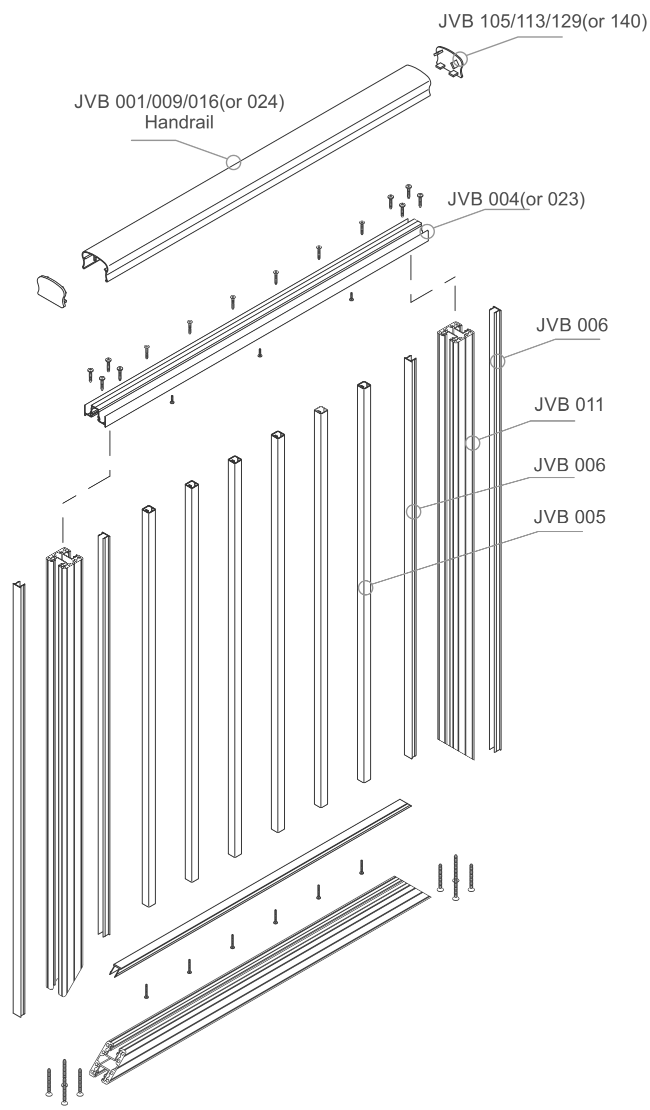
UNIVERSAL DROP LATCH
AND STRIKER
Nylon, Black. (External Access)
JEF/DLEA

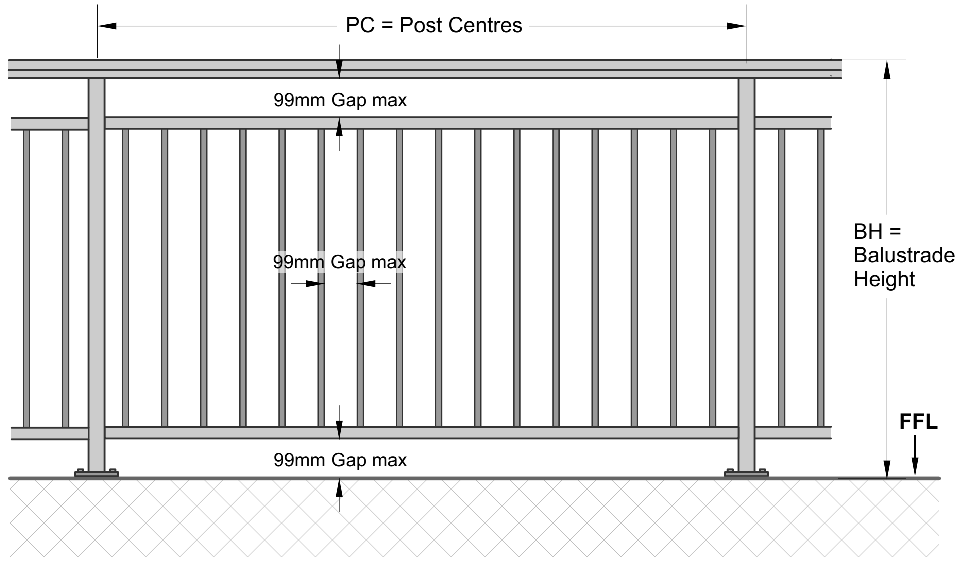


PARKING LATCH
Nylon, Black.
G21

Typical Gate Assembly - Baluster







BALUSTRADE DESIGN GUIDE - for building types below. All other applications please contact Juralco

	BUILDING TYPE	LOCATION	HEIGHT (mm)
BH = HEIGHT OF BALUSTRADE (Measured from FFL to top of Barrier)	Detached dwellings and within household units of multi-unit dwellings	Stairs and ramps and their landings	900
		Balconies and decks, and edges of internal floors or mezzanine floors	1000
	All other buildings, and common areas of multi-unit dwellings	Stairs or ramps (excluding landings)	900
		Barriers within 530mm of the front of fixed seating	800
		All other locations	1100
	Note: A Building Consent is required when installing or replacing a Swimming Pool Fence. Refer to NZBC Clause F9	Swimming Pools	1200

GAPS (max) as above		99

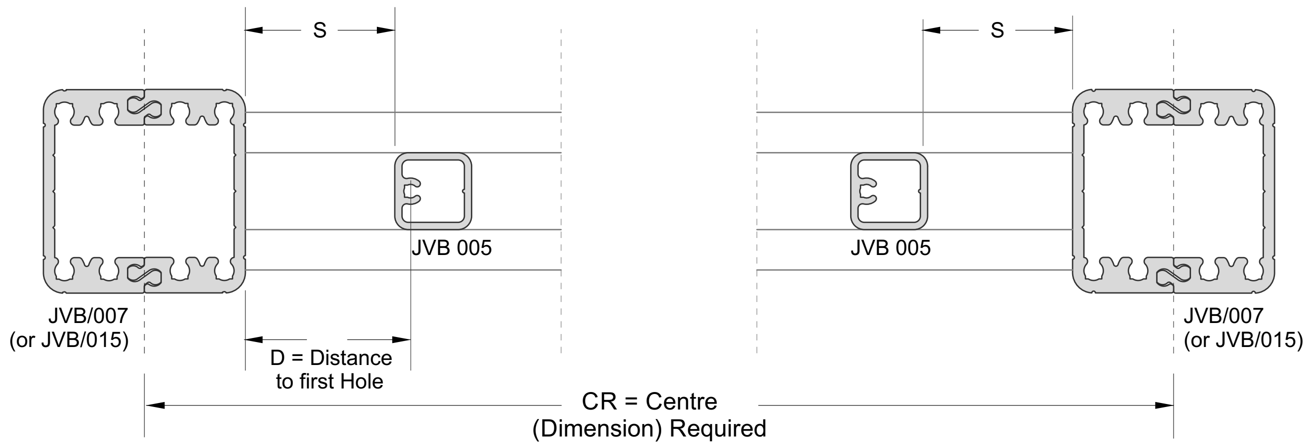
PC = POST CENTRES (max)	17mm Balusters	1400
	6mm Toughened glass infill panels	1400
	10mm Toughened glass for up to & including 1200mm High and High wind zone	1400

NZBC REQUIREMENTS - SECTION F4 : SAFETY FROM FALLING

NOTES:

1 - Heights are measured vertically from finished floor level (ignoring carpet or vinyl, or similar thickness coverings) on floors, landings and ramps. On stairs the height is measured vertically from the pitch line or stair nosings.
2 - A landing is a platform with the sole function of providing access.
3 - Household unit does not include a hostel, boarding house, motel or other specialized accommodation.
4 - Stairs or ramps for all other buildings does not include landings - use all other locations
5 - The triangular opening formed by the riser, tread, and bottom rail of the barrier on a stair shall be of such a size that a 150mm diameter sphere can not pass though it, except for swimming pool stairs where the diameter is 100mm.
6 - Barriers to swimming pools shall have in addition to performance F9 and NZS8500
   a) All gates and doors fitted with latching devices not readily operated by children, and constructed to automatically close and latch when released from any stationary position 150mm or more from the closed and secured position but excluding sliding and sliding-folding doors that give access tothe immediate pool surround from a building that forms part of the barrier and,
   b) No permanent objects on the outside or inside of the barrier that could provide a climbing step
7 - No toeholds between the heights of 150mm and 760mm above floor level (or stair nosing). .

Juralco Viking Balustrade System building code compliance documentation requires all balustrade installations are to be completed in accordance with the requirements of our authorised installer certification.



CM= Centre, Maximum (of Posts)	Max CRS for No of Balusters	No of Balusters
	260	1
	376	2
	492	3
	608	4
	724	5
	840	6
	956	7
	1072	8
	1188	9
	1304	10
	1420	11

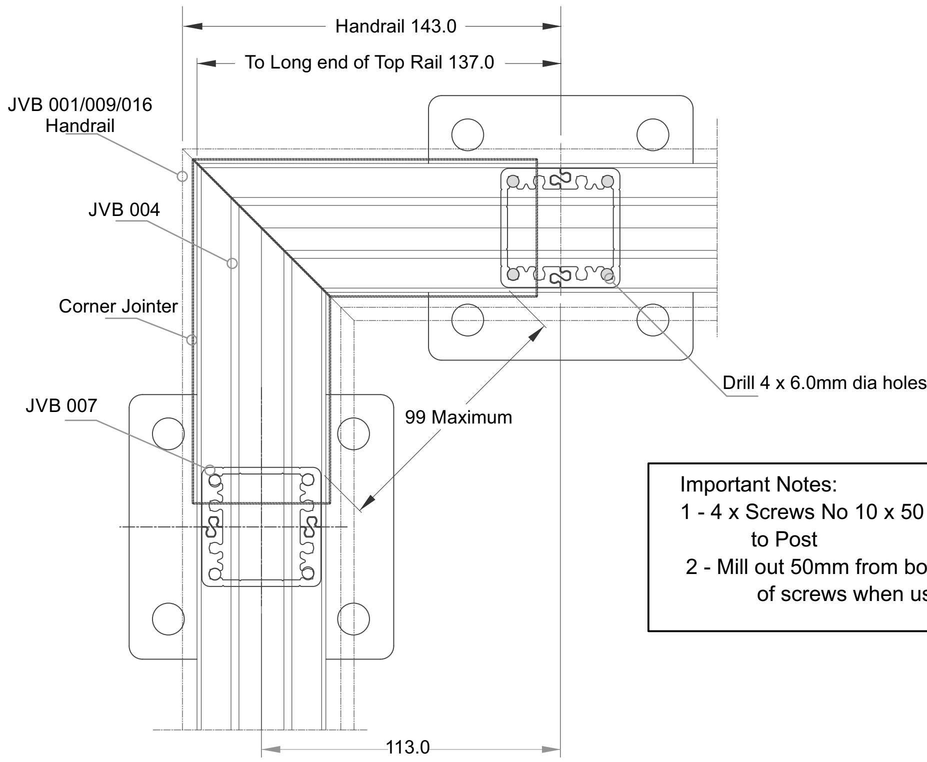
To achieve an equal distance 'S' from either end of panel use equation shown below.
= 103 - (CM - CR)/2

Example:
The distance required (CR) is 890mm centres,
Use CM = 956 as next value up.

Substitute into equation shown above
= 103 - [(956-890)/2]
= 103-33
= 70

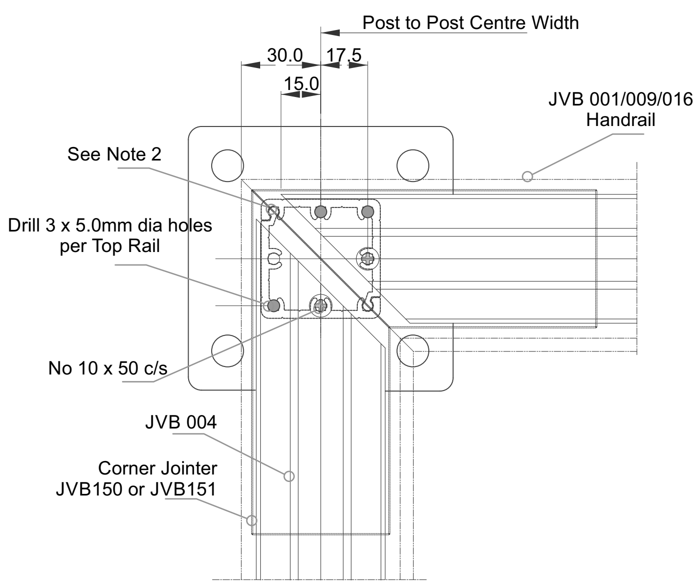
Therefore make first cut 70mm from first hole (as shown at the LH End)
Note: Balusters JVB005 must be positioned as shown.

Two Post 90 Deg Corner



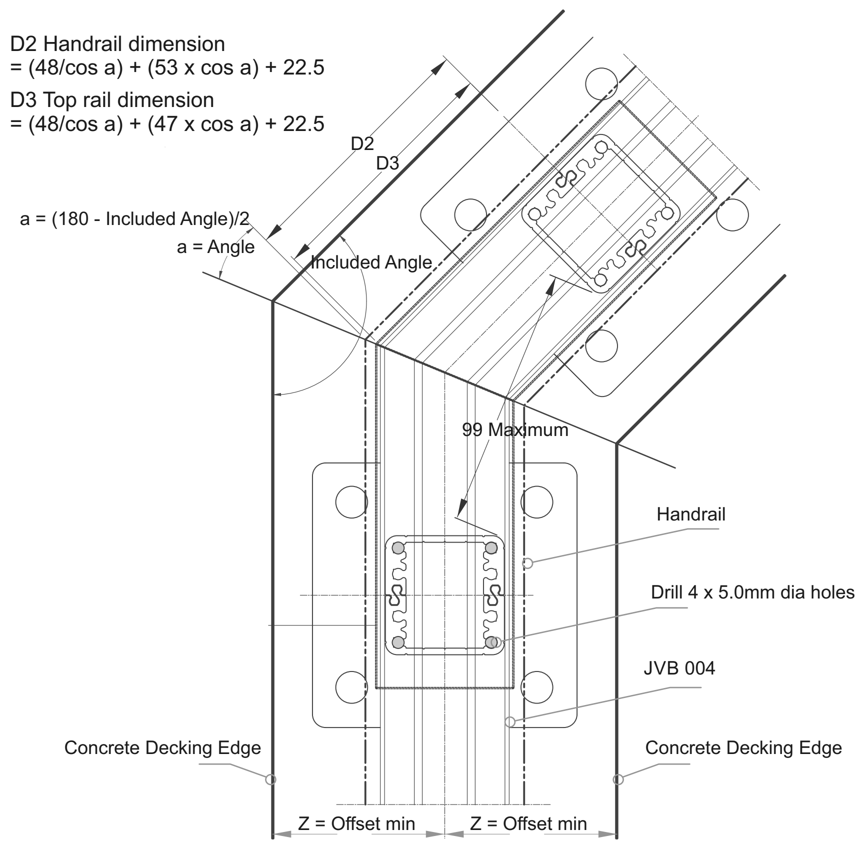
Important Notes:
1 - 4 x Screws No 10 x 50 pan required to fix Base Plate to Post
2 - Mill out 50mm from bottom of Post to allow fitment of screws when using Base Plate

Single Post 90 Deg Corner



Important Notes:
1 - 4 x Screws No 10 x 50 pan required to fix Base Plate to Post
2 - Mill out 50mm from bottom of Post to allow fitment of screws when using Base Plate

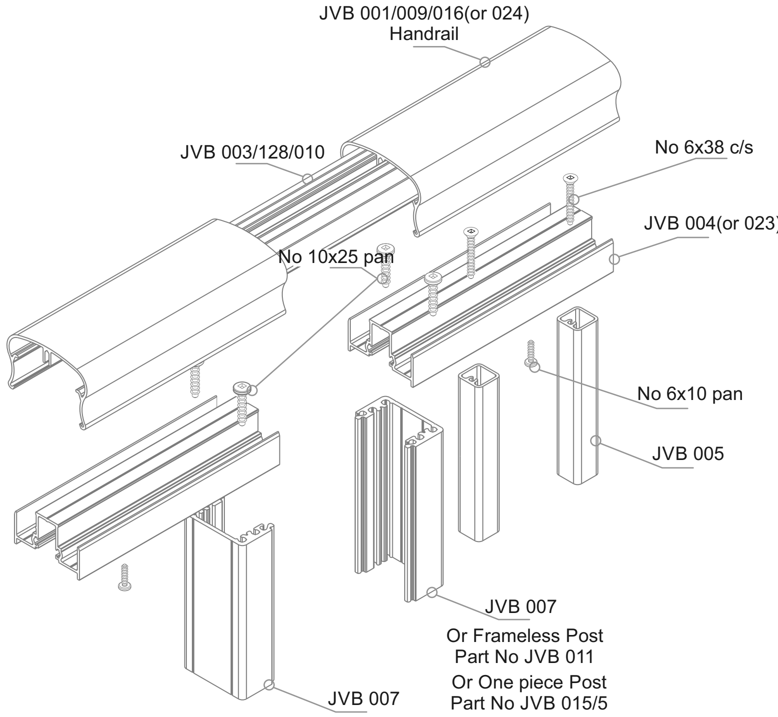
Angled Corner



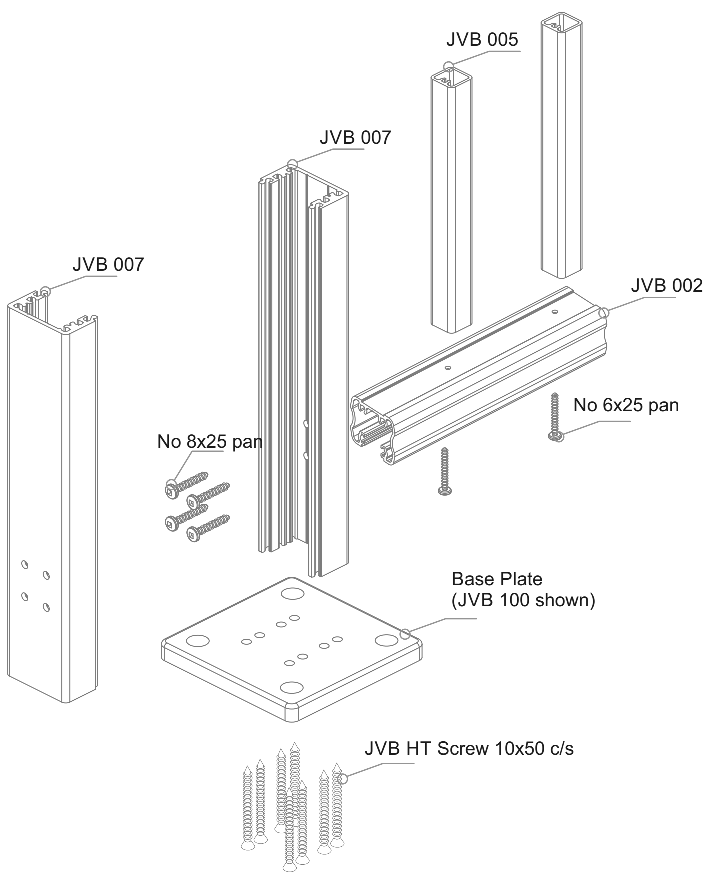
Minimum Edge distance when fixing to Concrete and anchor types.
JVB 101 Baseplate , Z = 50mm, 2 x M12 x 80mm deep Epcon or Chemset anchors
JVB 100 Baseplate , Z = 70mm, 4 x M10 x 90mm deep Epcon or Chemset anchors
These fastenings are based on a minimum concrete strength of 17.5 MPa

Corner Jointers to be Cut/Welded by installer

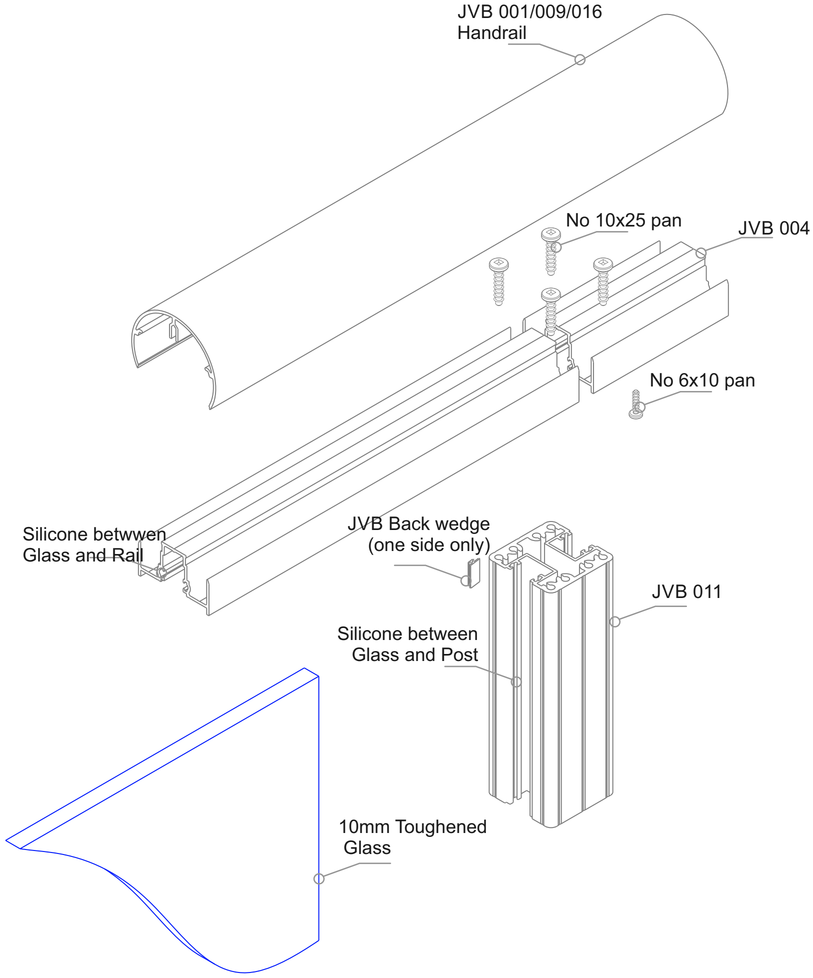
Typical Assembly - Top Handrail



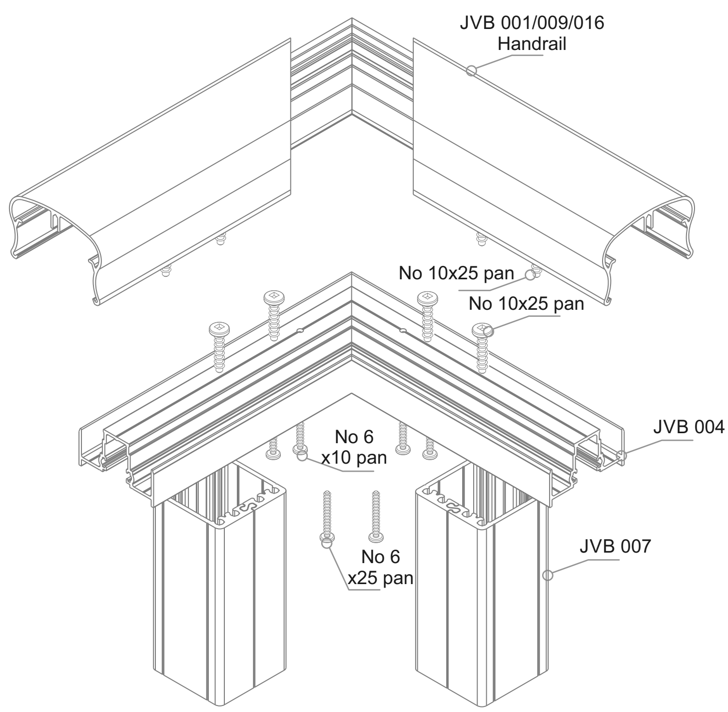
Typical Assembly - Lower Panel



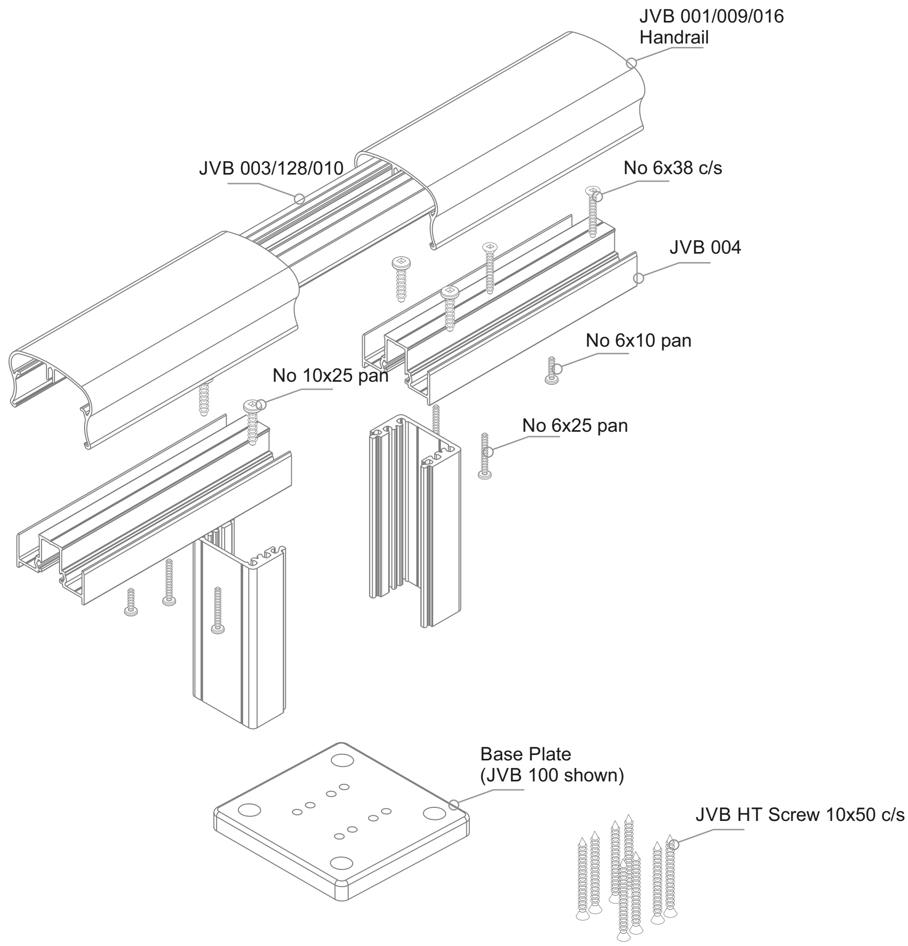
Typical Assembly - Semi Frameless Glass and Handrail



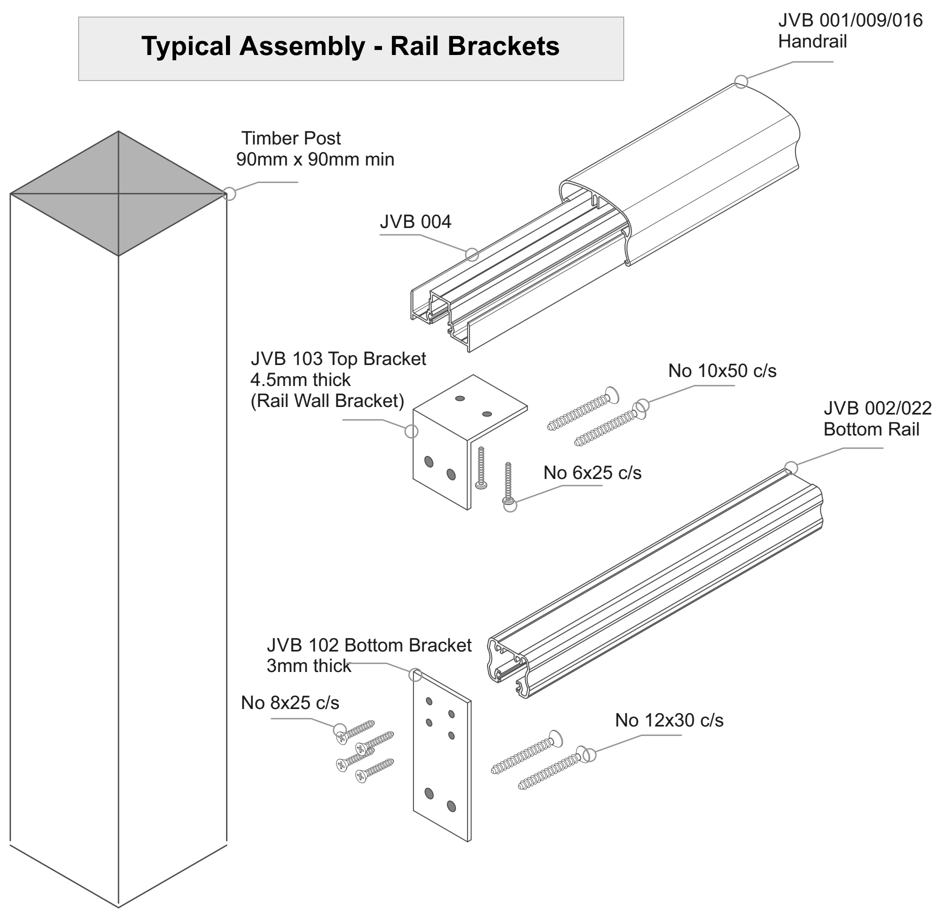
Typical Assembly - Corner



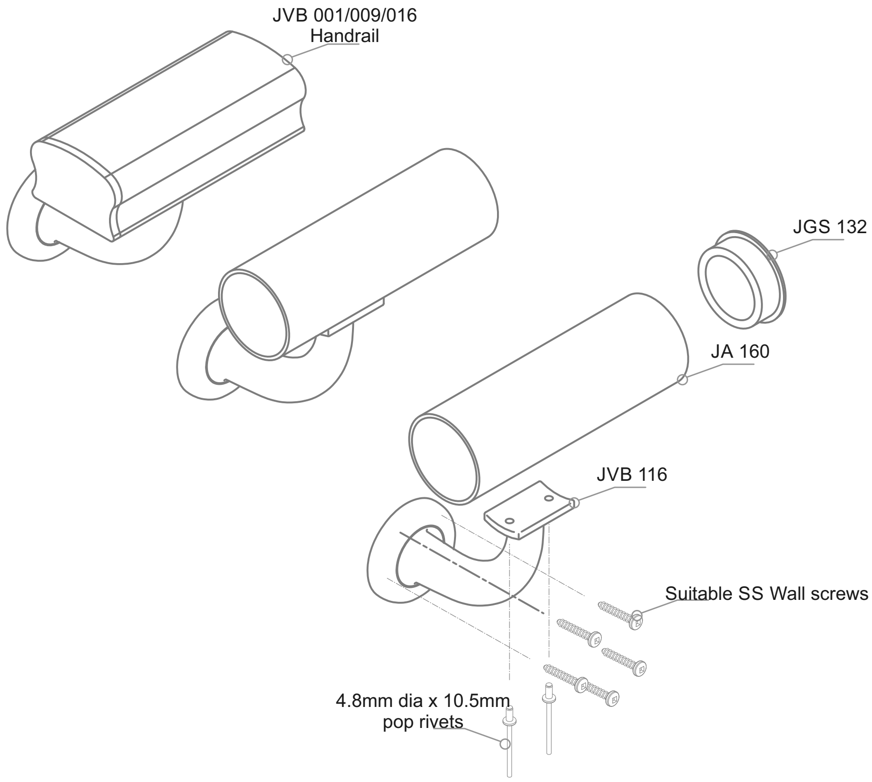
Typical Assembly - Nib Wall, Handrail



Typical Assembly - Rail Brackets

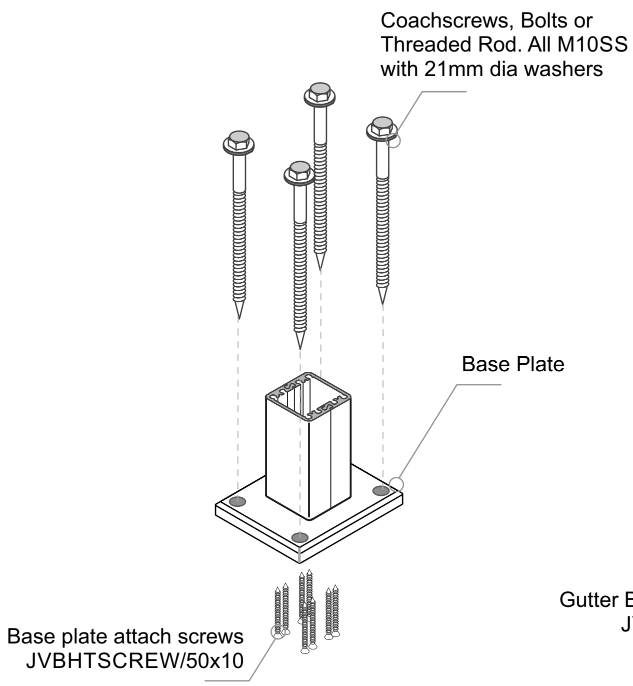


Typical Assembly - Wall Mounted Handrail

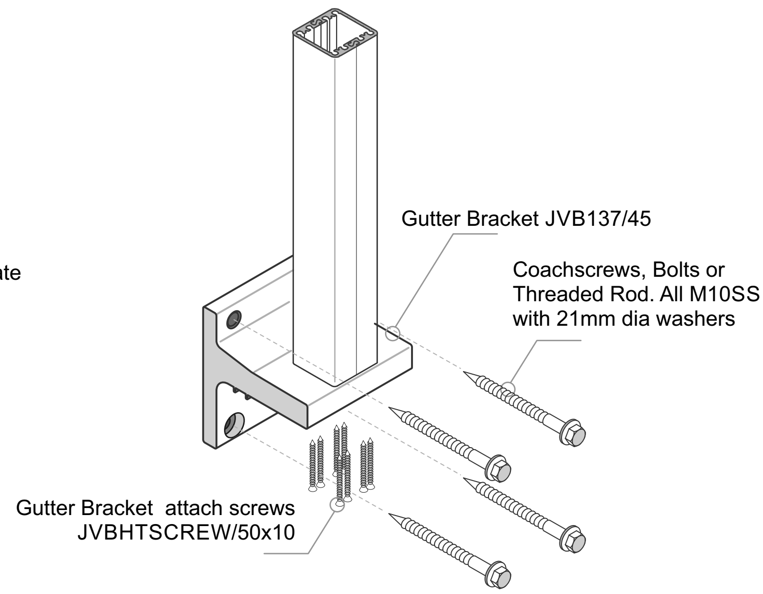


Juralco Viking® Balustrade System

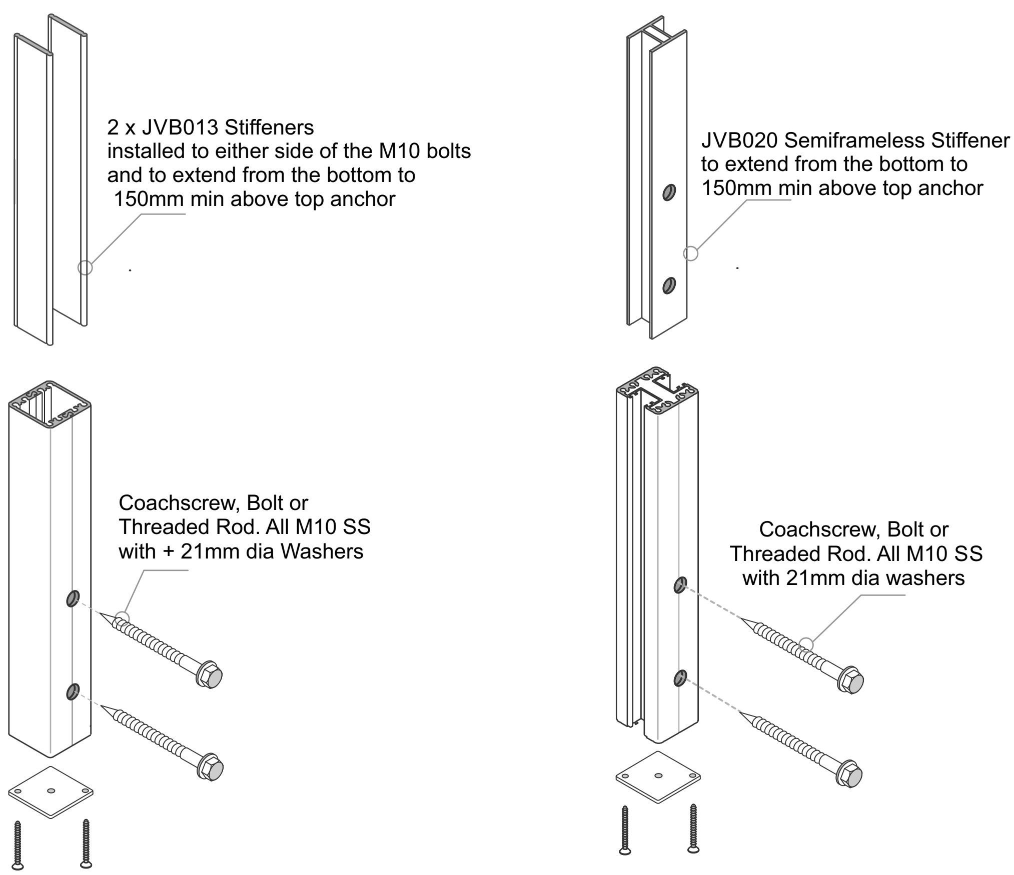
Post Mounting Options



Top Fix
Base Plates in a variety of sizes.
Different fasteners types depending
on the Building substrate.
Includes a 90deg Semi Frameless Corner Post



Gutter Bracket Face Fix
Different fasteners types depending
on the Building substrate.



Face Fix
Different fasteners types depending
on the Building substrate.

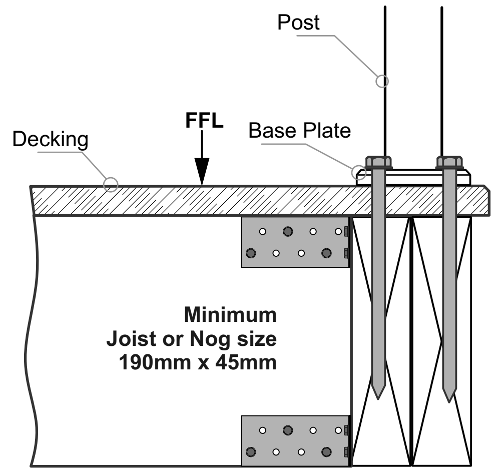
Typical TOP Fix to Timber - JVB121, 110mm x 90mm, 4 hole Base Plate - M10 SS Coachscrews

Balustrade Dimensions by Wind Zone

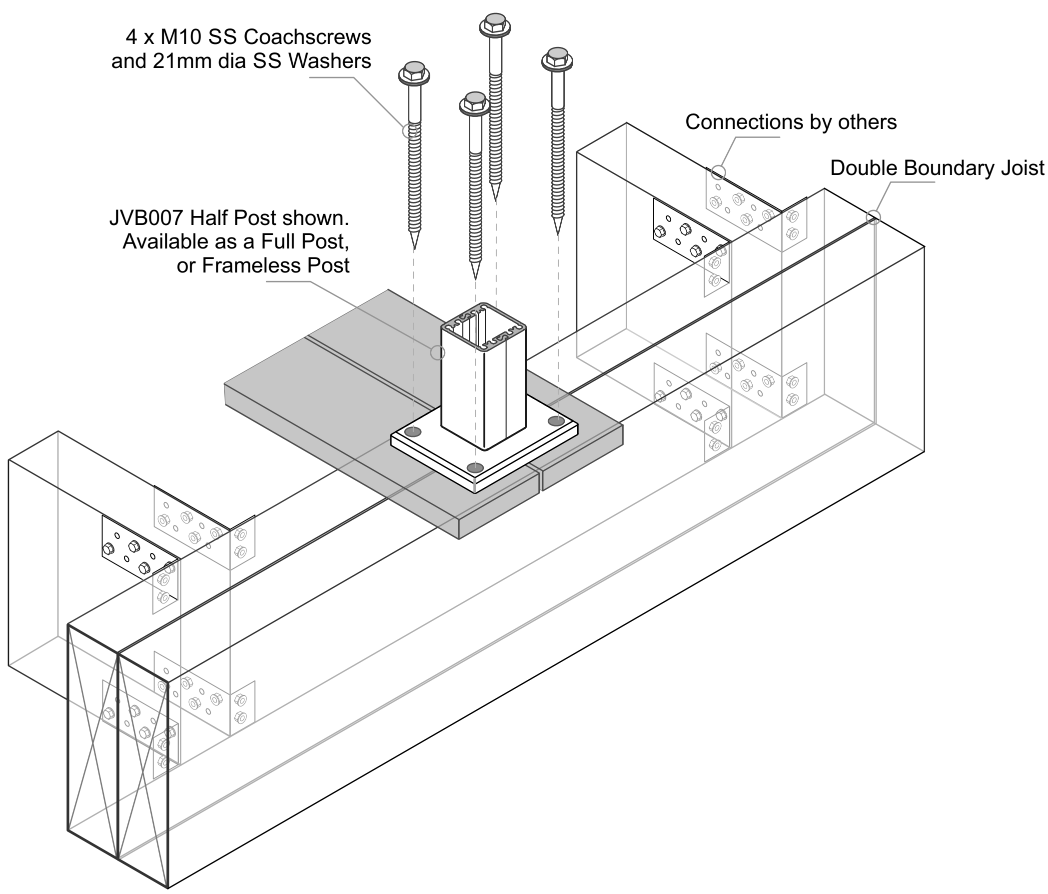
Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - Coachscrews 130mm min engagement into joists, predrill 6mm holes.
4 - Bond all coachscrews with SIKA Supergrip to full depth
5 - All Fixings must be Stainless steel



Typical FACE Fix Post to Timber - M10 SS Coachscrews

Balustrade Dimensions by Wind Zone

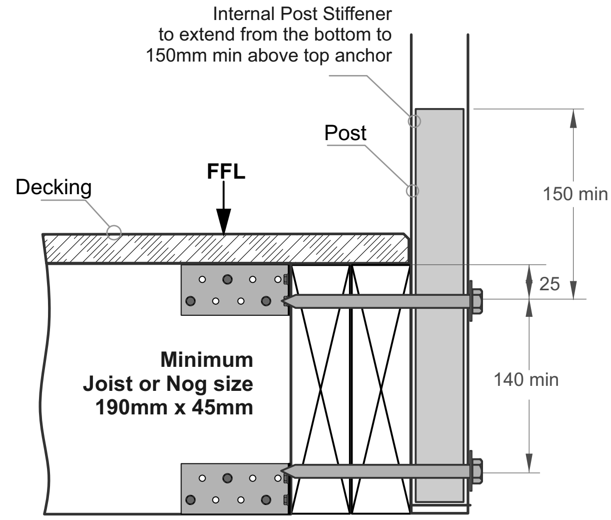
Up to and including Very High Wind Zone

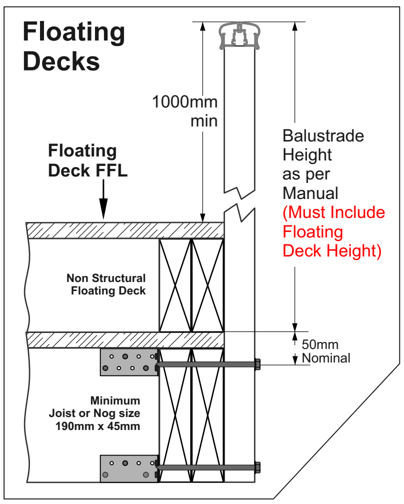
Balustrade Height, mm	1000	1050	1100	1150	1200	1250	1300 max
Post Spacing max, mm	1400	1350	1300	1250	1200	1150	1100

Extra High Wind Zone

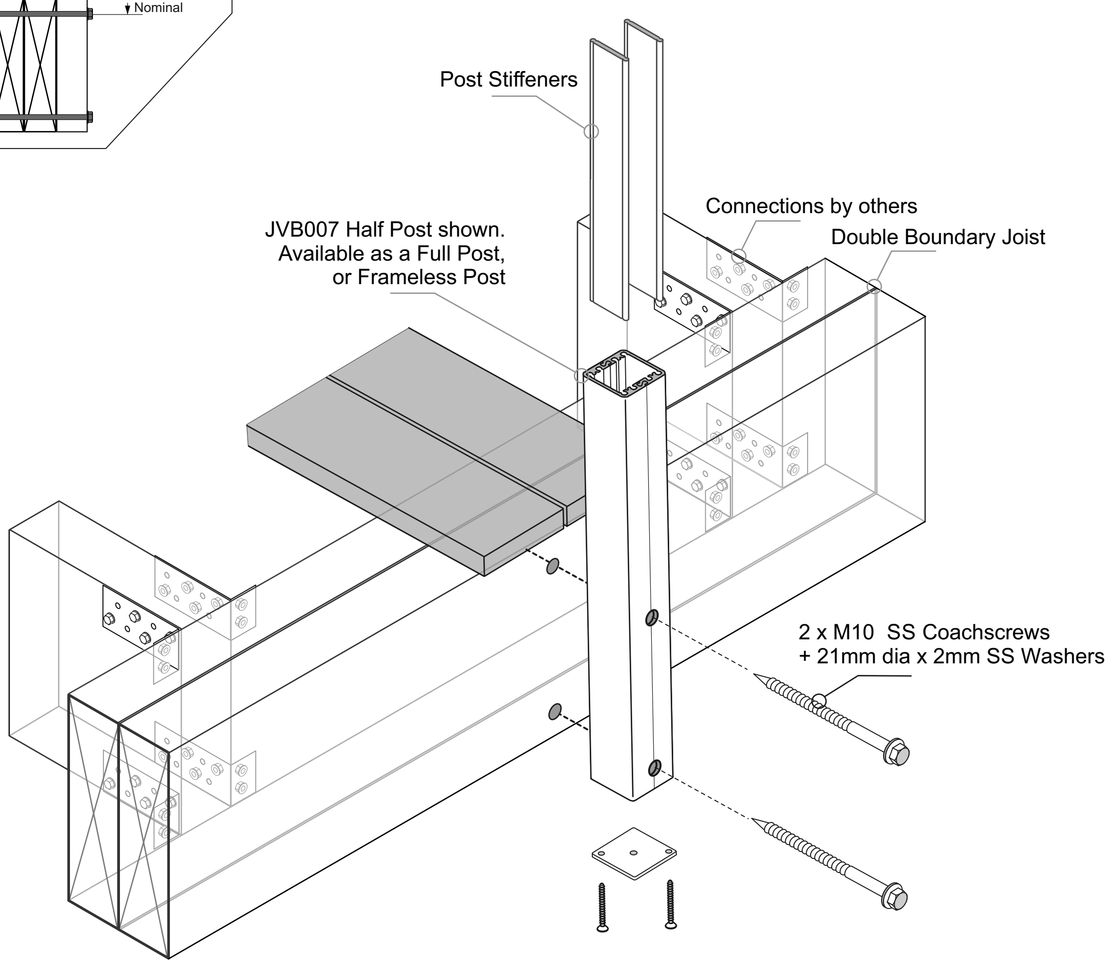
All Balustrades
Coachscrews as shown NOT SUITABLE. Must use Bolts

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011





Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - Coachscrews 90mm min engagement into joists, predrill 6mm holes.
4 - Bond all coachscrews with SIKA Supergrip to full depth
5 - All Fixings must be Stainless steel



Typical FACE Fix Post to Timber - M10 SS Bolts or Threaded Rod

Balustrade Dimensions by Wind Zone

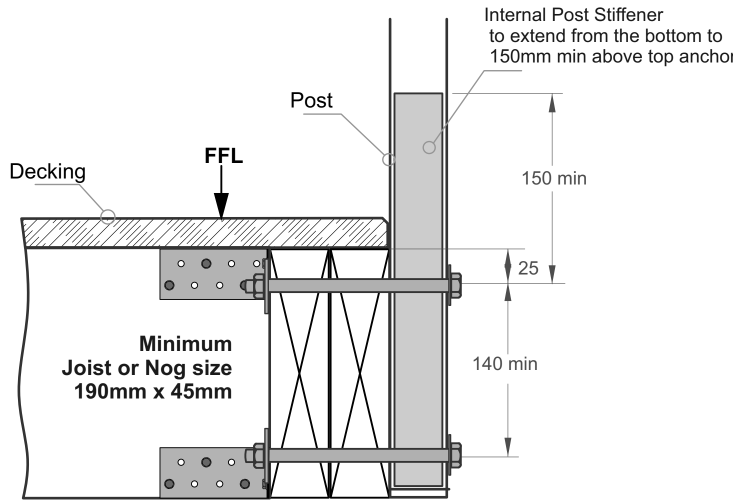
Up to and including Very High Wind Zone

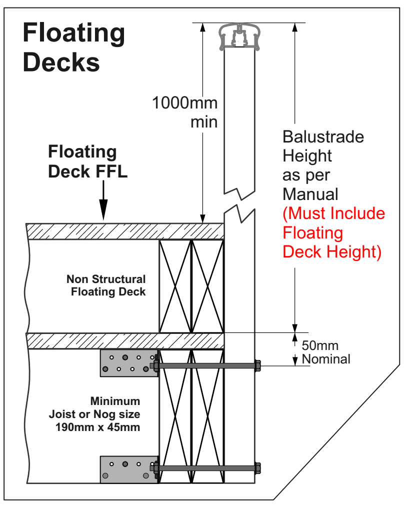
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

Extra High Wind Zone

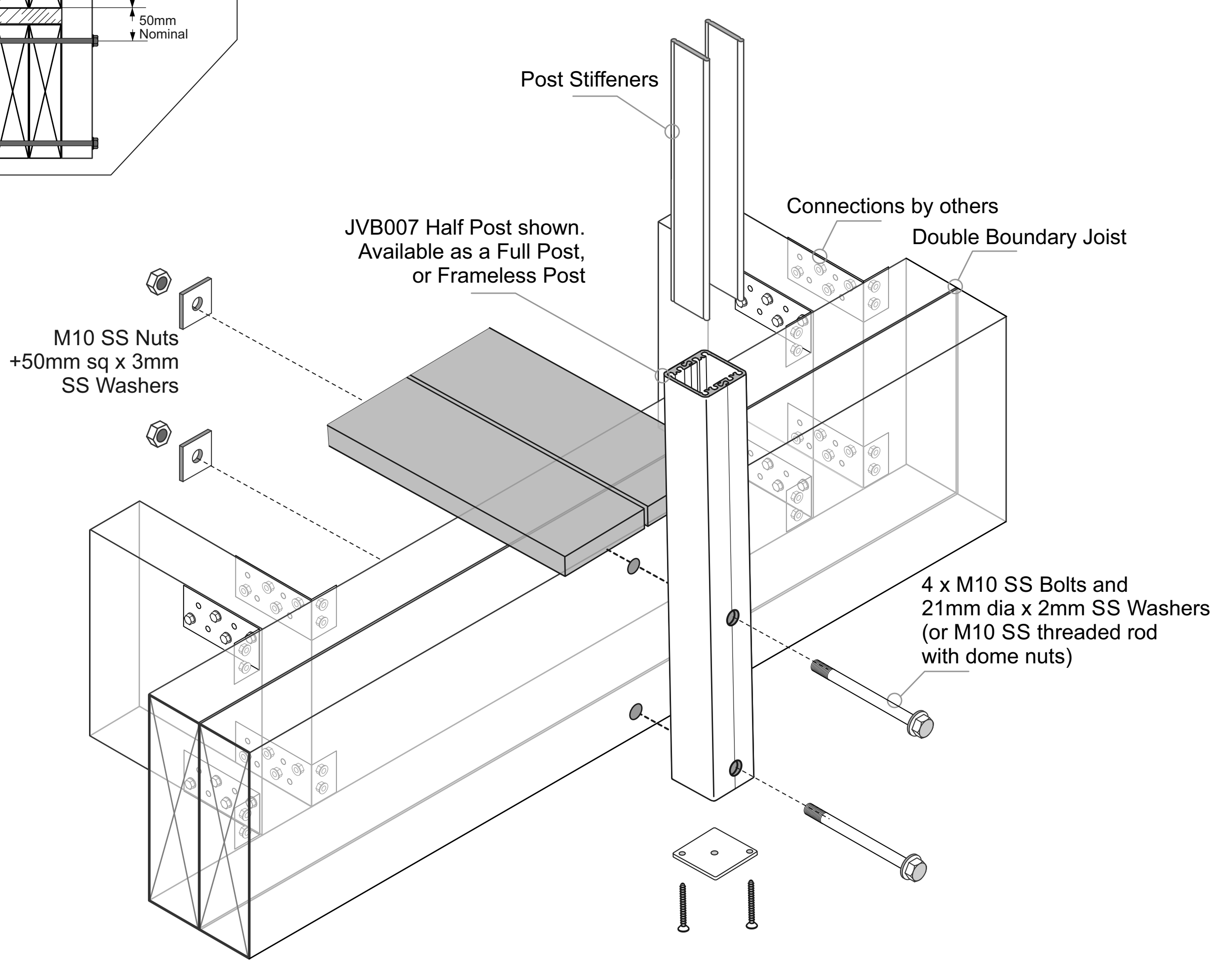
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011





Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - All Fixings must be Stainless steel



Juralco Viking® Balustrade Systems - Typical Post Fixings

Complies with NZS3604:2011 - Double Boundary Joists

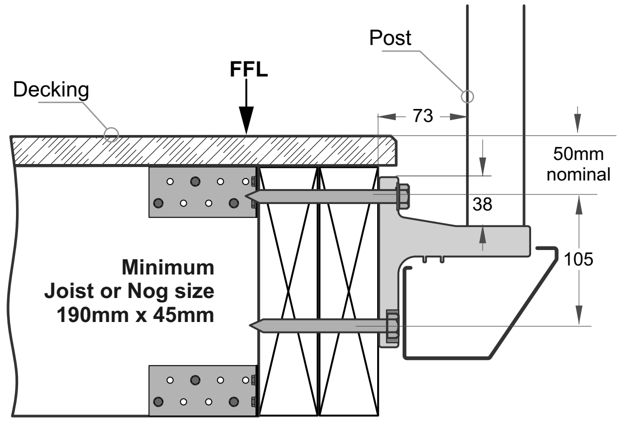
Typical FACE Fix to Timber - JVB137/45, Gutter Bracket - M10 SS Coachscrews

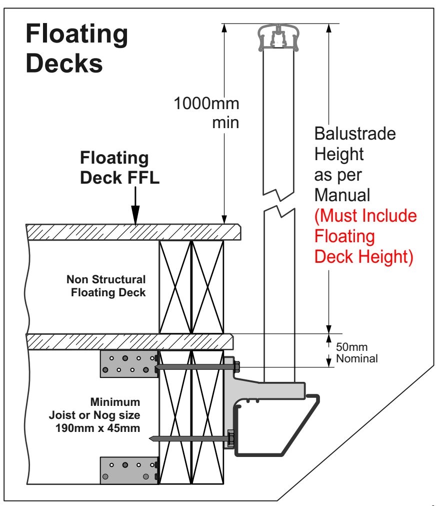
Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

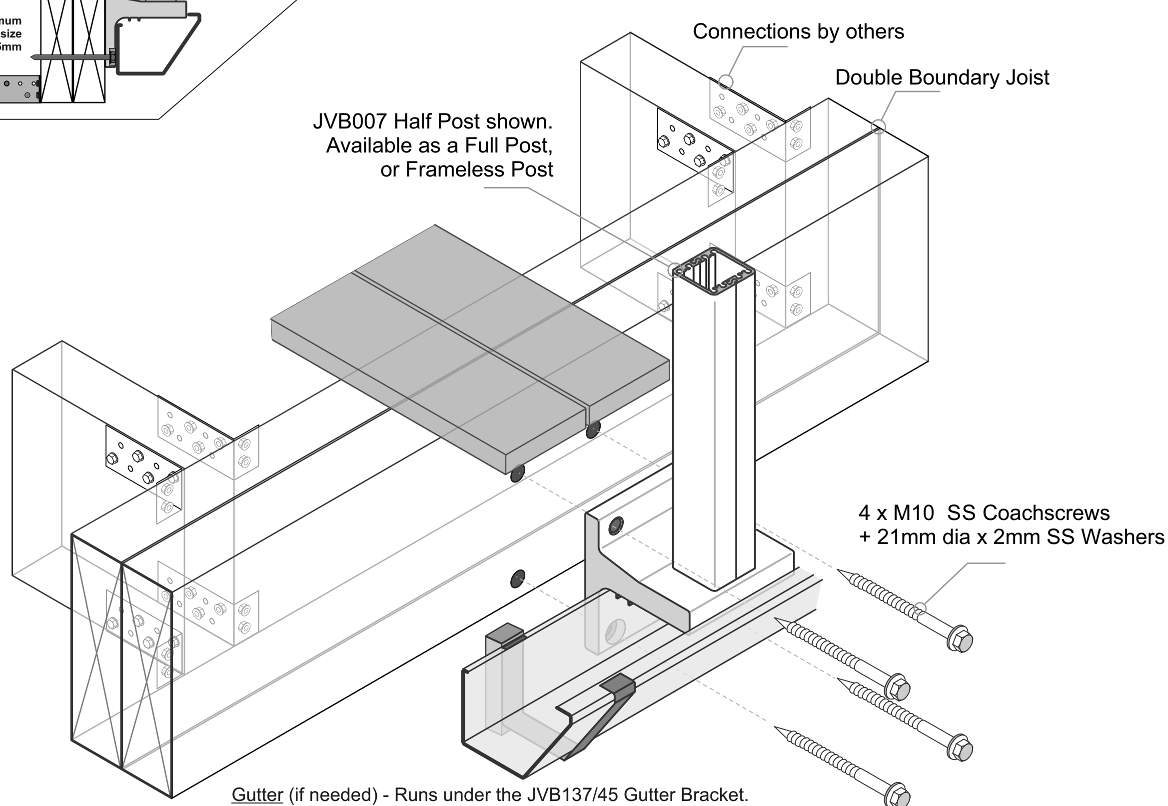
Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011





Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - Coachscrews 90mm min engagement into joists, predrill 6mm holes.
4 - Bond all coachscrews with SIKA Supergrip to full depth
5 - All Fixings must be Stainless steel



Gutter (if needed) - Runs under the JVB137/45 Gutter Bracket.
To be attached separately either side
of the Bracket to suit propriety guttering system.
Pack out to clear.

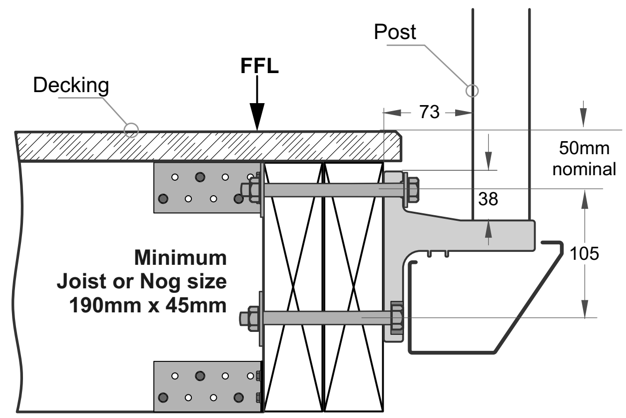
Typical FACE Fix to Timber - JVB137/45, Gutter Bracket - M10 SS Bolts

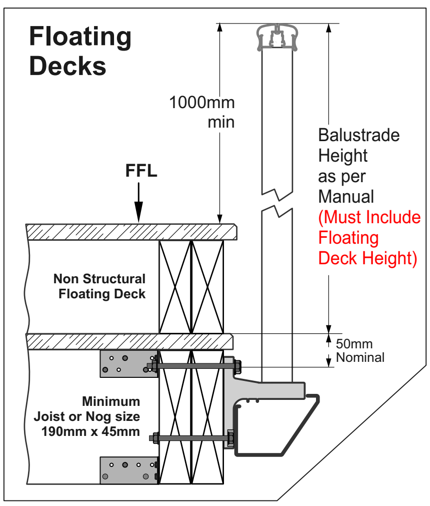
Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

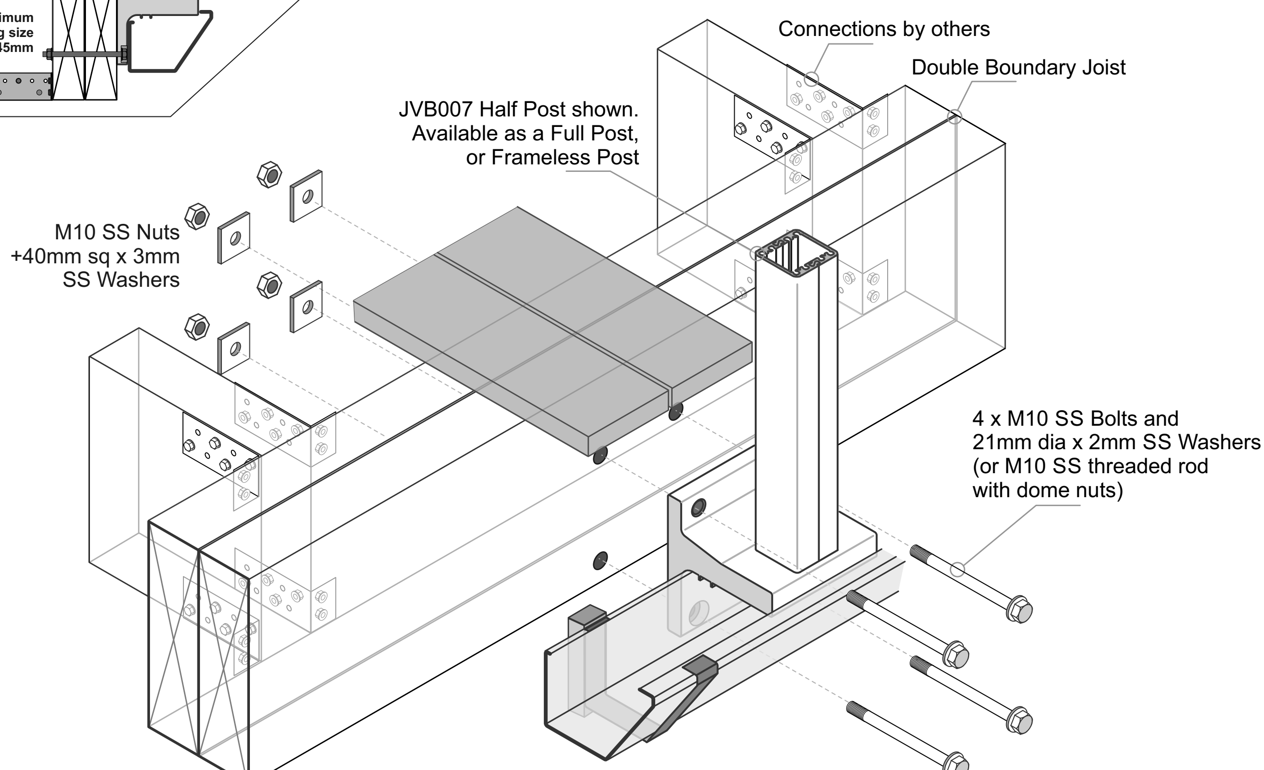
Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011





Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - All Fixings must be Stainless steel



Gutter (if needed) - Runs under the JVB137/45 Gutter Bracket.
To be attached separately either side
of the Bracket to suit propriety guttering system.
Pack out to clear.

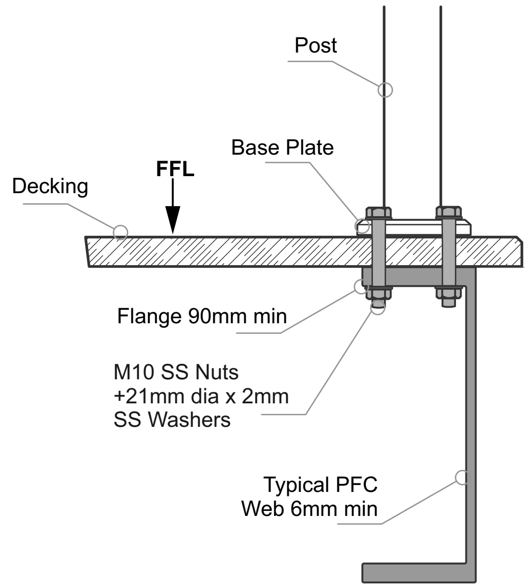
Typical TOP Fix to Steel with Timber Deck - JVB121, 110mm x 90mm, 4 hole Base Plate - M10 SS Bolts

Balustrade Dimensions by Wind Zone

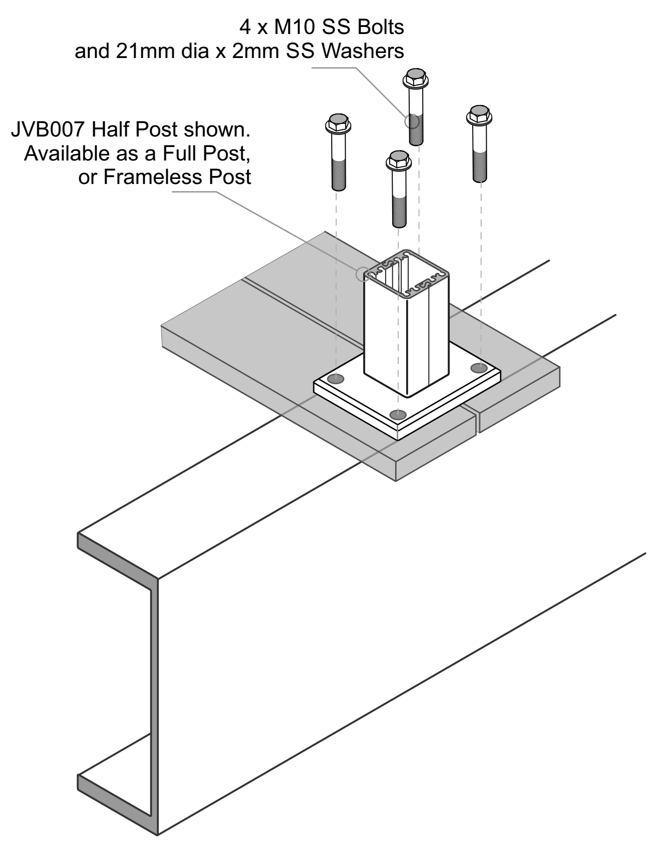
Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - All fixings must be Stainless steel



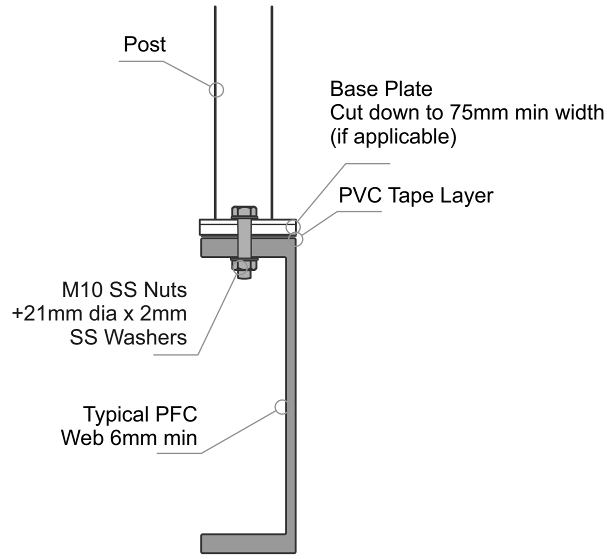
Typical TOP Fix to Timber Deck + Steel - JVB101, 110mm x 90mm, 2 hole Base Plate - M10 SS Bolts

Balustrade Dimensions by Wind Zone

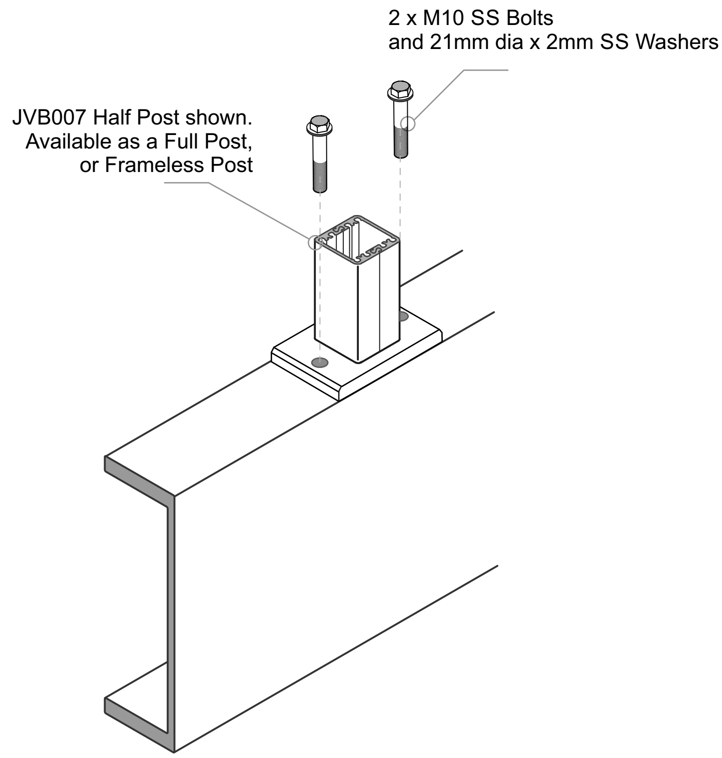
Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

Extra High Wind Zone
NOT SUITABLE.

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - The Baseplate can be cut down to 75mm wide
4 - Both Base plate and PFC must be aligned, with Bolt at C/L
5 - A PVC tape layer must be placed between Baseplate and Steel
6 - All fixings must be Stainless steel



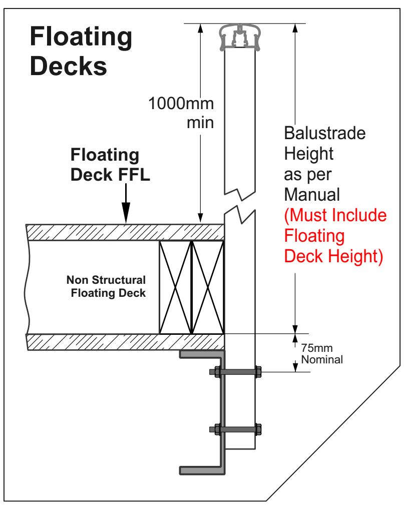
Typical FACE Fix Post to Steel - M10 SS Bolts

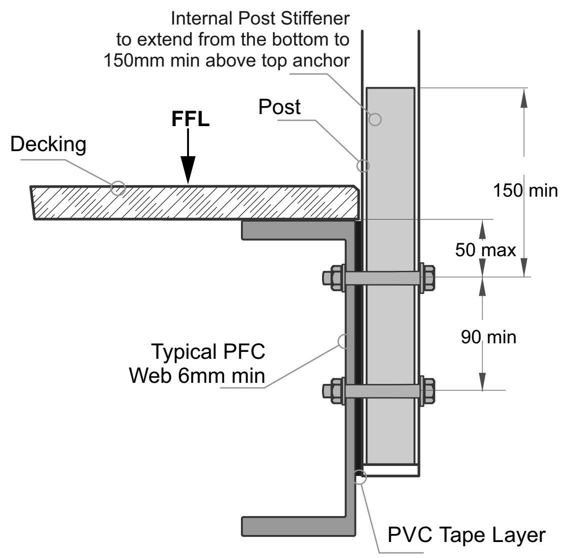
Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

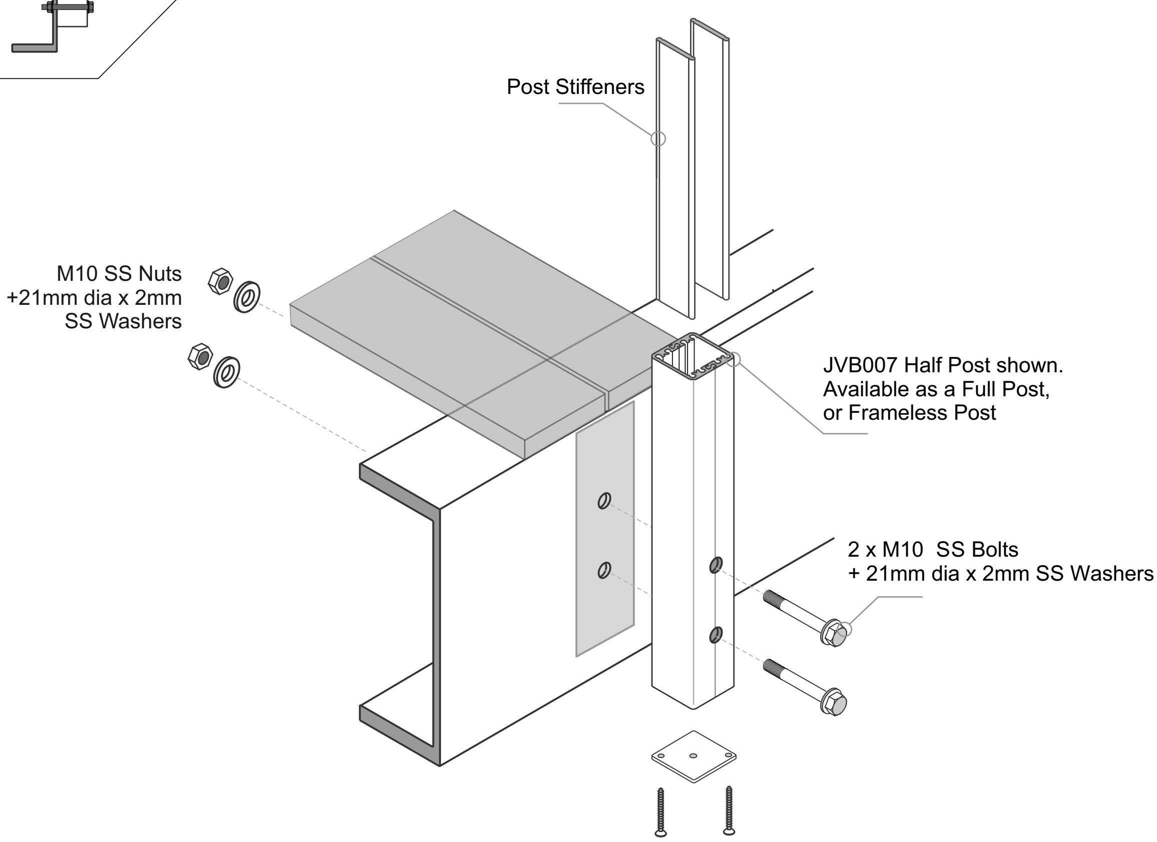
Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011





Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - A PVC tape layer must be placed between Post and Steel
4 - All fixings must be Stainless steel



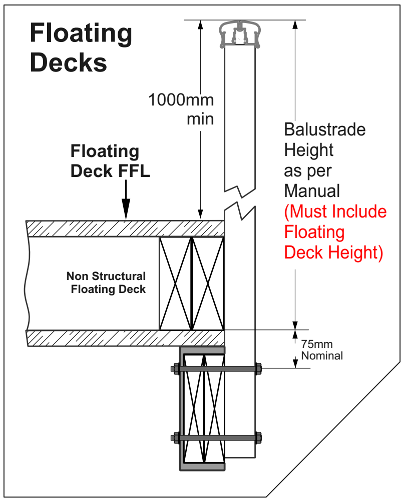
Typical FACE Fix Post to Steel + Wooden Packers - M10 SS Bolts

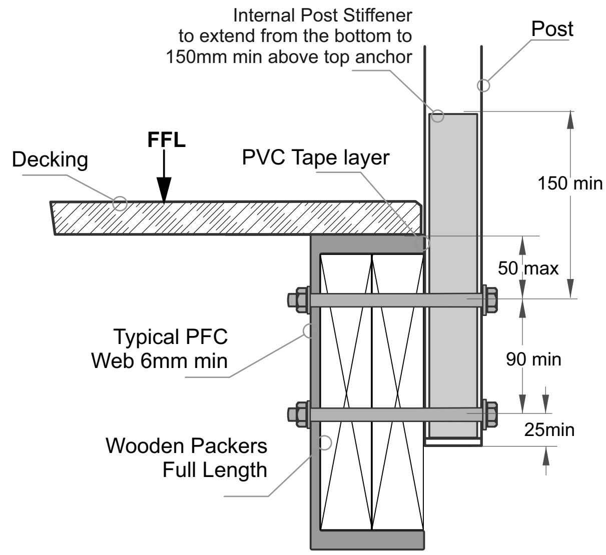
Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

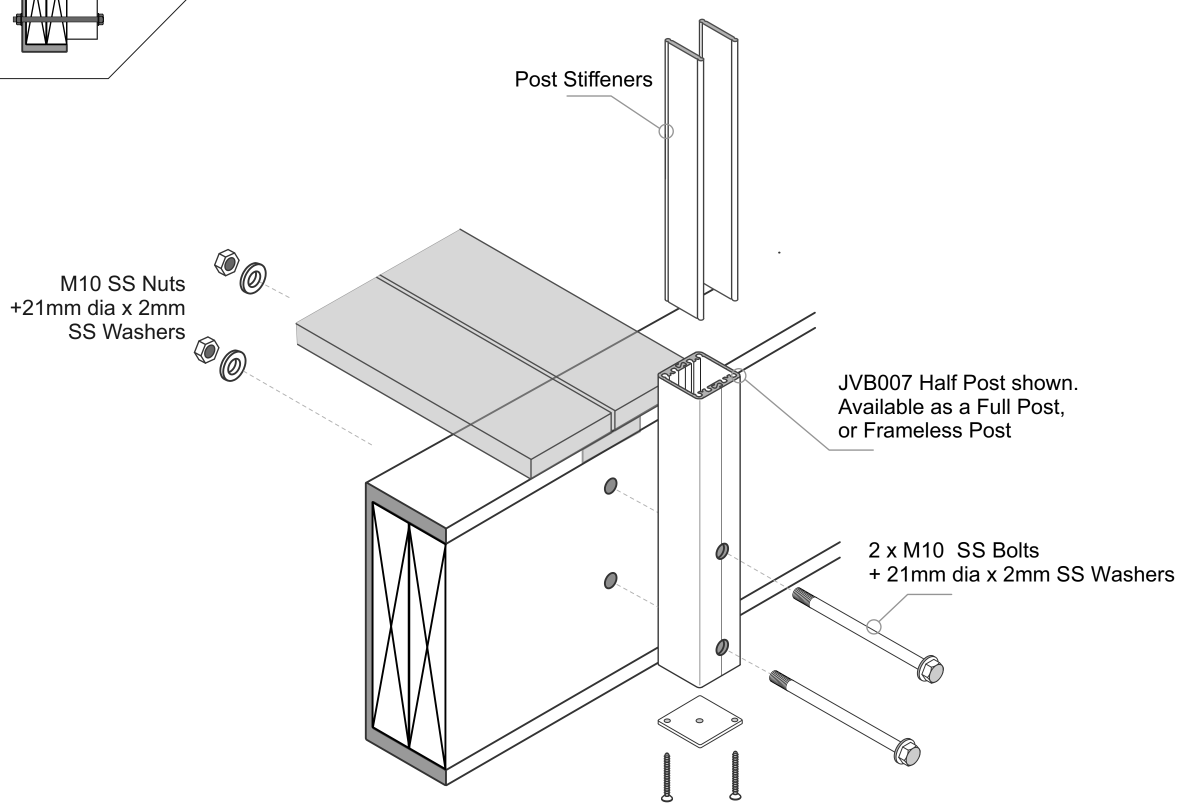
Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011





Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - A PVC Tape layer must be installed between the Post and Top Steel Flange
4 - All Fixings must be Stainless steel



Typical FACE Fix Post to Steel + Wooden Packers - M10 SS Bolts

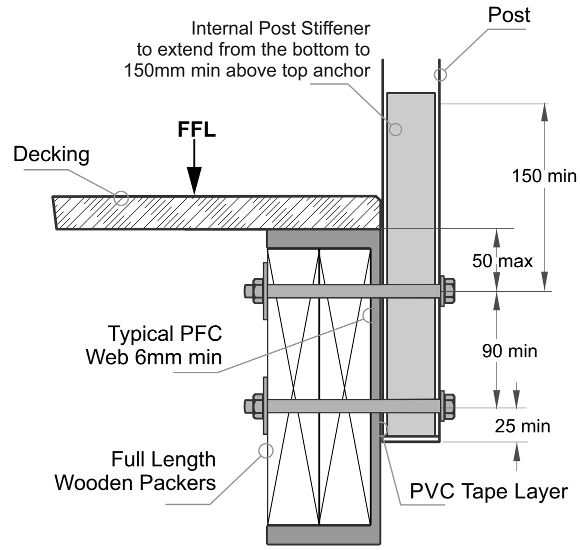
Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone

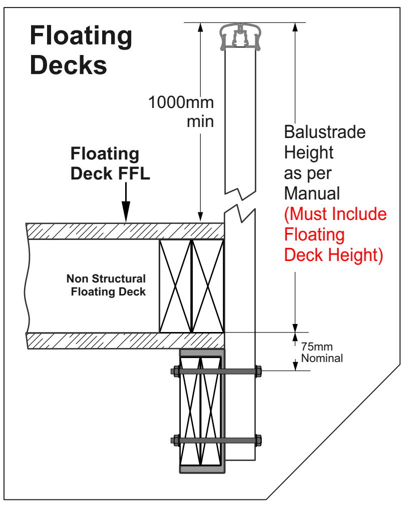
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

Extra High Wind Zone

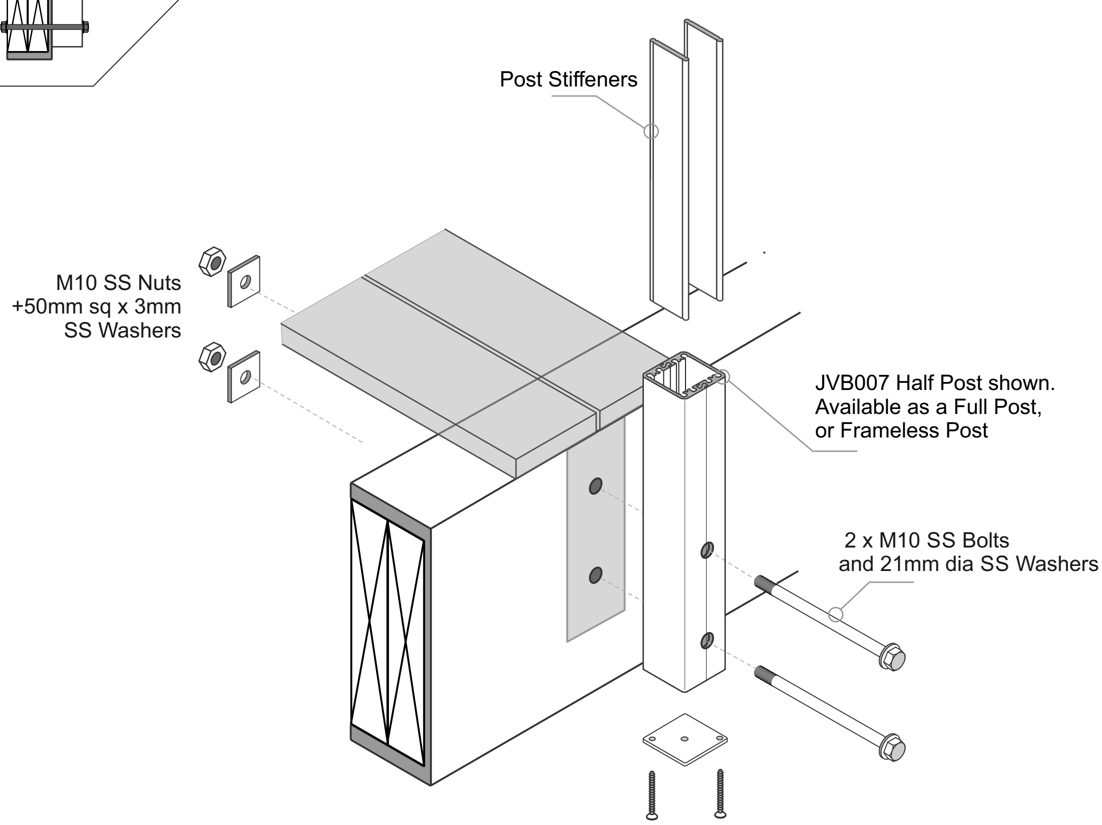
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm



General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - A PVC Tape layer must be installed between the Post and Steel
4 - All Fixings must be Stainless steel

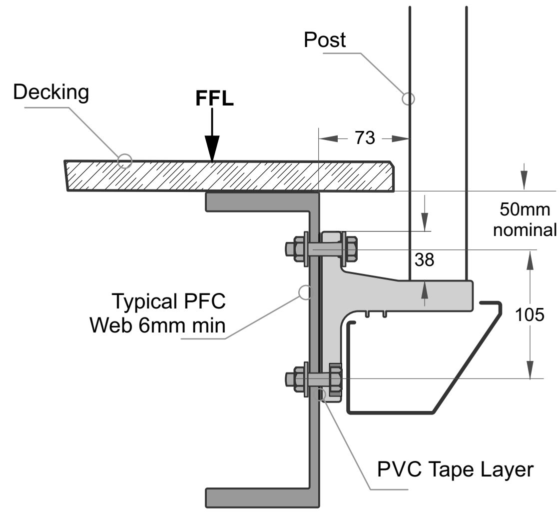


Typical FACE Fix to Steel - JVB137/45, Gutter Bracket - M10 SS Bolts

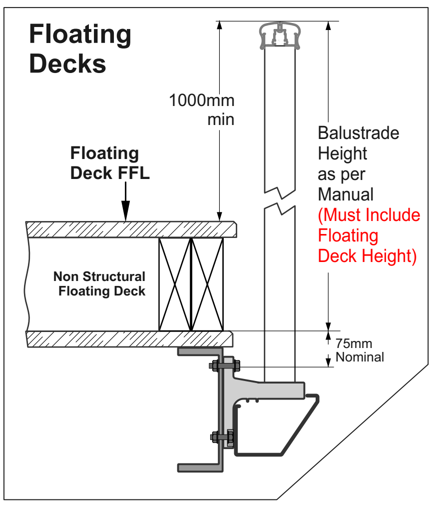
Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone							
Balustrade Height, mm	**1000**	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
Post Spacing max, mm	1400	1350	1300	1250	1200	1150	1100

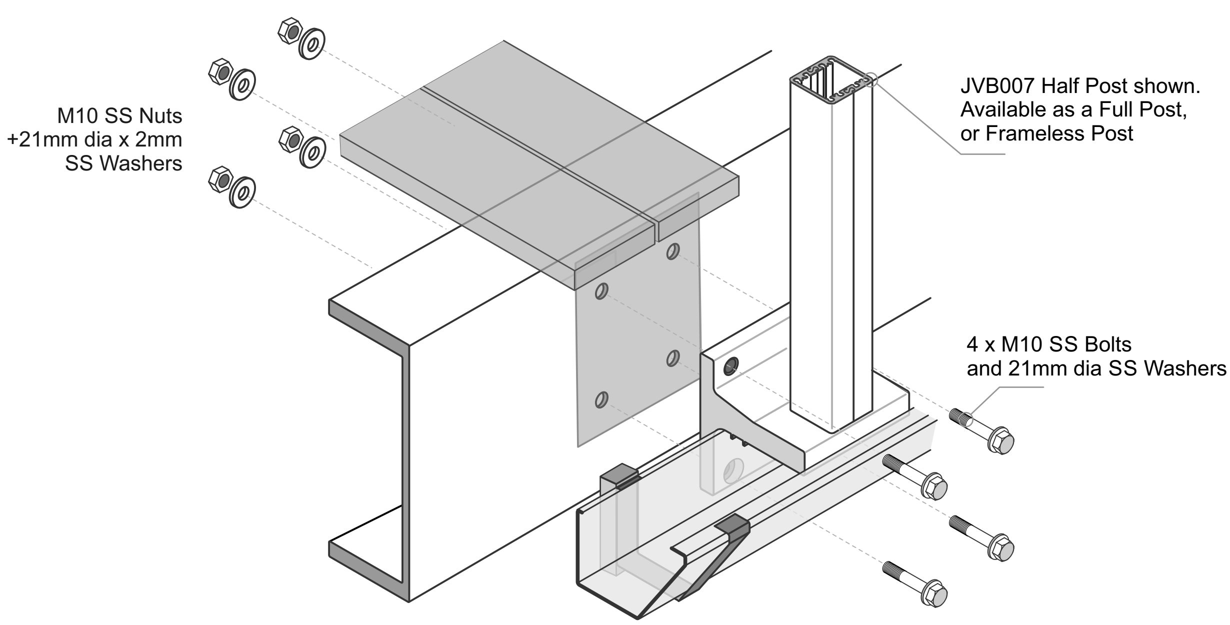
Extra High Wind Zone	
17mm Balusters only	
Balustrade Height, mm	**1275 max**
Post Spacing max, mm	1400



General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. Timber SG8 minimum strength
3 - A PVC Tape layer must be installed between the Gutter Bracket and Steel
4 - All Fixings must be Stainless steel



Gutter (if needed) - Runs under the JVB137/45 Gutter Bracket.
To be attached separately either side
of the Bracket to suit propriety guttering system.
Pack out to clear.

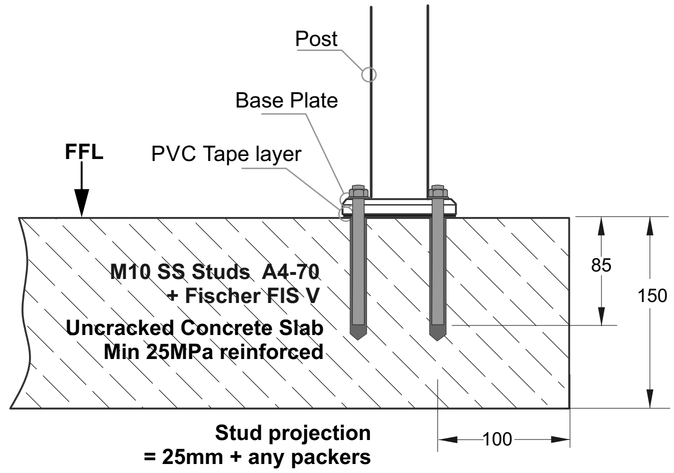
Typical TOP Fix to Concrete - JVB100, 100mm x 100mm, 4 hole Base Plate - M10 SS Studs

Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1150	1050	1000
Post Spacing max, mm						

Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1300
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011

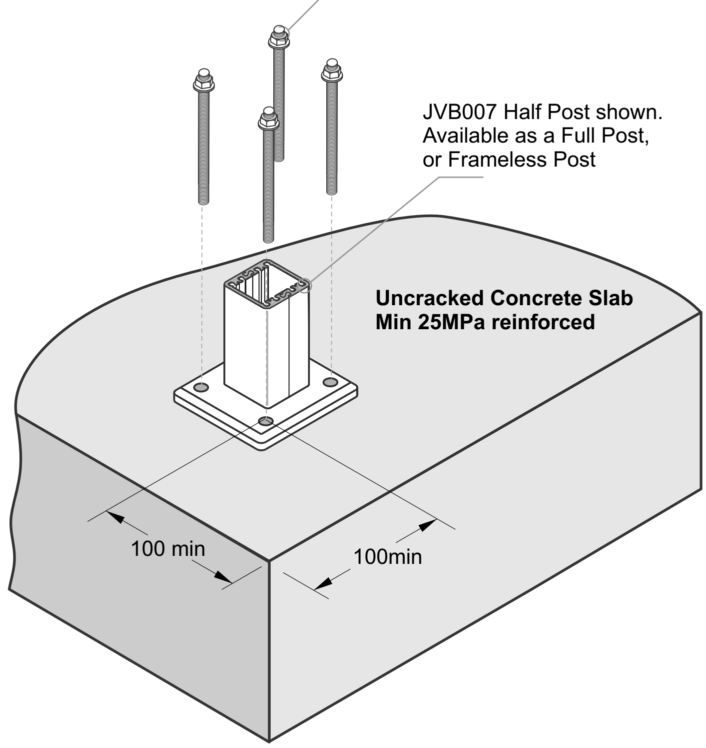


Installation details Fischer FIS V 300T

Thread diameter	M10
Drill hole diameter	= 12 mm
Drill hole depth	= 95 mm
Anchorage depth	= 85 mm
Drilling method	Hammer drilling
Drill hole cleaning	4 times blowing, 4 times brushing, 4 times blowing

No borehole cleaning required in case of using a hollow drill bit, e.g. fischer FHD.

Important Installation Notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - Fixings must engage into the structural slab
4 - A PVC Tape layer must be installed between the Baseplate and Concrete
5 - Use Threadlock on Nuts
6 - All fixings must be Stainless Steel



Typical TOP Fix to Thin Concrete - JVB100, 100mm x 100mm, 4 hole Base Plate - M10 SS Studs

Balustrade Dimensions - 17mm Balusters ONLY

Up to and including Extra High Wind Zone

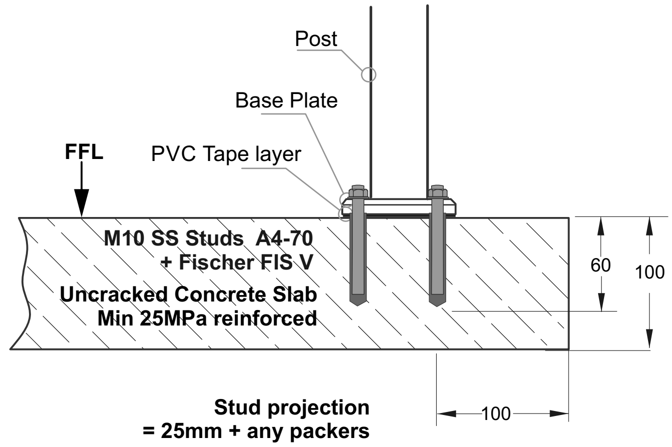
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1250	1200	1150	1100	1050	1050	1050
Post Spacing max, mm						

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011

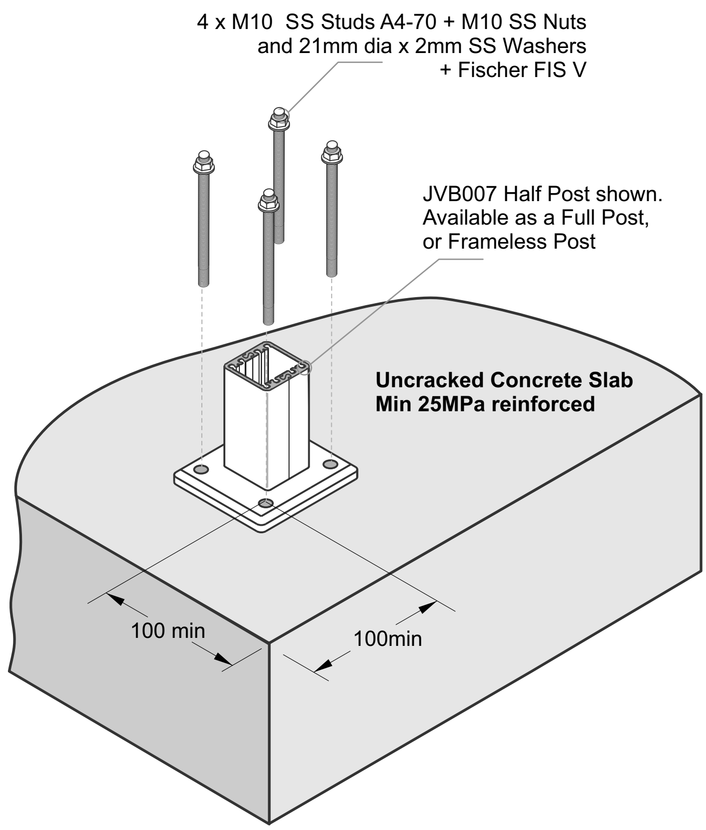
Installation details Fischer FIS V 300T

Thread diameter	M10
Drill hole diameter	= 12 mm
Drill hole depth	= 70mm
Anchorage depth	= 60 mm
Drilling method	Hammer drilling
Drill hole cleaning	4 times blowing, 4 times brushing, 4 times blowing

No borehole cleaning required in case of using a hollow drill bit, e.g. fischer FHD.



Important Installation Notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - Fixings must engage into the structural slab
4 - A PVC Tape layer must be installed between the Baseplate and Concrete
5 - Use Threadlok on Nuts
6 - All fixings must be Stainless Steel



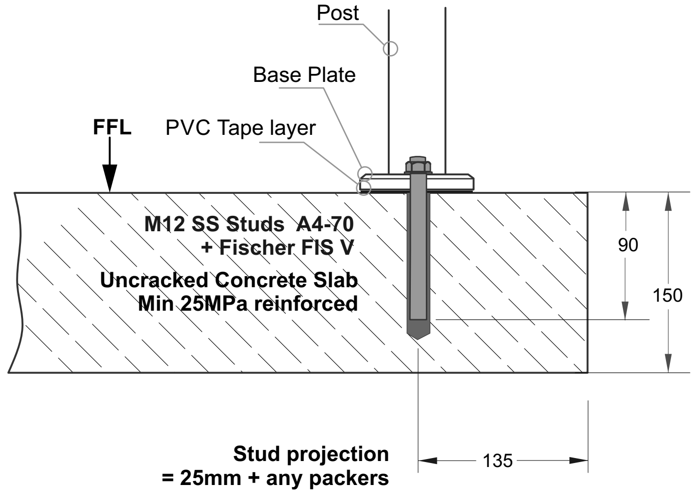
Typical TOP Fix to Concrete - JVB101, 110mm x 90mm, 2 hole Base Plate - M12 SS Studs

Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1100	1050	1000	900	800	750	700
Post Spacing max, mm						

Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
900
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011

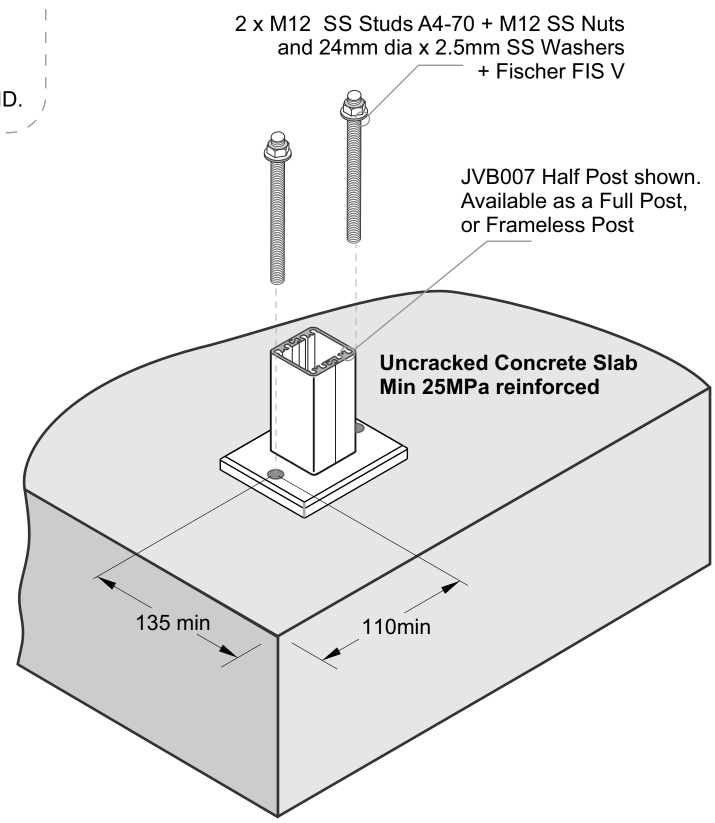


Installation details Fischer FIS V 300T

Thread diameter	M12
Drill hole diameter	= 14 mm
Drill hole depth	= 100 mm
Anchorage depth	= 90 mm
Drilling method	Hammer drilling
Drill hole cleaning	4 times blowing, 4 times brushing, 4 times blowing

No borehole cleaning required in case of using a hollow drill bit, e.g. fischer FHD.

Important Installation Notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - Fixings must engage into the structural slab
4 - A PVC Tape layer must be installed between the Baseplate and Concrete
5 - Use Threadlok on Nuts
6 - All fixings must be Stainless Steel



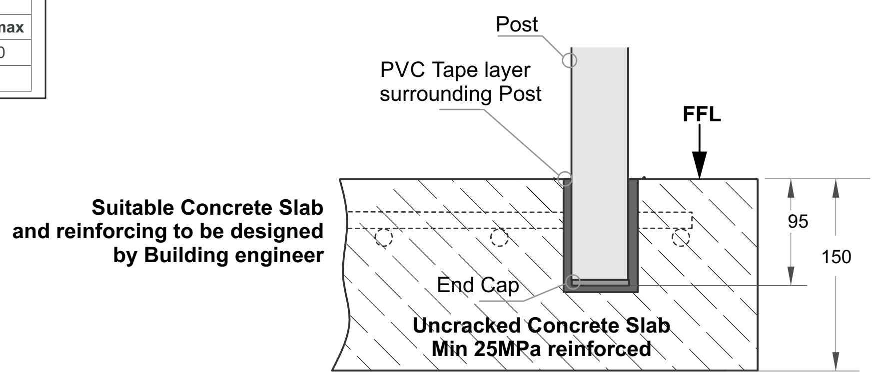
Typical TOP Fix to Concrete - Embed Post in Concrete Slab

Balustrade Dimensions by Wind Zone

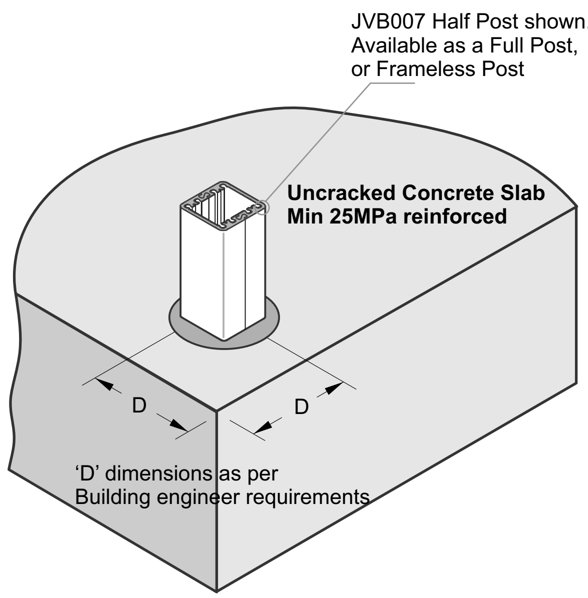
Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - A PVC Tape layer must completely surround the Post
4 - Mortar pocket 60mm sq or 80mm dia.
Avoid mortar splashes on exposed aluminium. Wash off immediately.



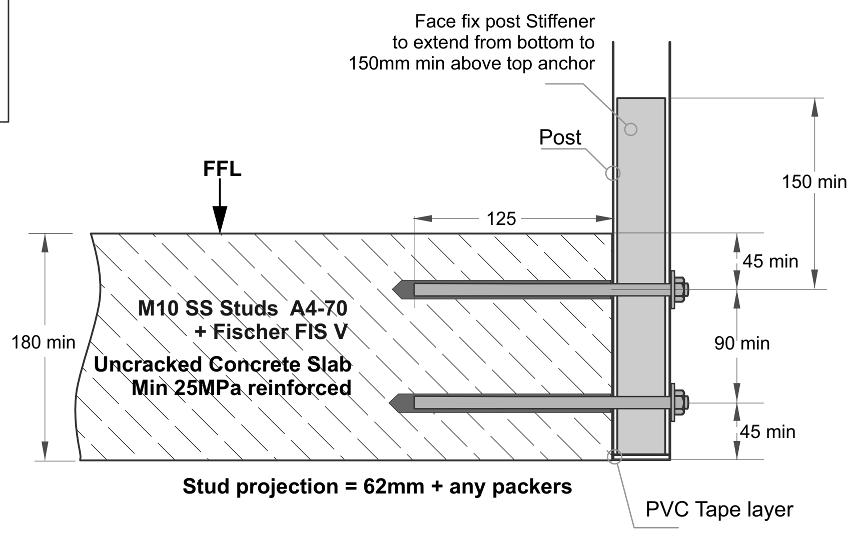
Typical FACE Fix Post to Concrete - M10 SS Studs

Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1200	1100	1000	900	850	800	750
Post Spacing max, mm						

Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1000
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011

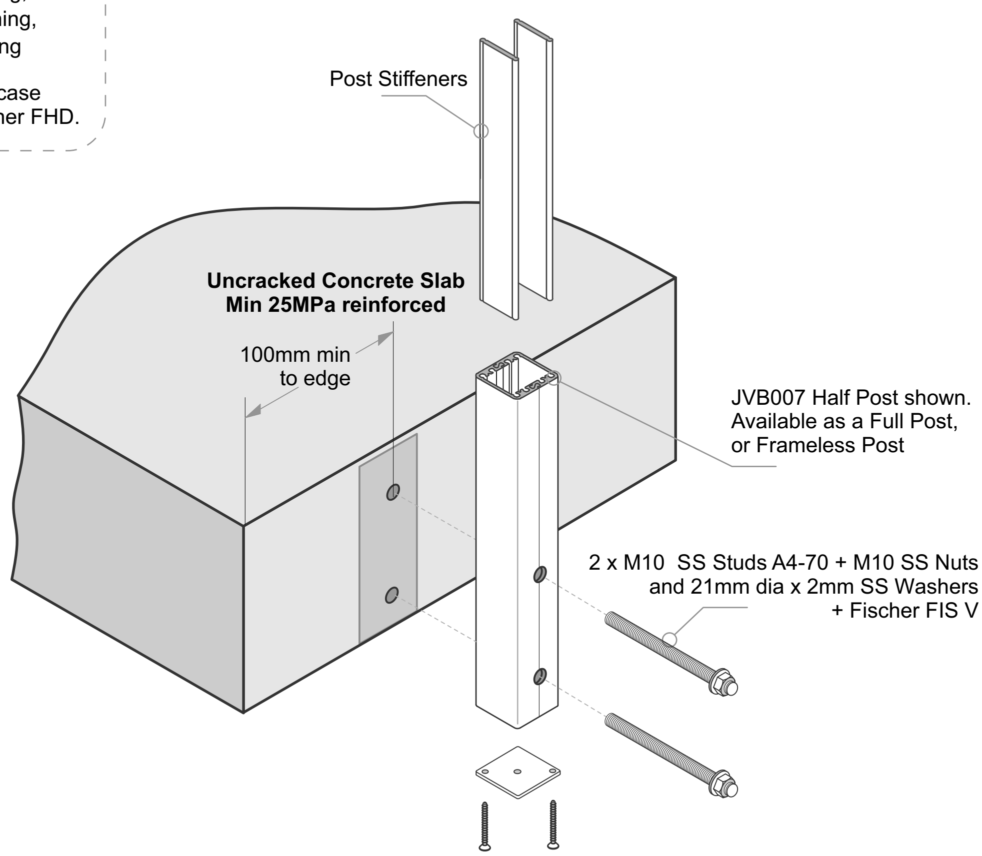


Installation details Fischer FIS V 300T

Thread diameter	M10
Drill hole diameter	= 12 mm
Drill hole depth	= 135 mm
Anchorage depth	= 125 mm
Drilling method	Hammer drilling
Drill hole cleaning	4 times blowing, 4 times brushing, 4 times blowing

No borehole cleaning required in case of using a hollow drill bit, e.g. fischer FHD.

Important Installation Notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - Fixings must engage into the structural slab
4 - A PVC Tape layer must be installed between the Post and Concrete
5 - Use Threadlok on Nuts
6 - All fixings must be Stainless Steel



Typical FACE Fix Post to Blockwall - M12 SS Studs

Balustrade Dimensions by Wind Zone

Up to and including Very High Wind Zone						
Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1200	1100	1000	900	850	800	750
Post Spacing max, mm						

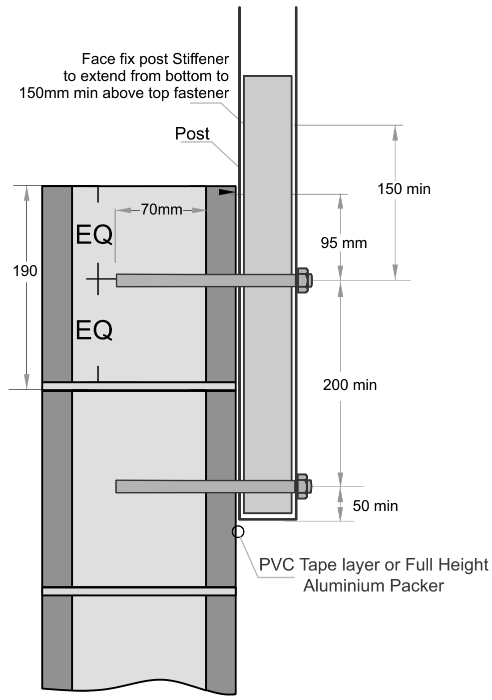
Extra High Wind Zone
17mm Balusters only
Balustrade Height, mm
1275 max
1050
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011

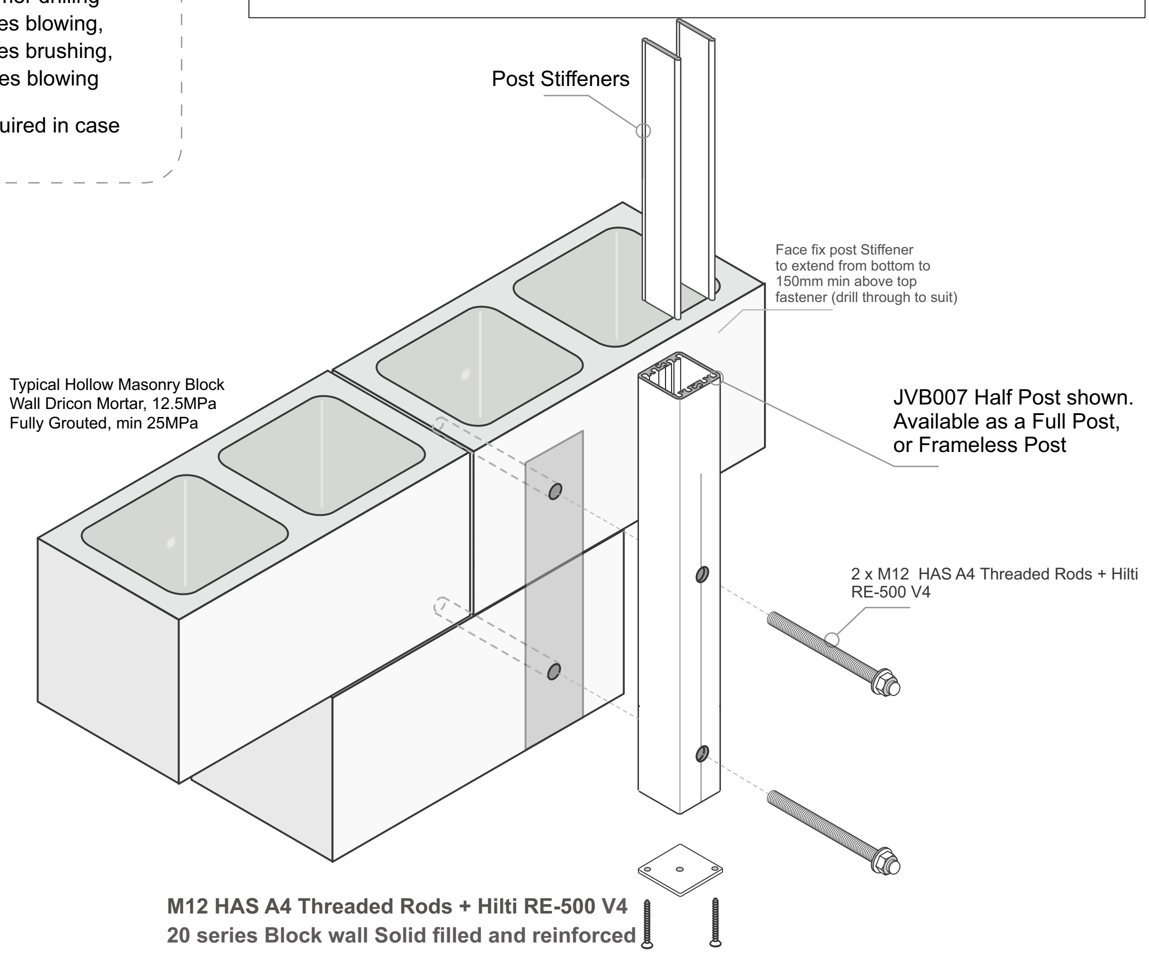
Installation details Hilti RE-500 V4

Thread diameter	M12
Drill hole diameter	= 14 mm
Drill hole depth	= 80mm
Anchorage depth	= 70mm
Drilling method	Hammer drilling
Drill hole cleaning	4 times blowing, 4 times brushing, 4 times blowing

No borehole cleaning required in case of using a hollow drill bit.



Important Installation Notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - Fixings must engage into the structural slab
4 - A PVC Tape layer must be installed between the Post and Concrete
5 - Use Threadlok on Nuts
6 - All fixings must be Stainless Steel



M12 HAS A4 Threaded Rods + Hilti RE-500 V4
20 series Block wall Solid filled and reinforced with 25MPa concrete

Typical FACE Fix to Concrete - JVB137/45, Gutter Bracket - M10 SS Studs

Balustrade Dimensions by Wind Zone

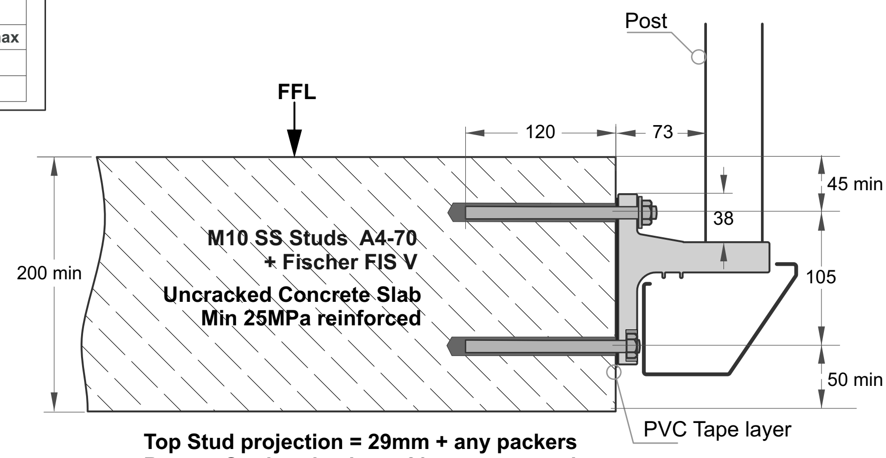
Up to and including Very High Wind Zone

Balustrade Height, mm						
1000	**1050**	**1100**	**1150**	**1200**	**1250**	**1300 max**
1400	1350	1300	1250	1200	1150	1100
Post Spacing max, mm						

Extra High Wind Zone

17mm Balusters only
Balustrade Height, mm
1275 max
1400
Post Spacing max, mm

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



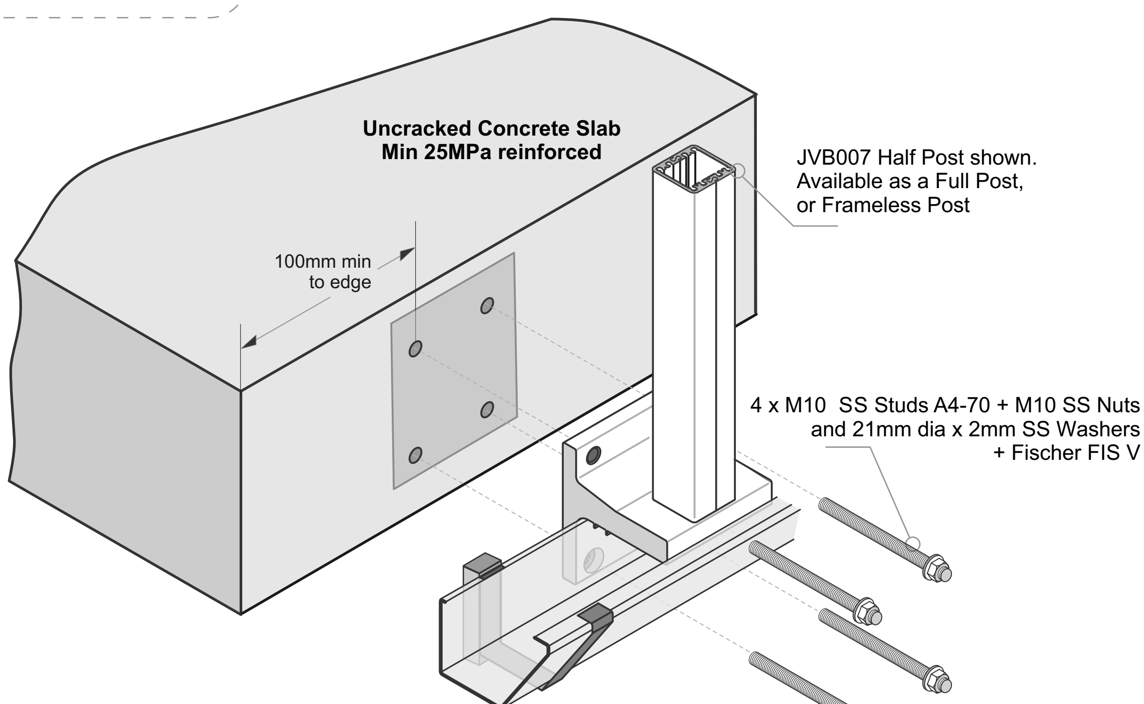
Top Stud projection = 29mm + any packers
Bottom Stud projection = 20mm + any packers

Installation details Fischer FIS V 300T

Thread diameter	M10
Drill hole diameter	= 12 mm
Drill hole depth	= 130 mm
Anchorage depth	= 120 mm
Drilling method	Hammer drilling
Drill hole cleaning	4 times blowing, 4 times brushing, 4 times blowing

No borehole cleaning required in case of using a hollow drill bit, e.g. fischer FHD.

Important Installation Notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only
3 - Fixings must engage into the structural slab
4 - A PVC Tape layer must be installed between the Gutter Bracket and Concrete
5 - Use Threadlok on Nuts
6 - All fixings must be Stainless Steel



Gutter (if needed) - Runs under the JVB137/45 Gutter Bracket.
To be attached separately either side
of the Bracket to suit propriety guttering system.
Pack out to clear.

Typical TOP Fix to Timber - JVB121, 110mm x 90mm, 4 hole Base Plate - M10 SS Coachscrews or Bolts

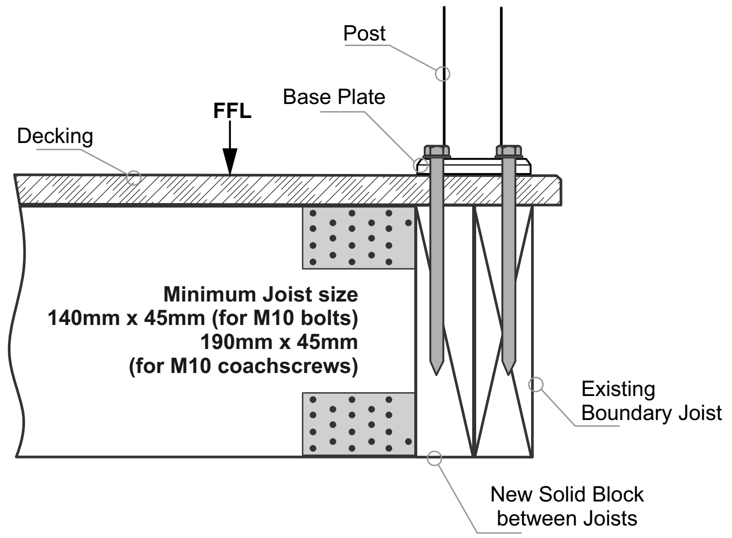
The pre NZS3604:2011 mounting details are included for older, existing buildings. New buildings must comply with NZS3604:2011- Double Boundary Joists

Balustrade Dimensions by Wind Zone

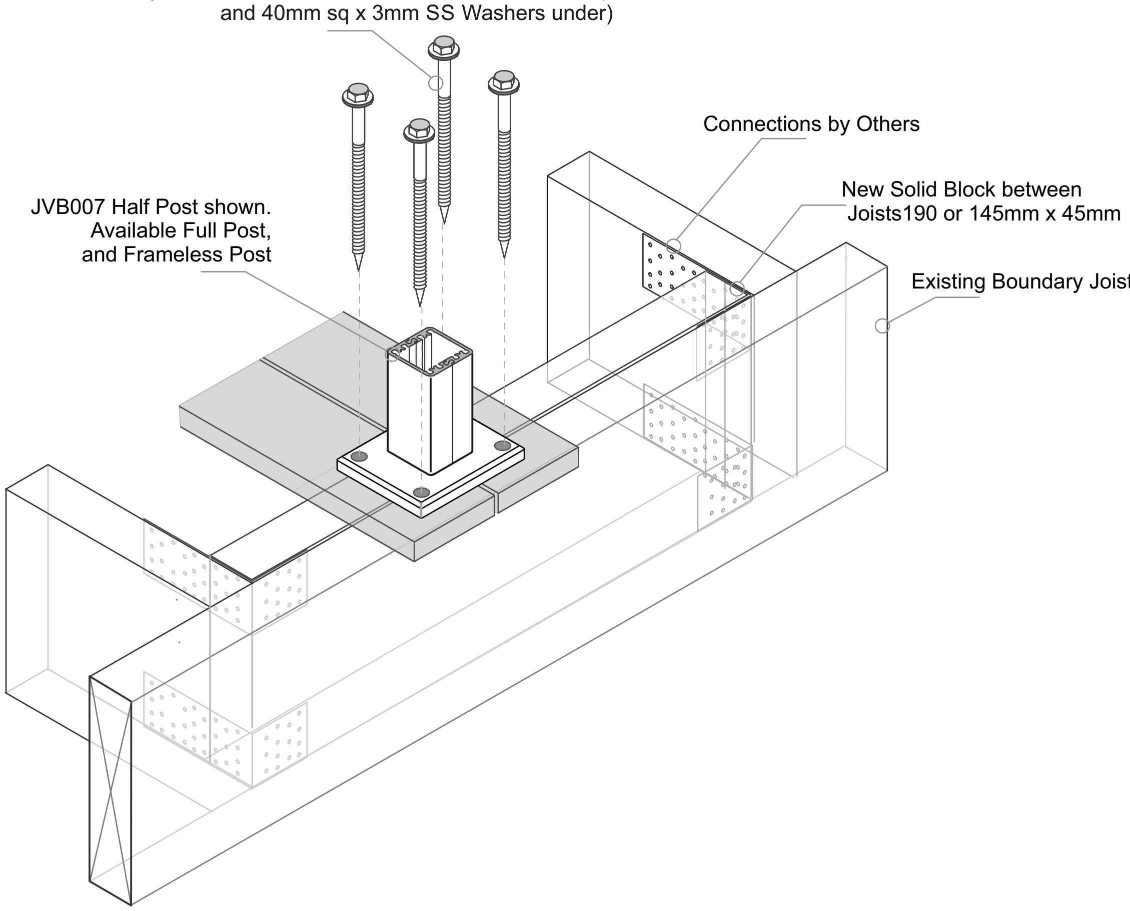
Up to and including Very High Wind Zone				
Balustrade Height, mm				
1000	**1050**	**1100**	**1150**	**1200**
1400	1350	1300	1250	1200
Post Spacing max, mm				

Extra High Wind Zone
NOT SUITABLE.

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. New Timber SG8 minimum strength
3 - Coachscrews 130mm min engagement into joists, predrill 6mm hole
4 - Bond all coachscrews with SIKA Supergrip to full depth
5 - All Fixings must be Stainless steel



Typical TOP Fix to Timber - JVB100, 100mm x 100mm, 4 hole Base Plate - M10 SS Coachscrews or Bolts

The pre NZS3604:2011 mounting details are included for older, existing buildings. New buildings must comply with NZS3604:2011- Double Boundary Joists

Balustrade Dimensions by Wind Zone

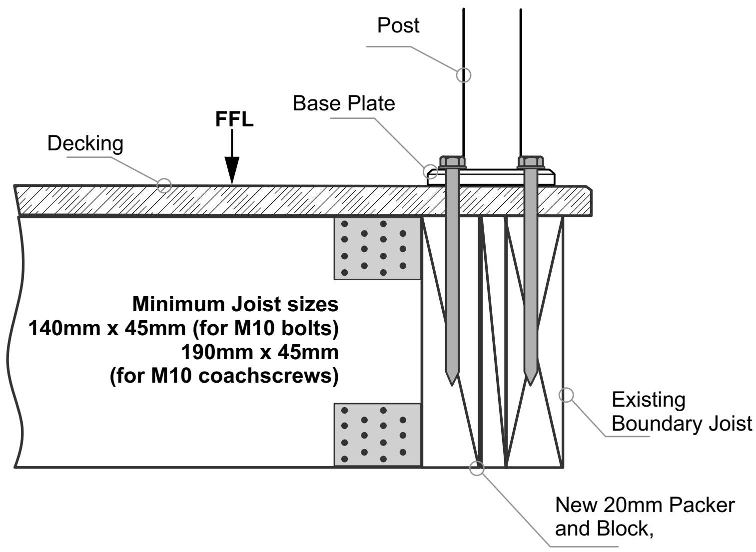
Up to and including Very High Wind Zone

Balustrade Height, mm				
1000	**1050**	**1100**	**1150**	**1200**
1400	1350	1300	1250	1200
Post Spacing max, mm				

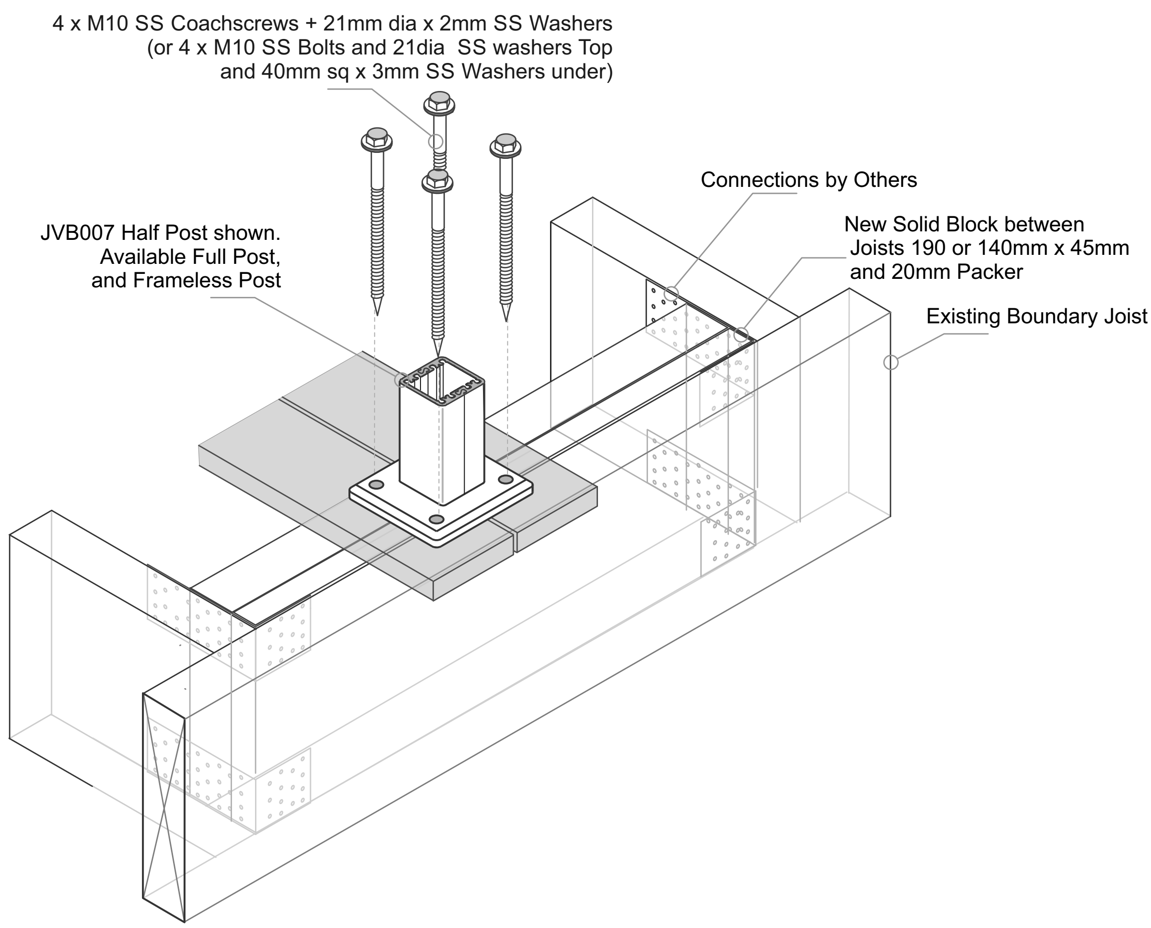
Extra High Wind Zone

NOT SUITABLE.

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. New Timber SG8 minimum strength
3 - Coachscrews 130mm min engagement into joists, predrill 6mm hole
4 - Bond all coachscrews with SIKA Supergrip to full depth
5 - All Fixings must be Stainless steel



Typical FACE Fix Post to Timber - M10 SS Bolts or Threaded Rod

The pre NZS3604:2011 mounting details are included for older, existing buildings. New buildings must comply with NZS3604:2011- Double Boundary Joists

Balustrade Dimensions by Wind Zone

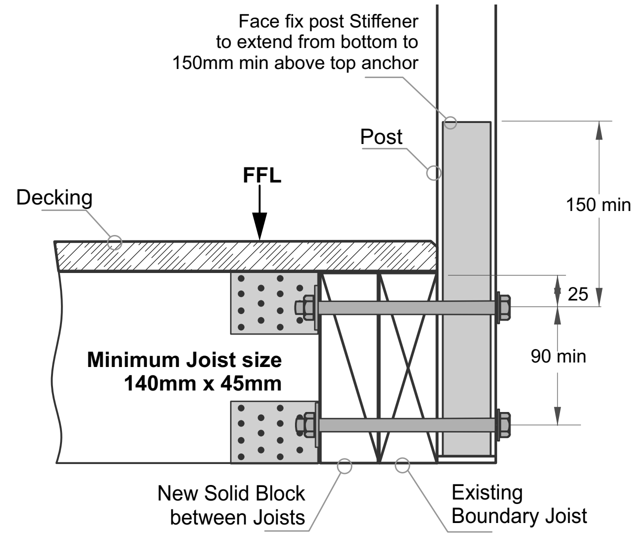
Up to and including Very High Wind Zone

Balustrade Height, mm				
1000	**1050**	**1100**	**1150**	**1200**
1400	1350	1300	1250	1200
Post Spacing max, mm				

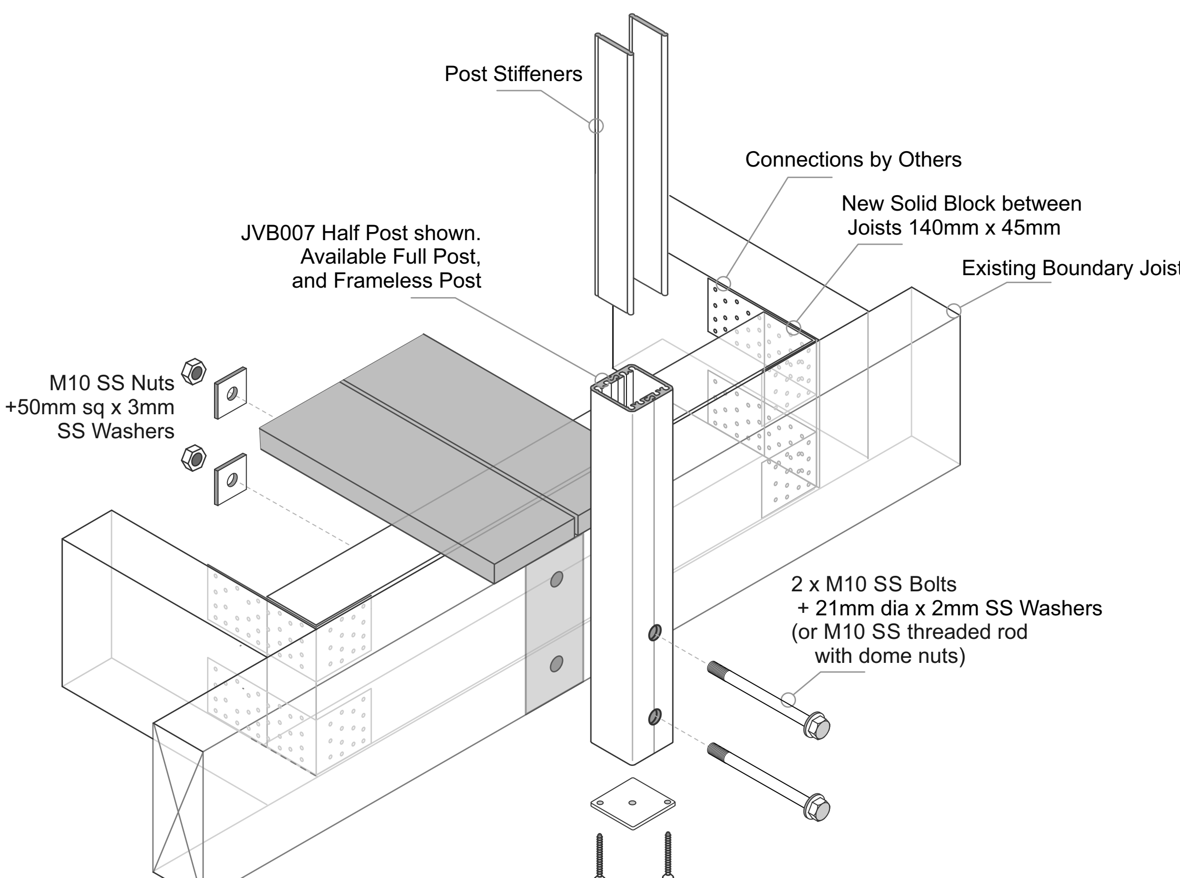
Extra High Wind Zone

NOT SUITABLE.

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. New Timber SG8 minimum strength
3 - All Fixings must be Stainless steel



Typical FACE Fix Post to Timber - M10 SS Coachscrews

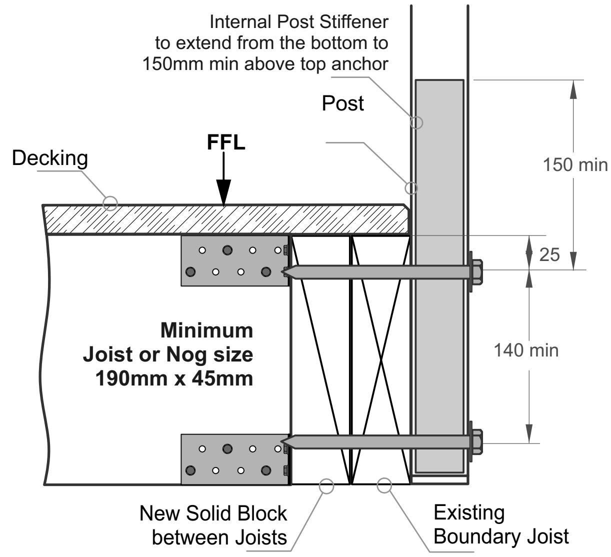
The pre NZS3604:2011 mounting details are included for older, existing buildings. New buildings must comply with NZS3604:2011- Double Boundary Joists

Balustrade Dimensions by Wind Zone

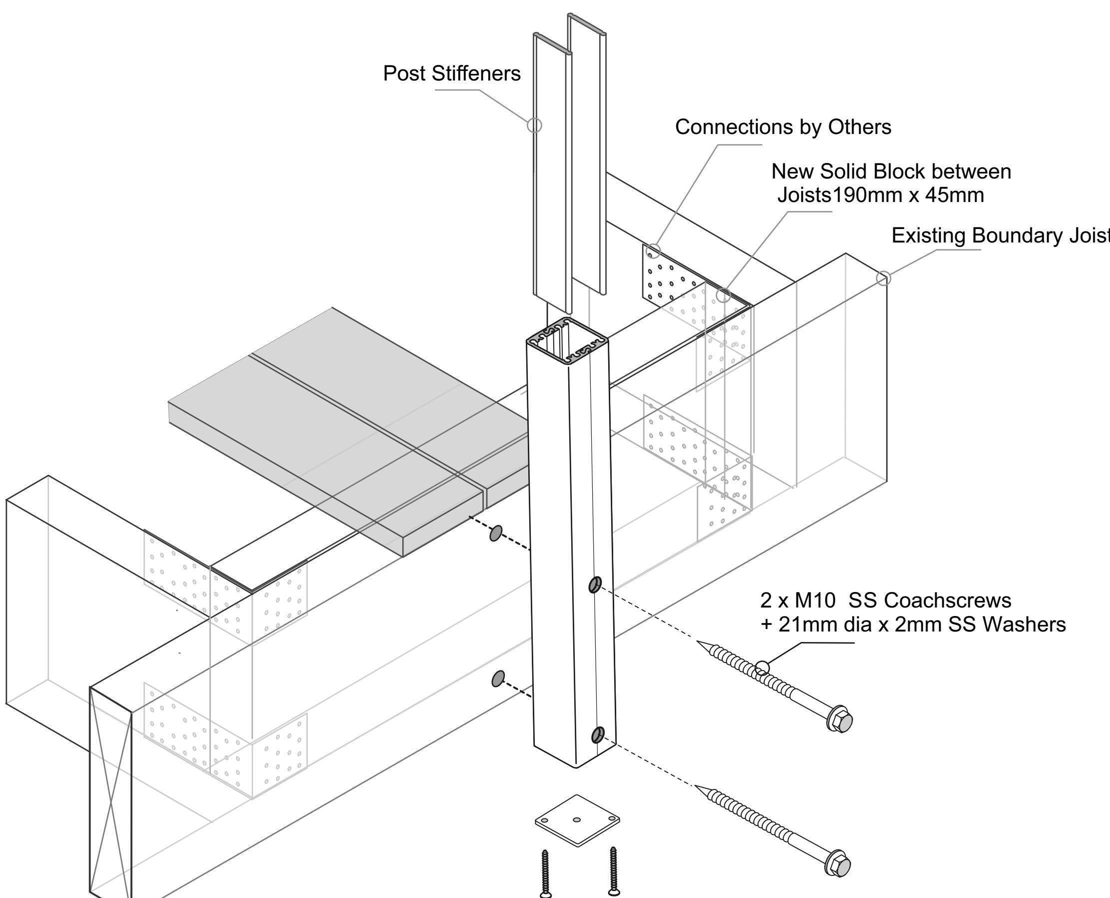
Up to and including Very High Wind Zone				
Balustrade Height, mm				
1000	**1050**	**1100**	**1150**	**1200**
1400	1350	1300	1250	1200
Post Spacing max, mm				

Extra High Wind Zone
NOT SUITABLE.

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. New Timber SG8 minimum strength
3 - Coachscrews 90mm min engagement into joists, predrill 6mm hole
4 - Bond all coachscrews with SIKA Supergrip to full depth
5 - All Fixings must be Stainless steel



Juralco Viking® Balustrade Systems - Typical Post Fixing
Pre NZS3604:2011 - Single Boundary Joist + Block

Typical FACE Fix to Timber - JVB137/45, Gutter Bracket - M10 SS Coachscrews

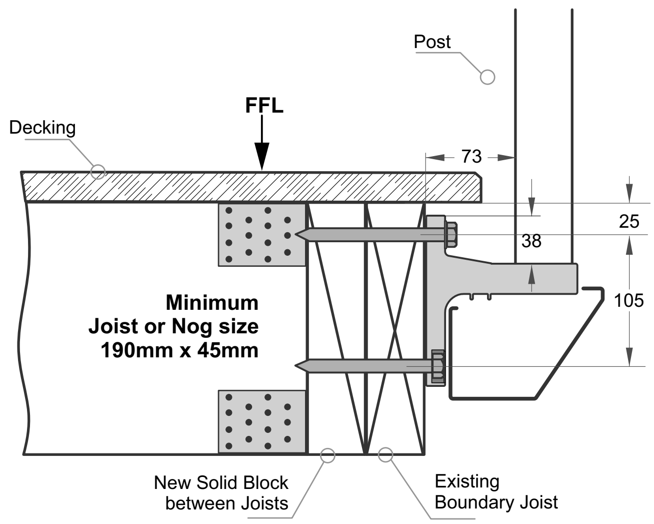
The pre NZS3604:2011 mounting details are included for older, existing buildings. New buildings must comply with NZS3604:2011- Double Boundary Joists

Balustrade Dimensions by Wind Zone.

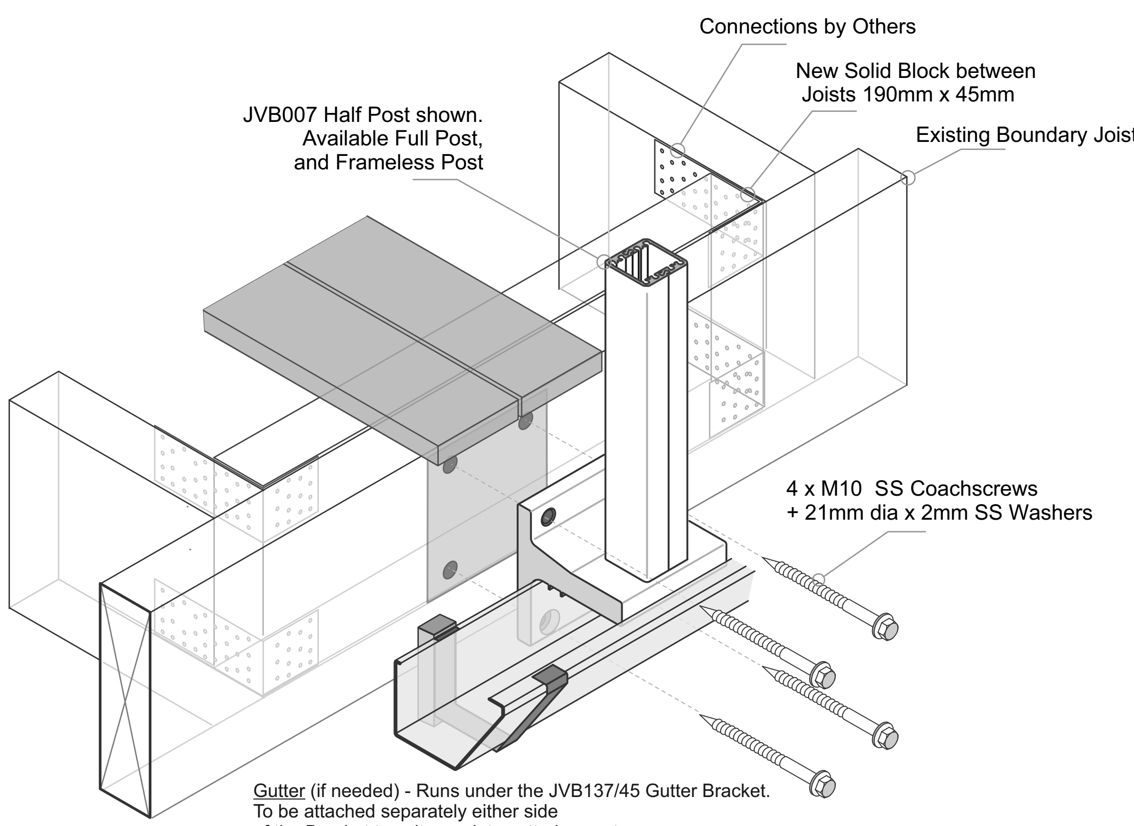
Up to and including Very High Wind Zone				
Balustrade Height, mm				
1000	**1050**	**1100**	**1150**	**1200**
1400	1350	1300	1250	1200
Post Spacing max, mm				

Extra High Wind Zone
NOT SUITABLE.

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011



Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. New Timber SG8 minimum strength
3 - Coachscrews 90mm min engagement into joists, predrill 6mm hole
4 - Bond all coachscrews with SIKA Supergrip to full depth
5 - All Fixings must be Stainless steel



Gutter (if needed) - Runs under the JVB137/45 Gutter Bracket.
To be attached separately either side
of the Bracket to suit propriety guttering system.
Pack out to clear.

Typical FACE Fix to Timber - JVB137/45, Gutter Bracket - M10 SS Bolts

The pre NZS3604:2011 mounting details are included for older, existing buildings. New buildings must comply with NZS3604:2011- Double Boundary Joists

Balustrade Dimensions by Wind Zone

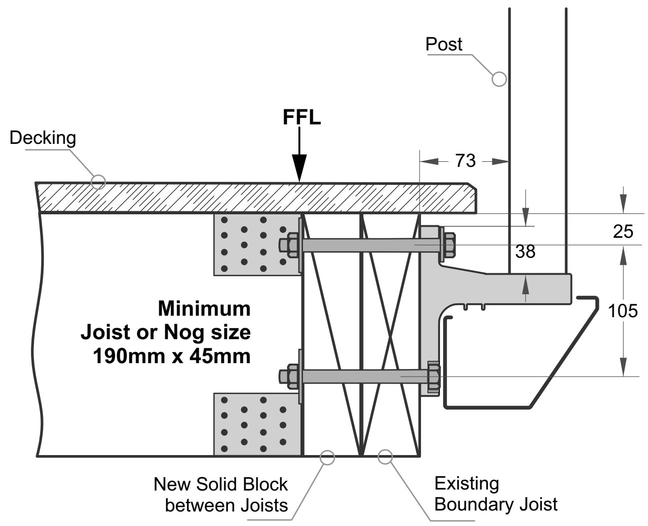
Up to and including Very High Wind Zone

Balustrade Height, mm				
1000	1050	1100	1150	1200
1400	1350	1300	1250	1200
Post Spacing max, mm				

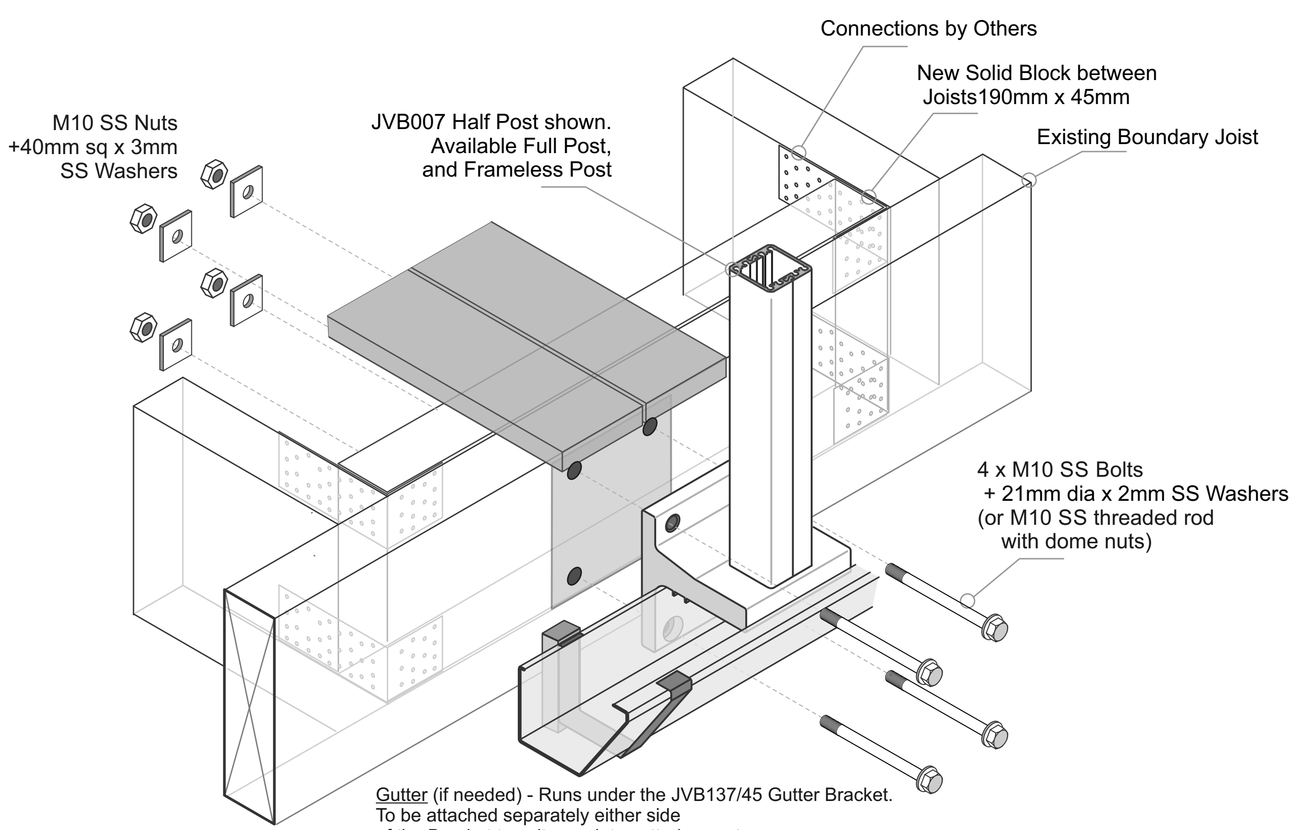
Extra High Wind Zone

NOT SUITABLE.

General Notes:
1 - All measurements mm
2 - Occupancy only A, A other and C3.
3 - Balustrade Height measured above Deck/FFL.
4 - Wind Zones as per NZS 3604:2011

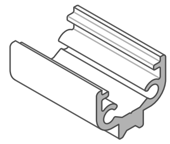


Important Installation notes:
1 - The Project Engineer must ensure the structure can support the appropriate loads
2 - Substructure shown indicatively only. New Timber SG8 minimum strength
3 - All Fixings must be Stainless steel

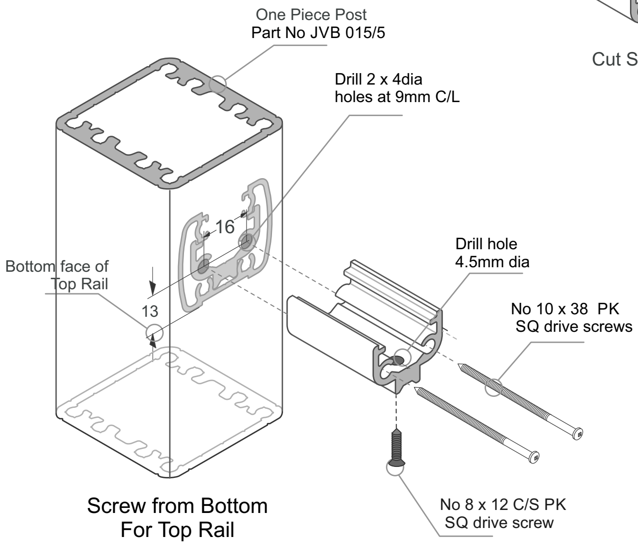


Gutter (if needed) - Runs under the JVB137/45 Gutter Bracket.
To be attached separately either side
of the Bracket to suit propriety guttering system.
Pack out to clear.

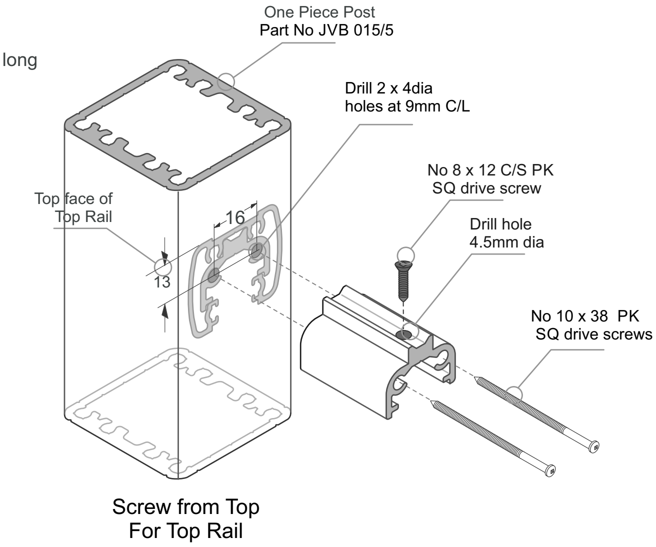
Typical Assembly - One Piece Post to Rail



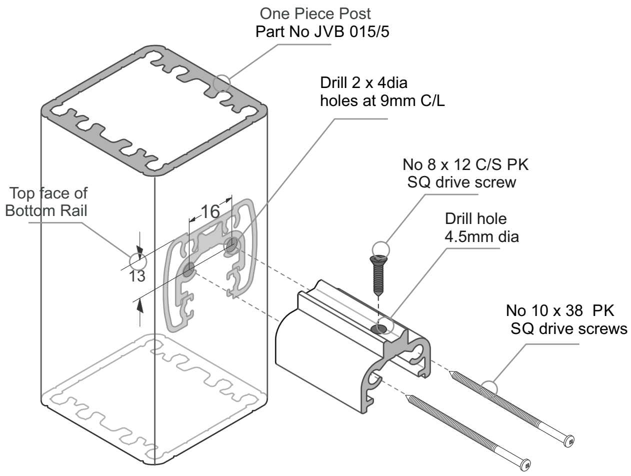
Cut Spigot 30mm long



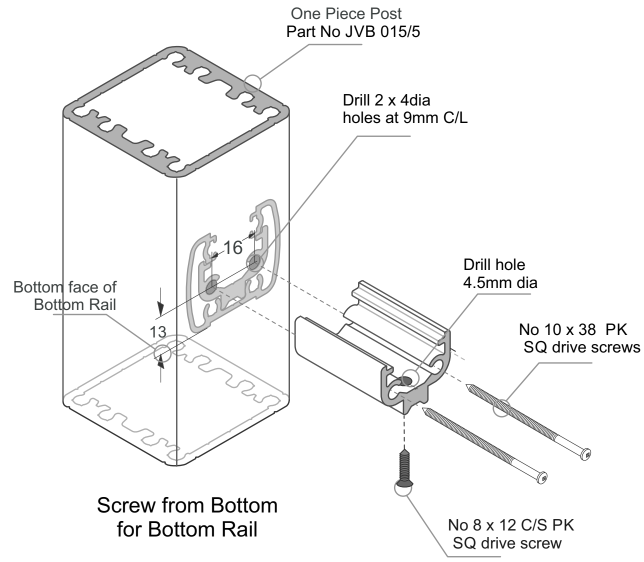
Screw from Bottom
For Top Rail



Screw from Top
For Top Rail



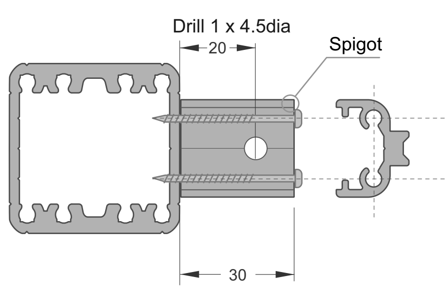
Screw from Top
for Bottom Rail



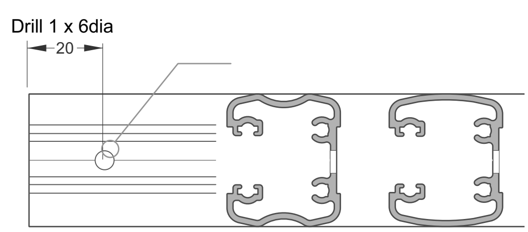
Screw from Bottom
for Bottom Rail

This Setout for Baluster Installations

This Setout for Glass Installations

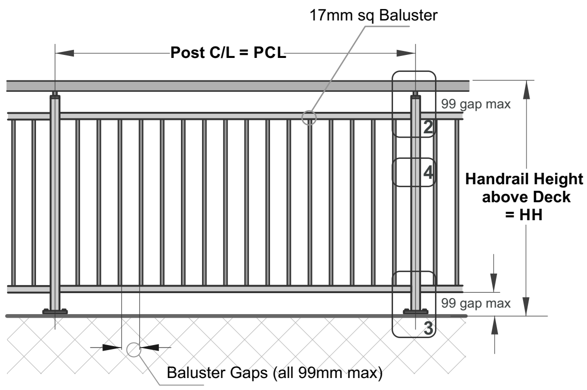


Attaching Bottom or Top Rail
to One Piece Post
Part No JVB 015/5

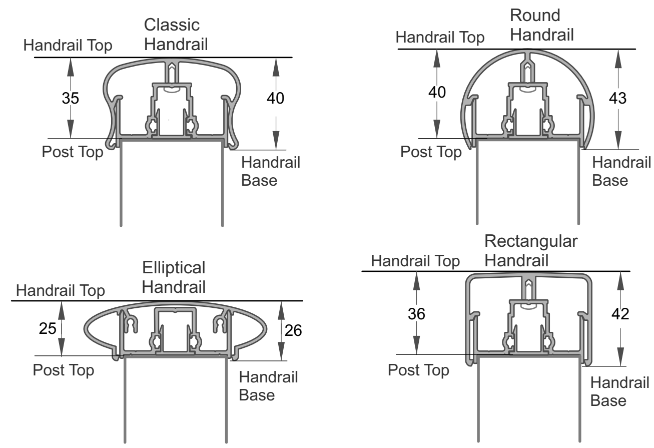


Drill Bottom Rail, Top rail if needed

1 - Refer Post Mounting type and installation Wind zone. Then choose Balustrade Height and max Post spacing.



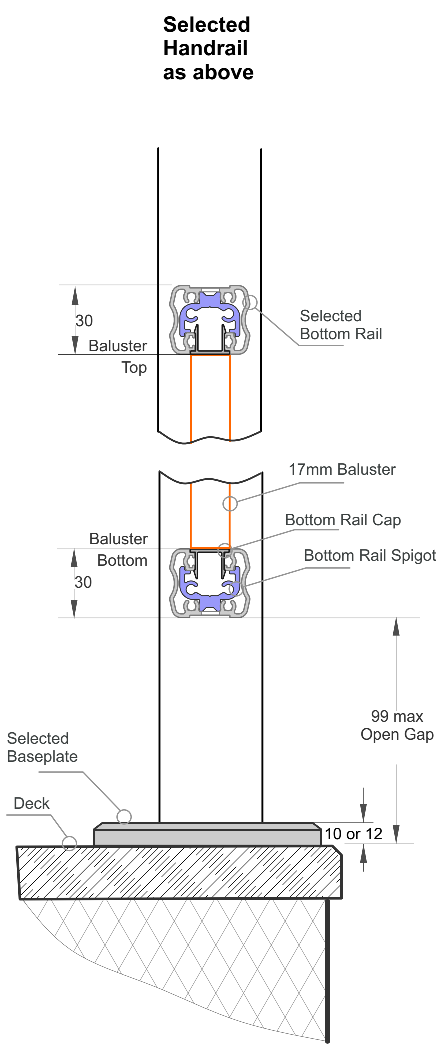
2 - Handrail - Offsets



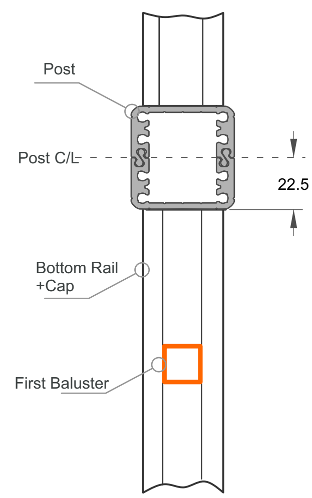
A
For Post Height.
Post Top to Handrail Top
25-40 Depending on
chosen Handrail

B
For Top Gap Height
Handrail Base to Handrail Top
26-43 Depending on
chosen Handrail

3 - Post Height offsets



4 - Width Offsets



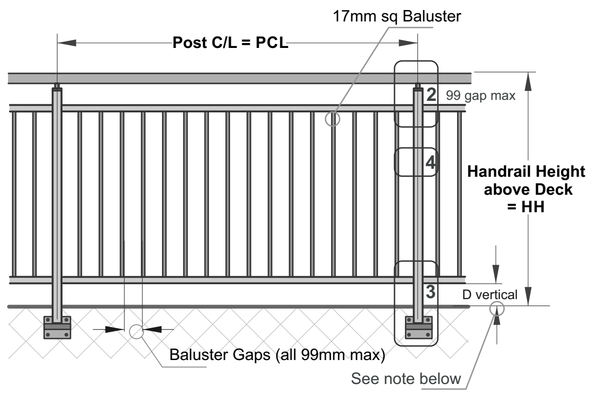
Viking post
JVB/007 shown
(or could be JVB/015)

5 - Cutting Guide

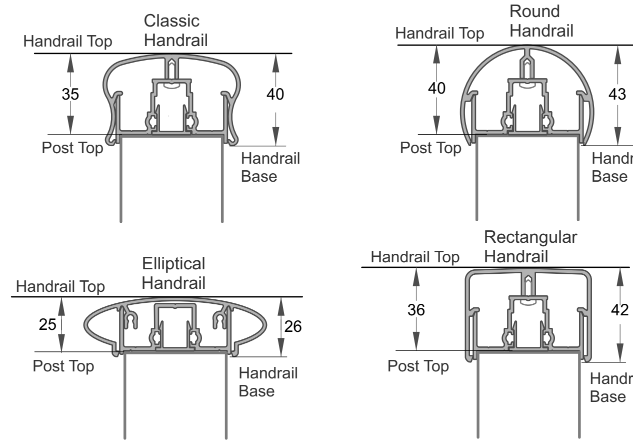
a - Hand Rail = Use maximum lengths
b - Post, Cut to
= HH - **A** - (10 or 12) = Post Height
c - Bottom Rail (x2), Cut to
= PCL- 2x22.5 = PCL - 45

d - 17mm Balustrade Height
= HH - **B** - 2x99 - 2x30

1 - Refer Post Mounting type and installation Wind zone. Then choose Balustrade Height and max Post spacing.



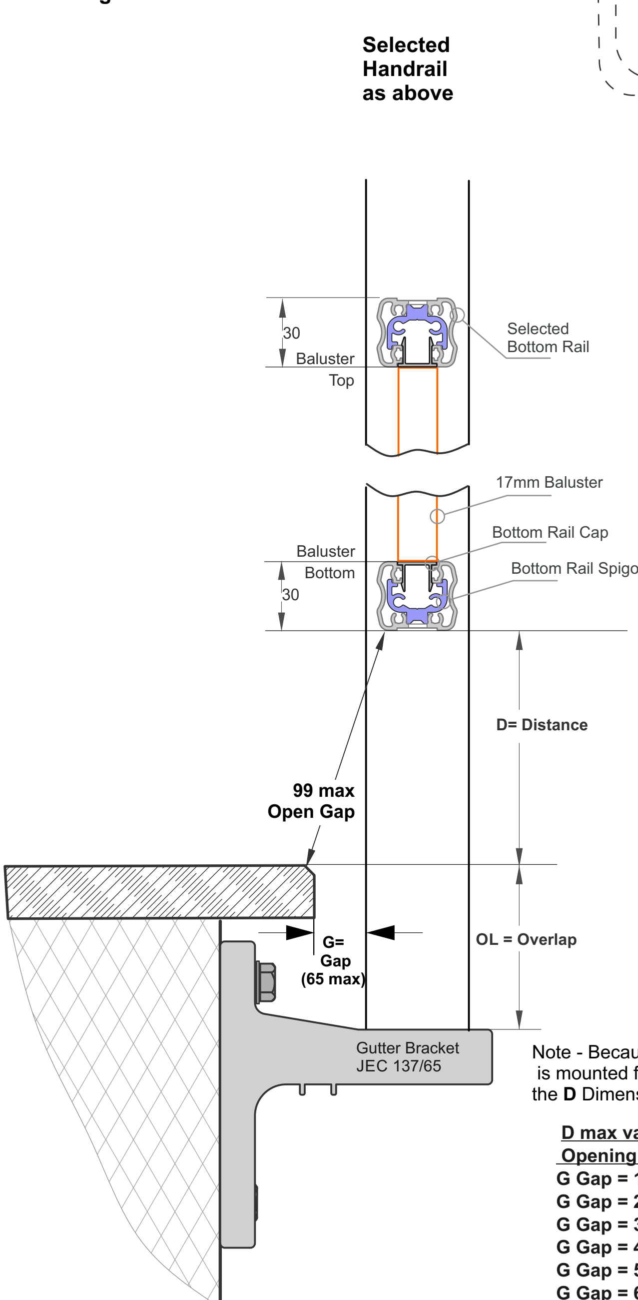
2 - Handrail - Offsets



A
For Post Height.
Post Top to Handrail Top
25-40 Depending on
chosen Handrail

B
For Top Gap Height
Handrail Base to Handrail Top
26-43 Depending on
chosen Handrail

3 - Post Height offsets

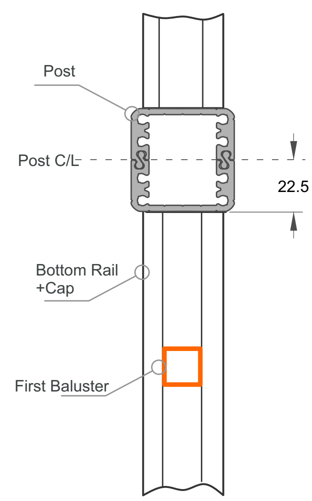


Note - Because the Bottom Rail is mounted forward of the Deck Edge the **D** Dimension is no longer 99mm.

D max values for a 99mm Opening to the deck

G Gap = 10mm D = 97mm
G Gap = 20mm D = 95mm
G Gap = 30mm D = 91mm
G Gap = 40mm D = 86mm
G Gap = 50mm D = 79mm
G Gap = 60mm D = 71mm
G Gap = 65mm max D = 66mm

4 - Width Offsets



Viking post
JVB/007 shown
(or could be JVB/015)

5 - Cutting Guide

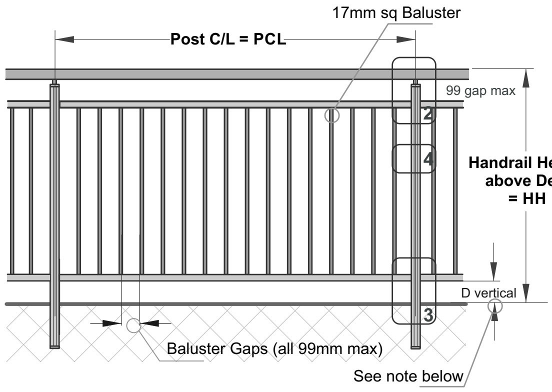
a - Hand Rail = Use maximum lengths
b - Post, Cut to
= HH - **A** + OL = Post Height
c - Bottom Rail (x2), Cut to
= PCL- 2x22.5 = PCL - 45

d - 17mm Balustrade Height
= HH - **B** - 99 - D - 2x30

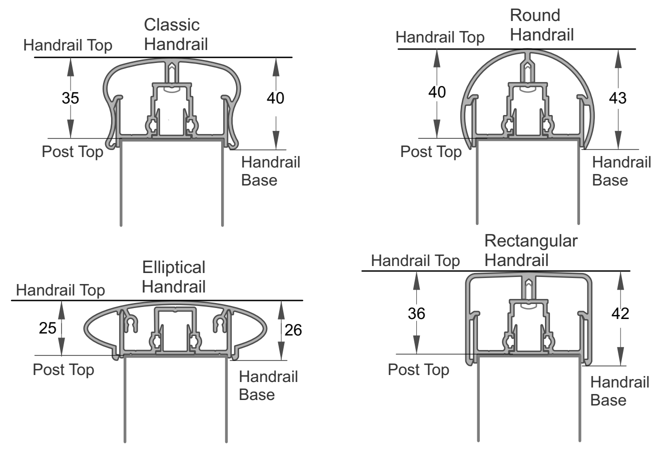
Juralco Viking® Balustrade System - Offset Details

17mm Baluster - Split Rail. Handrail + Top and Bottom Rail. Face Fix

1 - Refer Post Mounting type and installation Wind zone. Then choose Balustrade Height and max Post spacing.



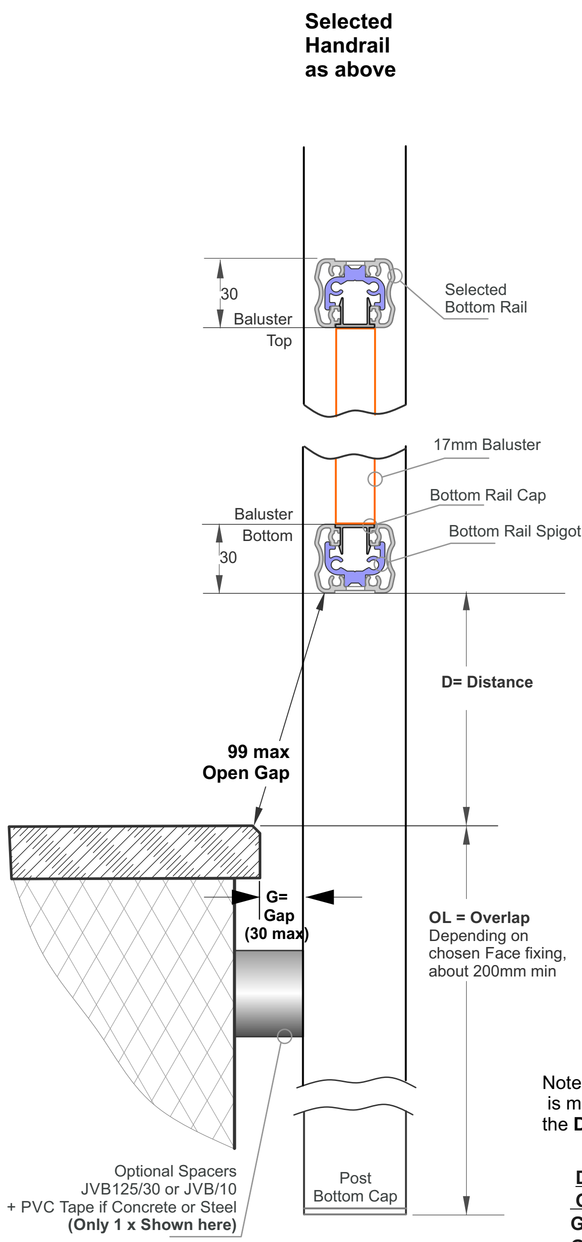
2 - Handrail - Offsets



A
For Post Height.
Post Top to Handrail Top
25-40 Depending on
chosen Handrail

B
For Top Gap Height
Handrail Base to Handrail Top
26-43 Depending on
chosen Handrail

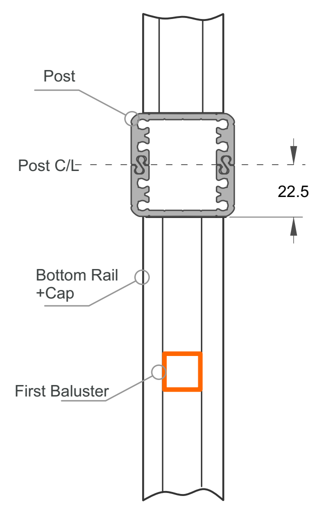
3 - Post Height offsets



Note - Because the Bottom Rail is mounted forward of the Deck Edge the **D** Dimension is no longer 99mm.

D max values for a 99mm Opening to the deck
G Gap = 10mm D = 97mm
G Gap = 20mm D = 95mm
G Gap = 30mm max D = 91mm

4 - Width Offsets



Viking post
JVB/007 shown
(or could be JVB/015)

5 - Cutting Guide

a - Hand Rail = Use maximum lengths

b - Post, Cut to
= HH - **A** + OL = Post Height

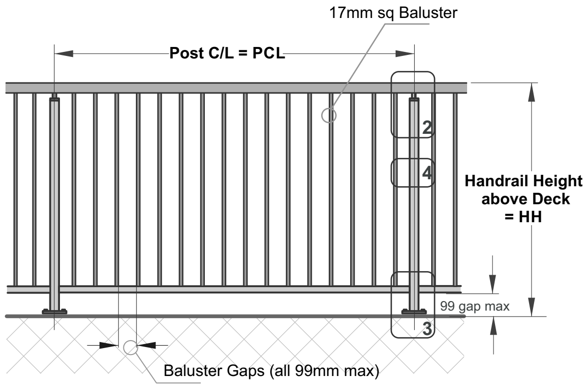
c - Bottom Rail (x2), Cut to
= PCL- 2x22.5 = PCL - 45

d - 17mm Balustrade Height
= HH - **B** - 99 - D - 2x30

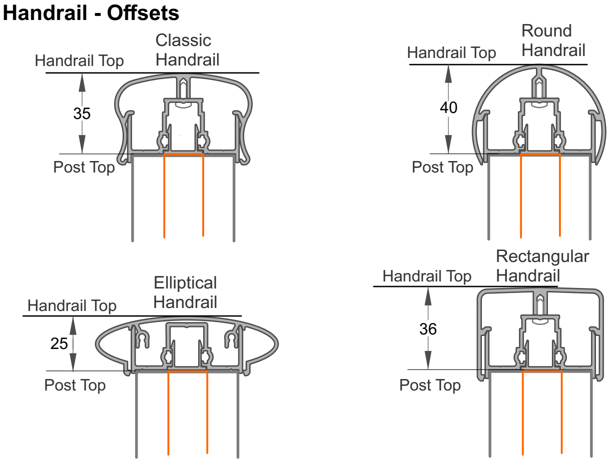
Juralco Viking® Balustrade System - Offset Details

17mm Baluster - Full Height. Handrail + Bottom Rail. Top Fix

1 - Refer Post Mounting type and installation Wind zone. Then choose Balustrade Height and max Post spacing.



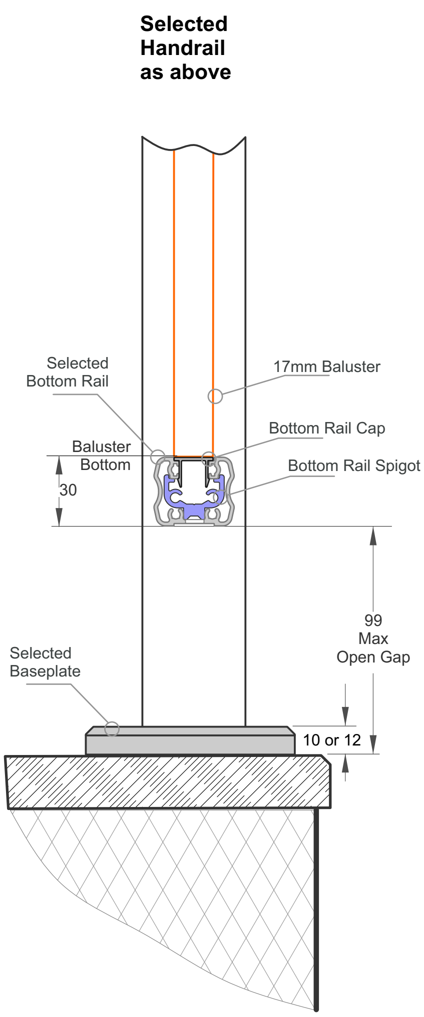
2 - Handrail - Offsets



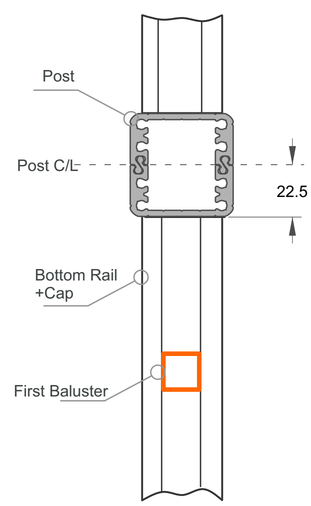
A

For Post and Baluster Top Height.
Post Top to Handrail Top
25-40 Depending on
chosen Handrail

3 - Post Height offsets



4 - Width Offsets



Viking post
JVB/007 shown
(or could be JVB/015)

5 - Cutting Guide

a - Hand Rail = Use maximum lengths

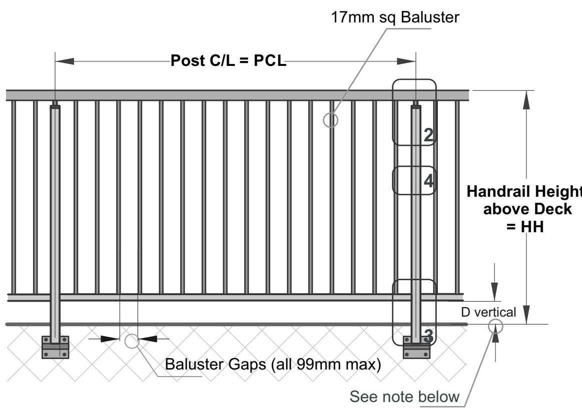
b - Post, Cut to
= HH - **A** -(10 or 12) = Post Height

c - Bottom Rail (x2), Cut to
= PCL- 2x22.5 = PCL - 45

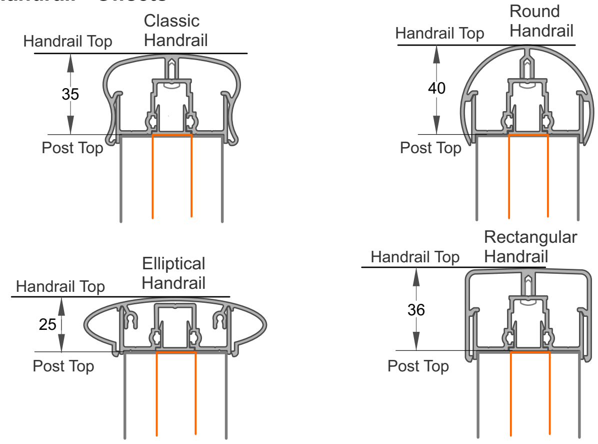
d - 17mm Balustrade Height
= HH - **A** - 99 - 30

17mm Baluster - Full Height. Handrail + Bottom Rail. Gutter Brkt Face Fix

1 - Refer Post Mounting type and installation Wind zone. Then choose Balustrade Height and max Post spacing.

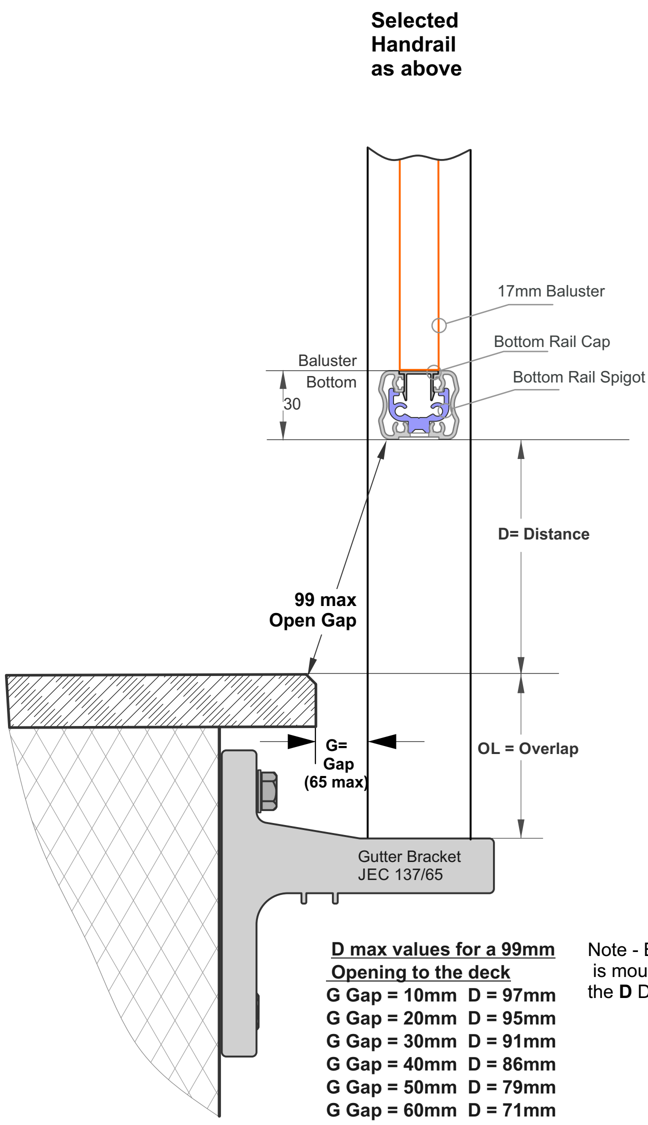


2 - Handrail - Offsets



A
For Post and Baluster Top Height. Post Top to Handrail Top 25-40 Depending on chosen Handrail

3 - Post Height offsets



D max values for a 99mm Opening to the deck
G Gap = 10mm D = 97mm
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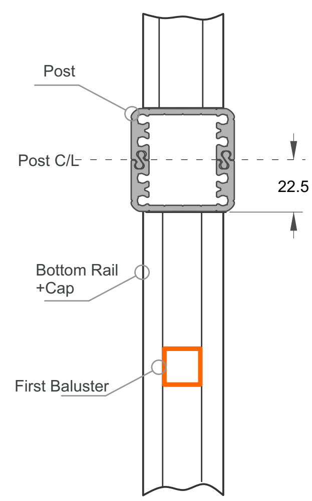
G Gap = 30mm D = 91mm
G Gap = 40mm D = 86mm
G Gap = 50mm D = 79mm
G Gap = 60mm D = 71mm
G Gap = 65mm max D = 66mm

Note - Because the Bottom Rail is mounted forward of the Deck Edge the **D** Dimension is no longer 99mm.

4 - Width Offsets



Viking post
JVB/007 shown
(or could be JVB/015)

5 - Cutting Guide

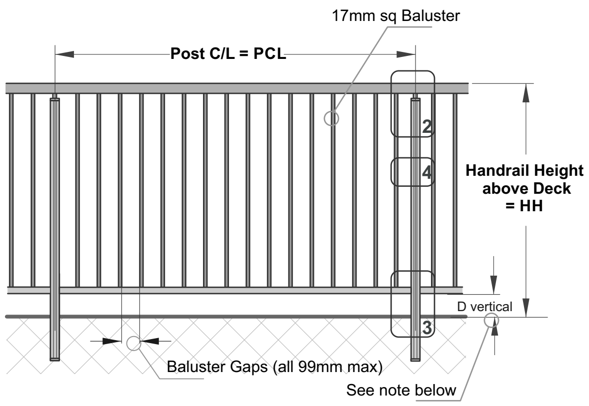
a - Hand Rail = Use maximum lengths
b - Post, Cut to
= HH - **A** + OL = Post Height
c - Bottom Rails (x2), Cut to
= PCL- 2x22.5 = PCL - 45

d - 17mm Balustrade Height
= HH - **A** - 99 - 30

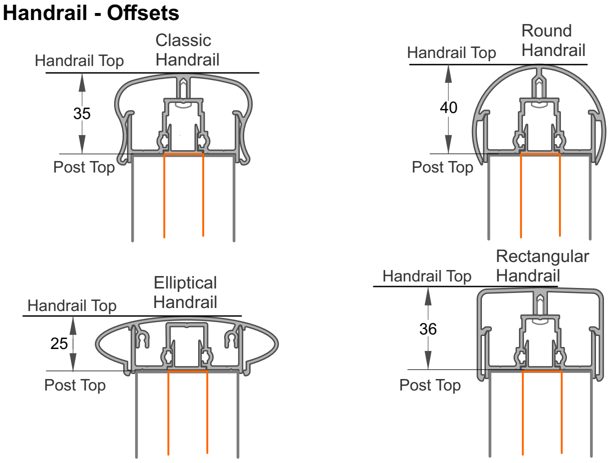
Juralco Viking® Balustrade System - Offset Details

17mm Baluster - Full Height. Handrail + Bottom Rail. Face Fix

1 - Refer Post Mounting type and installation Wind zone. Then choose Balustrade Height and max Post spacing.

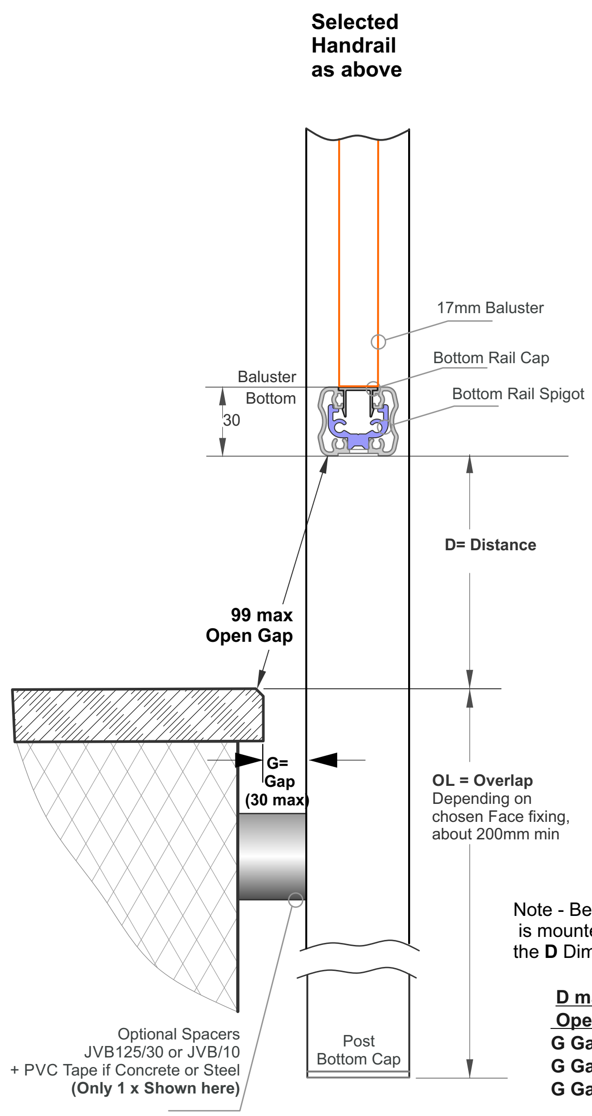


2 - Handrail - Offsets



A
For Post and Baluster Top Height.
Post Top to Handrail Top
25-40 Depending on
chosen Handrail

3 - Post Height offsets



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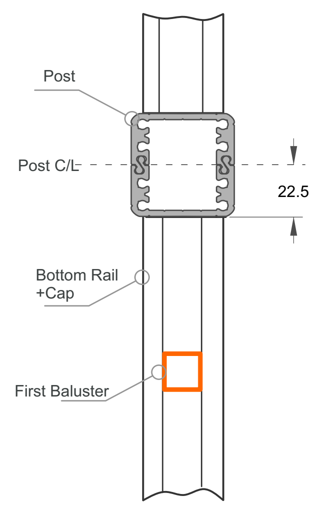

Viking post
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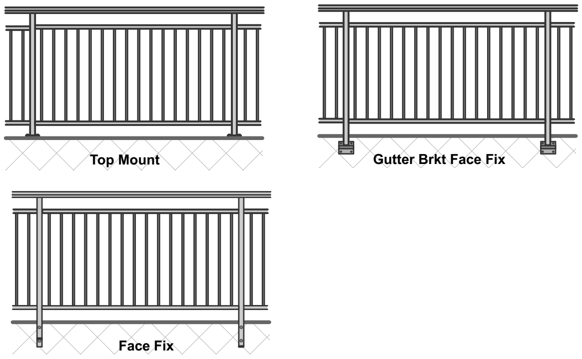
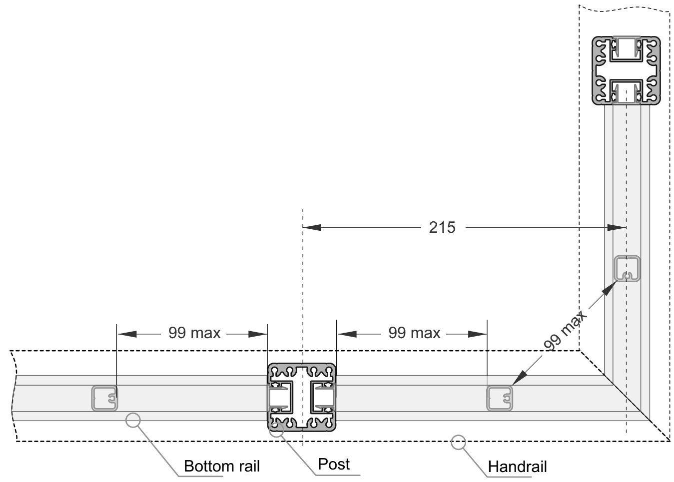
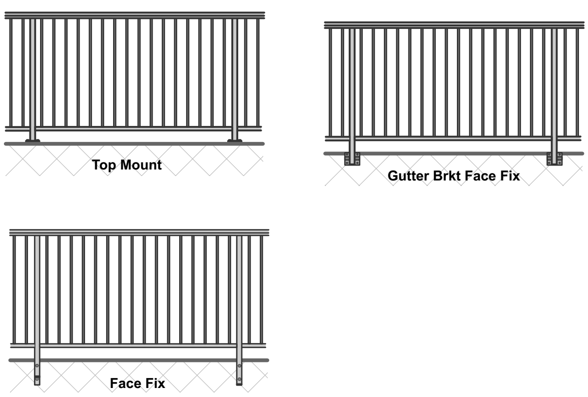
= HH - **A** - 99 - 30

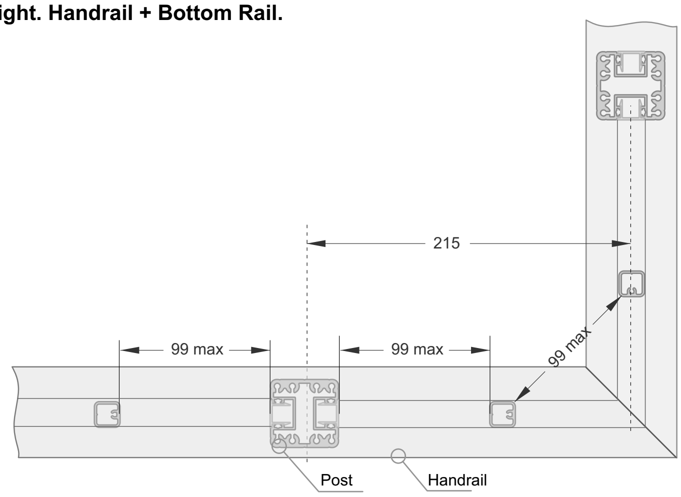
17mm Baluster - Split Rail. Handrail + Top and Bottom Rail.





17mm Baluster - Full Height. Handrail + Bottom Rail.





Powder Coating Installation Care

Warning re use of solvents:
- In some cases strong solvents are recommended for thinning various types of paints and also for cleaning up mastics and sealants.
- These can be harmful to the extended life of the powder coated surface, and must not be used for cleaning purposes.
- It is important to note that the damage will not be visible immediately and may take up to l2 months to develop.

If paint splashes or sealants and mastics need to be removed then the following may be safely used:
Methylated Spirits, Ethyl Alcohol, Isopropanol or preferably a mild detergent in warm water.

Joinery Protection during Installation:
All the activity on a construction site means that your powder coated items may get knocked or scratched, splattered with mortar, plaster, textured coating or paint during the later stages of construction.

Please ensure that all powder coated articles are masked or covered at this time.
It is far easier to prevent accidents than to try and correct them.
Should your joinery receive mortar or paint splashes see that these are removed before cure and follow the instructions contained in this brochure.

Typical sticker used to warn other trades of the need to protect and mask off powder coated joinery (applies to anodised joinery also)

"IMPORTANT ALL TRADES"
This valuable aluminium joinery will suffer permanent damage from: plaster, mortar and paint splashes - Protect if splashes occur - Immediately wash down joinery with water or meths - Do not allow splashes to harden! ~ Do not use solvents! - Do not remove this label until final clean completed.

This photograph display damage that has occurred on site, post installation. The photo of the masked joinery displays clear signs of damage that could have occurred were it not masked. Please ensure that your joinery is protected right through the entire construction process.

Powder Coating Maintenance

External - Maintenance Program:
To extend the life of external powder coated articles and to comply with warranty requirements for powder coated aluminium joinery, a simple, regular maintenance program must be implemented.
The effects of ultra violet light, atmospheric pollution, dirt, grime and airborne salt deposits will all accumulate over time and must be removed or surface staining and weathering will occur, leading to an unsightly appearance.

For external coatings, cleaning should take place every six months. In areas where pollutants are more prevalent, such as beachfront houses and industrial or geothermal areas, then a cleaning program should be carried out on a more frequent basis ie. every one to three months.

Fences or Balustrades in close proximity to swimming pools must be washed down every six months, to clean off chlorine and salt deposits.

Cleaning your powder coating:
1. Carefully remove any loose surface deposits with a wet sponge.
2. Use a soft brush (non abrasive) and a mild household detergent (do not use solvents) in warm water, remove dust, salt and other deposits.
3. Rinse off with clean fresh water.

Restoring weathered or scratched surfaces:
Repair of Scuffed or Scratched surfaces
Dulux Spray Cans are available in all colour card colours.

Repair of Small Scratches or Chips.
Dulux Dabsticks are ideally suited for the repair of small scratches.
Dabsticks may not be available in all colour card colours.

Repair of Weathered areas .
Dulux Gloss Up is a light to medium cutting cream ideally suited for gloss restoration and has been specifically designed for this purpose.
Gloss Up contains no waxes or silicone and is a one step system.

Contact Dulux Powder Coatings , ph 0064 9 441 8244